

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\A150NRC11CM001K0 Drum Syngas Compressor7percent.pvdb

Date of Analysis : Sep 18, 2026 1:33pm

PV Elite 28 U1, March 2026

Vessel Design Summary:

ASME Code, Section VIII Division 2, 2025

Diameter Spec : 2750.000 x 2750.000 mm. ID	
Vessel Design Length, Tangent to Tangent	5810.00 mm.
Distance of Bottom Tangent above Grade	3300.00 mm.
Distance of Base above Grade	300.00 mm.
Specified Datum Line Distance	3000.03 mm.
Internal Design Temperature	200 °C
Internal Design Pressure	191.000 bars
External Design Temperature	200 °C
External Design Pressure	1.034 bars
Maximum Allowable Working Pressure	191 bars
External Max. Allowable Working Pressure	128 bars
Shop Test Pressure	276.960 bars
Required Minimum Design Metal Temperature	-10.0 °C
Warmest Computed Minimum Design Metal Temperature	34.0 °C

Warning: Computed overall MDMT was higher than the required value !

Wind Design Code	ASCE/SEI 7-22
Earthquake Design Code	ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+N
Head	SA-516 70	...	...	K02700	+N
Flange	SA-266 2	...	...	K03506	
Skirt	SA-285 C	...	t <= 50	K02801	
Skirt	SA-516 70	...	...	K02700	+N
Nozzle	SA-105	...	...	K03504	
Nozzle Flg	SA-105	...	...	K03504	
Basing	SA-285 C	...	t <= 50	K02801	
Flg Bolting	SA-36	...	...	K02600	
Base Bolting	SA-36	...	...	K02600	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt 1	...	...	...	1.5000	N/A	No
Skirt 2	...	...	...	1.5000	N/A	No
Bottom D end	191.235	1.03	193.000	3.0000	N/A	No
Shell 1	191.235	1.03	193.000	3.0000	N/A	No
Shell 2	191.000	1.03	193.000	3.0000	N/A	No
Top D End	191.000	1.03	194.000	3.0000	N/A	No
body flange 2	191.000	1.03	No Calc	3.0000	N/A	No
Body flange 1	191.000	1.03	No Calc	3.0000	N/A	No

Liquid Level: 2400.03 mm. Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:



Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Skirt	-1000.0	2000.0	20.0	20.0	...	...	0.70	0.70
Skirt	-0.0	1000.0	20.0	20.0	...	...	0.70	0.70
Sphere	-0.0	0.0	104.0	98.0	96.8	7.1	1.00	1.00
Cylinder	3000.0	3000.0	192.0	192.0	190.4	16.2	1.00	1.00
Cylinder	5810.0	2810.0	192.0	192.0	190.1	16.2	1.00	1.00
Sphere	5810.0	0.0	104.0	98.0	96.7	7.1	1.00	1.00
Body Flg	7917.5	370.0	280.0	280.0	0.0	243.3	1.00	1.00
Body Flg	3224.9	344.0	260.0	260.0	0.0	251.1	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N3_3"	WRC-107/537	0.986	n/a	Passed
N1_28"	WRC-297	0.993	Shell	Passed
N4_2"	WRC-297	0.993	Shell	Passed
N2_28"	WRC-107/537	0.985	n/a	Passed

Wind/Earthquake Shear, Bending:

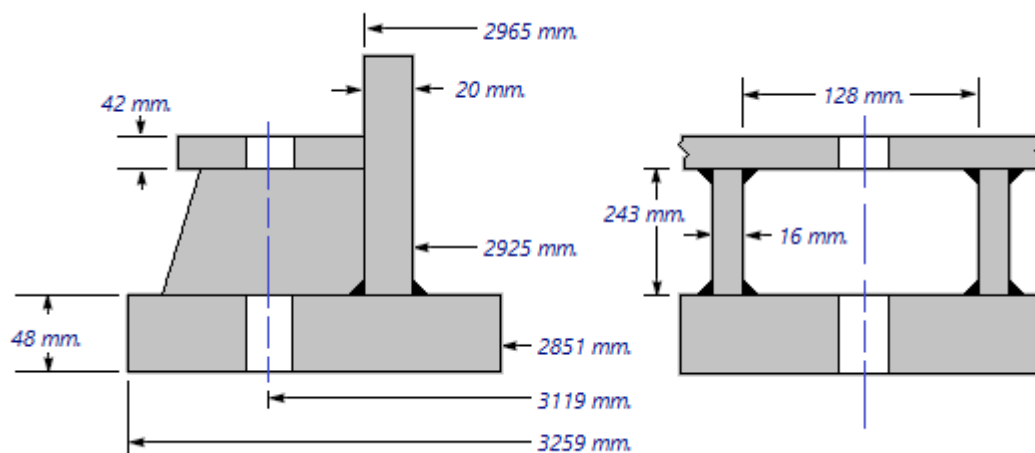
From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1000	10845.9	38588.3	76304.4	259430
20	30	2500	9967.91	38432.1	55490.1	182408
30	40	2999.98	9528.92	38299	45741.8	144043
40	50	4500	9528.91	36630.4	45741.5	144042
50	60	7405	7390.63	24289.9	20361.7	52659.6
60	70	8809.98	4866.84	9039.21	3139.77	5831.5

Abs Max of the all of the Stress Ratio's : 0.9356

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	48.0000	mm.
Inside Diameter of Basing	2850.9998	mm.
Outside Diameter of Basing	3258.9998	mm.
Nominal Diameter of Bolts	48.0000	mm.
Diameter of Bolt Circle	3119.0000	mm.
Number of Bolts	16	
Thickness of Gusset Plates	16.0000	mm.
Average Width of Gusset Plates	147.0000	mm.
Height of Gussets	243.0000	mm.
Distance between Gussets	128.0000	mm.
Thickness of Top Plate or Ring	42.0000	mm.
Radial Width of the Top Plate	147.0000	mm.
Number of Gussets	32	

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	29692.	Kgf
Total Earthquake Shear on Support	57434.	Kgf
Wind Moment on Support	307603.	Kg-m.
Earthquake Moment on Support	490728.	Kg-m.
Maximum Initial Bolt Load	27058.	Kgf

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	144763.3	kg.
Shop Test - Fabricated + water (full)	199461.8	kg.
Shipping - Fab. + rem. intls.+ shipping app.	155426.7	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	163304.7	kg.
Empty - Fab. + intls. + details + wghts.	163304.7	kg.
Operating - Empty + operating liquid (no ca)	177553.6	kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	Tbl 4.5.2	Weld	Pass/Fail
N3_3"	194	OK	...	Failed	OK	Passed
N5A_2"	193	OK	...	OK	OK	Passed
N5B_2"	193	OK	...	OK	OK	Passed
N7A_2"	193	OK	...	OK	OK	Passed
N8A_2"	193	OK	...	OK	OK	Passed
N9A_2"	193	OK	...	OK	OK	Passed
N7B_2"	193	OK	...	OK	OK	Passed
N8B_2"	193	OK	...	OK	OK	Passed
N9B_2"	193	OK	...	OK	OK	Passed
N1_28"	193	OK	...	OK	OK	Passed
N4_2"	193	OK	...	OK	OK	Passed
M1_24"	193	OK	...	...	OK	Passed
N6_3"	191	OK	...	OK	OK	Passed
N2_28"	194	OK	...	OK	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 191 Nozzle : N63"
 Minimum MAWP Shells/Flanges : 191 Element : Shell 1
 Minimum MAPnc Shells/Flanges : 229 Element : Shell 2

Computed Vessel M.A.W.P. : 191 bars

VIII-2 Nozzle to Gross Structural Discontinuity Check:

No Nozzles were too close to any gross structural discontinuities per 4.5.13.1

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm.	Layout Angle deg	Inside Radius mm.
30	N3_3"	0.000	0.000	41.100
40	N5A_2"	1249.970	180.000	28.400
40	N5B_2"	649.975	90.000	28.400
40	N7A_2"	2199.970	270.000	28.400
40	N8A_2"	1999.970	180.000	28.400
40	N9A_2"	1249.970	270.000	28.400
40	N7B_2"	649.975	270.000	28.400
40	N8B_2"	449.975	180.000	28.400
40	N9B_2"	249.975	270.000	28.400
50	N1_28"	3224.975	180.000	358.600
50	N4_2"	3599.975	270.000	28.400
50	M1_24"	3999.975	90.000	307.800
60	N6_3"	0.000	0.000	41.100
60	N2_28"	0.000	0.000	358.600

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

4.5.13 spacing violation detected between : N128" and N8A2".

The Centerline Dist. is 1225.00 while the sum of the inside Radii is 1238.14 mm..

4.5.13 spacing violation detected between : N228" and N63".

The Centerline Dist. is 1039.86 while the sum of the inside Radii is 1083.83 mm..





Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N5A_2"	2.000 in	Actual	RTJWn	2.787	10.000	...	...	394.65	...
N5B_2"	2.000 in	Actual	RTJWn	2.787	10.000	...	...	394.65	...
N7A_2"	2.000 in	Actual	RTJWn	2.787	10.000	...	...	394.65	...
N8A_2"	2.000 in	Actual	RTJWn	2.787	10.000	...	...	394.65	...
N9A_2"	2.000 in	Actual	RTJWn	2.787	10.000	...	...	394.65	...
N7B_2"	2.000 in	Actual	RTJWn	2.787	10.000	...	...	394.65	...
N8B_2"	2.000 in	Actual	RTJWn	2.787	10.000	...	...	394.65	...
N9B_2"	2.000 in	Actual	RTJWn	2.787	10.000	...	...	394.65	...
N4_2"	2.000 in	Actual	RTJWn	2.787	10.000	...	...	394.65	...
N3_3"	3.000 in	Actual	RTJWn	3.787	10.000	...	...	301.39	1500
N6_3"	3.000 in	Actual	RTJWn	3.787	10.000	...	...	425.52	1500
M1_24"	24.000 in	Actual	RTJWn	27.937	50.000	...	...	588.02	...
N1_28"	28.000 in	Actual	None	32.409	56.000	...	...	772.94	...
N2_28"	28.000 in	Actual	None	32.409	56.000	...	...	888.22	...

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grv Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grv Weld	Inside Weld
N5A_2"	SA-105	192.000	10.000	...	...	...
N5B_2"	SA-105	192.000	10.000	...	...	...
N7A_2"	SA-105	192.000	10.000	...	...	...
N8A_2"	SA-105	192.000	10.000	...	...	...
N9A_2"	SA-105	192.000	10.000	...	...	...
N7B_2"	SA-105	192.000	10.000	...	...	...
N8B_2"	SA-105	192.000	10.000	...	...	...
N9B_2"	SA-105	192.000	10.000	...	...	...
N4_2"	SA-105	91.000	10.000	...	...	...
N3_3"	SA-105	98.000	10.000	...	...	...
N6_3"	SA-105	98.000	10.000	...	...	...
M1_24"	SA-105	192.000	10.000	...	...	...
N1_28"	SA-105	192.000	10.000	...	...	...
N2_28"	SA-105	98.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
N5A_2"	1249.945	180.0	200.00	0.00	Shell 1
N5B_2"	649.950	90.0	200.00	0.00	Shell 1
N7A_2"	2199.944	270.0	200.00	0.00	Shell 1
N8A_2"	1999.945	180.0	200.00	0.00	Shell 1
N9A_2"	1249.945	270.0	200.00	0.00	Shell 1
N7B_2"	649.950	270.0	200.00	0.00	Shell 1
N8B_2"	449.950	180.0	200.00	0.00	Shell 1
N9B_2"	249.950	270.0	200.00	0.00	Shell 1
N4_2"	3599.950	270.0	200.00	0.00	Shell 2

N3_3"	...	0.0	200.00	0.00	Bottom D end
N6_3"	...	0.0	200.00	0.00	Top D End
M1_24"	3999.949	90.0	300.00	0.00	Shell 2
N1_28"	3224.950	180.0	450.00	0.00	Shell 2
N2_28"	...	0.0	685.50	0.00	Top D End



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	SPHERICAL HEAD: 104.0mm. THK X 2844.0mm. ID X 0.0mm.	SA-516 70
1	WELD NECK FLANGE: 280.0mm. THK X 1279.9mm. OD	SA-266 2
1	WELD NECK FLANGE: 260.0mm. THK X 1279.9mm. OD	SA-266 2
2	PLATFORM: 1000mm X 1500mm WIDE	
1	PLATFORM: 1000mm X 5114mm WIDE	
1	LADDER: 10150.0mm	...
1	24.0" BLIND FLANGE(S)	SA-105
1	CLASS 1500 GR 1.1, 3.0" BLIND FLANGE(S)	SA-105
1	BASERING: 3259.0mm. OD X 2851.0 ID X 48.0mm.	SA-285 C
16	BASERING BOLTS: 48.0mm. DIA	SA-36
16	BASERING BOLT NUTS: 48.0mm. DIA	...
1	BASERING TOP RING: 3259.0mm. OD X 2965.0 ID X 42.0mm.	...
32	BASERING GUSSET PLATES: 147.0mm. X 147.0mm. X 243.0mm. X 16.0mm.	...
1	GASKET: 800.1mm. OD X 743.0mm. ID	...
16	BODY FLANGE BOLTS: 100.0mm. DIA	SA-36
32	NUTS FOR BODY FLANGE BOLTS: 100.0mm. DIA	...
2	CYLINDRICAL SKIRT SEGMENT 20.0mm. THK X 3195.4mm. X 2000.0mm.	SA-285 C
1	CYLINDRICAL SKIRT SEGMENT 20.0mm. THK X 2798.1mm. X 2000.0mm.	SA-285 C
1	CYLINDRICAL SKIRT SEGMENT 20.0mm. THK X 3190.6mm. X 1000.0mm.	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 20.0mm. THK X 3216.2mm. X 1000.0mm.	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 20.0mm. THK X 2782.2mm. X 1000.0mm.	SA-516 70
2	CYLINDRICAL SEGMENT 192.0mm. THK X 3431.7mm. X 3000.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 192.0mm. THK X 1775.8mm. X 3000.0mm.	SA-516 70
2	CYLINDRICAL SEGMENT 192.0mm. THK X 3048.0mm. X 2810.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 192.0mm. THK X 2543.3mm. X 2810.0mm.	SA-516 70
1	NAMEPLATE	...

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm.	E*
Notes							
Bottom D end	[9] B	21	13	Yes	0.842	98.000	1.00
Shell 1	[8][17] Impact Testing reqd			Yes			
Shell 2	[8][17] Impact Testing reqd			Yes			
Top D End	[9] !	-10		Yes			
N33"	[1] A	1	-15	Yes	0.720	10.000	1.00
Nozzle Flg	[5] A	-18	-32	Yes			
N5A2"	[1] A	1	-36	Yes	0.497	10.000	1.00
N5B2"	[1] A	1	-36	Yes	0.498	10.000	1.00
N7A2"	[1] A	1	-36	Yes	0.497	10.000	1.00
N8A2"	[1] A	1	-36	Yes	0.497	10.000	1.00
N9A2"	[1] A	1	-36	Yes	0.497	10.000	1.00
N7B2"	[1] A	1	-36	Yes	0.498	10.000	1.00
N8B2"	[1] A	1	-36	Yes	0.498	10.000	1.00
N9B2"	[1] A	1	-36	Yes	0.498	10.000	1.00
N128"	[1]			Yes			
N42"	[1] A	1	-36	Yes	0.497	10.000	1.00
M124"	[1] A	30	19	Yes	0.802	50.000	1.00
N63"	[1] A	1	-15	Yes	0.719	10.000	1.00
Nozzle Flg	[5] A	-18	-32	Yes			
N228"	[1] A	42	34	Yes	0.840	98.000	1.00
Nozl. Bolting[22]		-48		No	1.000	64.000	1.00

Warmest MDMT: 42 34

Required Minimum Design Metal Temperature -10.0 °C
 Warmest Computed Minimum Design Metal Temperature 34.0 °C

Warning: Computed overall MDMT was higher than the required value !

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld



Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.

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Units used in this Analysis (bars):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	191 bars
Design Internal Temperature	200.0 °C
Projection of Nozzle from Vessel Top	500 mm.
Projection of Nozzle from Vessel Bottom	500 mm.
Minimum Design Metal Temperature	-10.0 °C
Type of Construction	Welded
Special Service	None
Use Higher Longitudinal Stresses (Flag)	N
Select t for Internal Pressure (Flag)	N
Select t for External Pressure (Flag)	N
Select t for Axial Stress (Flag)	N
Select Location for Stiff. Rings (Flag)	N
Consider Vortex Shedding	N

Shop Pressure Test:

Type of Pressure Test	Hydrostatic
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW



Wind Design Code	ASCE/SEI 7-22	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 180	Km/hr
Surface Roughness Category	D: Flat, unobstructed	
Type of Surface	0	
Base Elevation	300	mm.
Percent Wind for Hydrotest	33.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	H or Hh 0	mm.
Distance Upwind of Crest	Lh 0	mm.
Distance from Crest to the Vessel	x 0	mm.
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm.

Seismic Design Code	ASCE/SEI 7-22	
Importance Factor	1.500	
Force Modification Factor R	1.500	
Site Class	D	
Component Elevation Ratio	z/h 0.000	
Amplification Factor	Ap 0.000	
Force Factor	0.000	
Consider Vertical Acceleration	No	
Minimum Acceleration Multiplier	0.000	
User Value of Sds (used if > 0)	0.000	
User Value of Sd1 (used if > 0)	0.000	
Component resonance duct. fact.	[CAR] 0.000	
Component strength factor	[Rpo] 0.000	
Overstrength factor	[Omega0] 0.000	
Response Acceleration Factor	[SMS] 0.326	
Response Acceleration Factor	[SM1] 0.139	
Building Importance Factor	[Ie] 1.000	
Site Latitude	0.000	deg
Site Longitude	0.000	deg

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Using Metric Material Databases, ASME II D	Yes
Using ASME VIII-1 Appendix 46	No
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/8
Element From Node	10
Element To Node	20
Element Type	Skirt Support
Description	Skirt 1
Distance "FROM" to "TO"	2000 mm.
Skirt Outside Diameter	2965 mm.
Diameter of Skirt at Base	2965 mm.
Skirt Thickness	20 mm.
Internal Corrosion Allowance	0.75 mm.
Nominal Thickness	20 mm.
External Corrosion Allowance	0.75 mm.
Design Temperature	200 °C
Internal Pressure	



Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.3200001	
Material Name	SA-285 C	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	118	N./mm ²
Allowable Stress, Hydrotest	194.75	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	0	mm.
Yield Stress, Operating	177	N./mm ²
MDMT Curve	A	
External Pressure Chart Name	CS-2	
UNS Number	K02801	
Class / Thickness / Grade	::t <= 50	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	

Element#	2/8
Element From Node	20
Element To Node	30
Element Type	Skirt Support
Description	Skirt 2
Distance "FROM" to "TO"	1000 mm.
Skirt Outside Diameter	2965 mm.
Diameter of Skirt at Base	2965 mm.
Skirt Thickness	20 mm.
Internal Corrosion Allowance	0.75 mm.
Nominal Thickness	20 mm.
External Corrosion Allowance	0.75 mm.
Design Temperature Internal Pressure	200 °C
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.3200001
Material Name	SA-516 70 +N
Allowable Stress, Ambient	175 N./mm ²
Allowable Stress, Operating	150 N./mm ²
Allowable Stress, Hydrotest	247 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	0 mm.
Yield Stress, Operating	225 N./mm ²
MDMT Curve	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No

Element#	3/8
Element From Node	30
Element To Node	40
Element Type	Spherical
Description	Bottom D end
Distance "FROM" to "TO"	0 mm.
Inside Diameter	2844 mm.
Element Thickness	98 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	104 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	191 bars
Design Temperature Internal Pressure	200 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.3200001
Material Name	SA-516 70 +N
Efficiency, Longitudinal Seam	1.0



Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid 30	
Dist. from "FROM" Node / Offset dist	-1419	mm.
Height/Length of Liquid	0.02529	mm.
Liquid Density	0.0009984	kg./cm ³
Element From Node	30	
Detail Type	Nozzle	
Detail ID	N33"	
Dist. from "FROM" Node / Offset dist	0	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	None	
Nozzle Class	1500	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	69.382	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	30	
Detail Type	Weight	
Detail ID	Btm Dish SF	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	3900	Kgf
Offset from Element Centerline	0	mm.

Element#	4/8	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	Shell 1	
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	2750	mm.
Element Thickness	192	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	192	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	191	bars
Design Temperature Internal Pressure	200	°C
Design External Pressure	1.0342	bars
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.3200001	
Material Name	SA-516 70 +N	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Platform	
Detail ID	PLAT 1	
Dist. from "FROM" Node / Offset dist	0	mm.
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	360.0	
Platform Wind Area	60000	cm ²
Platform Weight	3850.5	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	1500	mm.
Platform Height	1000	mm.
Platform Clearance or End Offset	85	mm.
Platform Force Coefficient	2.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm.
Ladder End Elevation	0	mm.



Unit Weight of Ladder 0 Kg/mm.
 Platform Length (top head platform) 0 mm.

Element From Node 40
 Detail Type Liquid
 Detail ID Liquid: 40
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Liquid 2400 mm.
 Liquid Density 0.0009984 kg/cm³

Element From Node 40
 Detail Type Nozzle
 Detail ID N5A2"
 Dist. from "FROM" Node / Offset dist 1250 mm.
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 1500
 Layout Angle 180.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 59.48 Kg
 Grade of Attached Flange None
 Nozzle Matl SA-105

Element From Node 40
 Detail Type Nozzle
 Detail ID N5B2"
 Dist. from "FROM" Node / Offset dist 649.97 mm.
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 1500
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 59.48 Kg
 Grade of Attached Flange None
 Nozzle Matl SA-105

Element From Node 40
 Detail Type Nozzle
 Detail ID N7A2"
 Dist. from "FROM" Node / Offset dist 2200 mm.
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 1500
 Layout Angle 270.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 59.48 Kg
 Grade of Attached Flange None
 Nozzle Matl SA-105

Element From Node 40
 Detail Type Nozzle
 Detail ID N8A2"
 Dist. from "FROM" Node / Offset dist 2000 mm.
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 1500
 Layout Angle 180.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 59.48 Kg
 Grade of Attached Flange None
 Nozzle Matl SA-105

Element From Node 40
 Detail Type Nozzle
 Detail ID N9A2"
 Dist. from "FROM" Node / Offset dist 1250 mm.
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 1500
 Layout Angle 270.0



Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	59.48	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-105	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N7B2"	
Dist. from "FROM" Node / Offset dist	649.97	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	1500	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	59.48	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-105	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N8B2"	
Dist. from "FROM" Node / Offset dist	449.98	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	1500	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	59.48	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-105	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N9B2"	
Dist. from "FROM" Node / Offset dist	249.98	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	1500	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	59.48	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-105	
Element From Node	40	
Detail Type	Weight	
Detail ID	Lifting WT	
Dist. from "FROM" Node / Offset dist	2900	mm.
Miscellaneous Weight	3000	Kgf
Offset from Element Centerline	0	mm.
Element From Node	40	
Detail Type	Weight	
Detail ID	Clip WT	
Dist. from "FROM" Node / Offset dist	2905	mm.
Miscellaneous Weight	1010	Kgf
Offset from Element Centerline	0	mm.
Element From Node	40	
Detail Type	Weight	
Detail ID	Miscellaneous W	
Dist. from "FROM" Node / Offset dist	2910	mm.
Miscellaneous Weight	10000	Kgf
Offset from Element Centerline	0	mm.



Element Type	Cylinder
Description	Shell 2
Distance "FROM" to "TO"	2810 mm.
Inside Diameter	2750 mm.
Element Thickness	192 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	192 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	191 bars
Design Temperature Internal Pressure	200 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.3200001
Material Name	SA-516 70 +N
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Platform
Detail ID	PLAT 2
Dist. from "FROM" Node / Offset dist	1000 mm.
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	180.0
Platform Wind Area	60000 cm²
Platform Weight	1857.1 Kg
Platform Railing Weight	0 Kg/mm.
Platform Grating Weight	0.017 Kgs/cm²
Platform Width	1500 mm.
Platform Height	1000 mm.
Platform Clearance or End Offset	0 mm.
Platform Force Coefficient	2.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm.
Ladder End Elevation	0 mm.
Unit Weight of Ladder	0 Kg/mm.
Platform Length (top head platform)	0 mm.

Element From Node	50
Detail Type	Nozzle
Detail ID	N128"
Dist. from "FROM" Node / Offset dist	224.98 mm.
Nozzle Diameter	28 in.
Nozzle Schedule	None
Nozzle Class	None
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	3011.2 Kg
Grade of Attached Flange	None
Nozzle Matl	SA-105

Element From Node	50
Detail Type	Nozzle
Detail ID	N42"
Dist. from "FROM" Node / Offset dist	599.97 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	1500
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	None
Nozzle Matl	SA-105

Element From Node	50
Detail Type	Nozzle
Detail ID	M124"
Dist. from "FROM" Node / Offset dist	999.97 mm.
Nozzle Diameter	24 in.
Nozzle Schedule	None



Nozzle Class	1500	
Layout Angle	90.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	4749.3	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-105	

Element From Node	50	
Detail Type	Weight	
Detail ID	Pipe Davit WT	
Dist. from "FROM" Node / Offset dist	2500	mm.
Miscellaneous Weight	800	Kgf
Offset from Element Centerline	1400	mm.

Element From Node	50	
Detail Type	Weight	
Detail ID	Manway davit	
Dist. from "FROM" Node / Offset dist	1000	mm.
Miscellaneous Weight	50	Kgf
Offset from Element Centerline	1600	mm.

Element From Node	50	
Detail Type	Weight	
Detail ID	Demister WT	
Dist. from "FROM" Node / Offset dist	2400	mm.
Miscellaneous Weight	430	Kgf
Offset from Element Centerline	0	mm.

Element From Node	50	
Detail Type	Weight	
Detail ID	Demister suppor	
Dist. from "FROM" Node / Offset dist	2400	mm.
Miscellaneous Weight	30	Kgf
Offset from Element Centerline	0	mm.

Element From Node	50	
Detail Type	Weight	
Detail ID	N10VH	
Dist. from "FROM" Node / Offset dist	225	mm.
Miscellaneous Weight	3930	Kgf
Offset from Element Centerline	2413	mm.

Element#	6/8	
Element From Node	60	
Element To Node	70	
Element Type	Spherical	
Description	Top D End	
Distance "FROM" to "TO"	0	mm.
Inside Diameter	2844	mm.
Element Thickness	98	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	104	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	191	bars
Design Temperature Internal Pressure	200	°C
Design External Pressure	1.0342	bars
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.3200001	
Material Name	SA-516 70 +N+IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Platform	
Detail ID	PLAT 3	
Dist. from "FROM" Node / Offset dist	1800	mm.
Platform Start Angle (degrees)	0.0	



Platform End Angle (degrees)	0.0	
Platform Wind Area	184779	cm ²
Platform Weight	4955.8	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	5114	mm.
Platform Height	1000	mm.
Platform Clearance or End Offset	2557	mm.
Platform Force Coefficient	2.0	
Ladder Layout Angle	45.0	
Ladder Start Elevation	-3500	mm.
Ladder End Elevation	6650	mm.
Unit Weight of Ladder	0.05	Kgf/mm.
Platform Length (top head platform)	5114	mm.
Element From Node	60	
Detail Type	Nozzle	
Detail ID	N63"	
Dist. from "FROM" Node / Offset dist	950	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	None	
Nozzle Class	1500	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	82.732	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	60	
Detail Type	Nozzle	
Detail ID	N228"	
Dist. from "FROM" Node / Offset dist	0	mm.
Nozzle Diameter	28	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	1690.7	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-105	
Element From Node	60	
Detail Type	Weight	
Detail ID	Top Dish SF	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	3900	Kgf
Offset from Element Centerline	0	mm.
Element From Node	60	
Detail Type	Weight	
Detail ID	N20VH	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	3948	Kgf
Offset from Element Centerline	2913	mm.
Element From Node	60	
Detail Type	For./Mom.	
Detail ID	N2FXFZ	
Dist. from "FROM" Node / Offset dist	1400	mm.
Force in X Direction	18846	Kgf
Force in Y Direction	-18846	Kgf
Force in Z Direction	0	Kgf
Moment about X Axis	34286	Kg-m.
Moment about Y Axis	0	Kg-m.
Moment about Z Axis	34286	Kg-m.
Force/Moment Combination Method	SRSS	



Element From Node	70	
Element To Node	80	
Element Type	Flange	
Description	body flange 2	
Distance "FROM" to "TO"	370	mm.
Flange Inside Diameter	711.2	mm.
Element Thickness	280	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	280	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	191	bars
Design Temperature Internal Pressure	200	°C
Design External Pressure	1.0342	bars
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.3200001	
Material Name	SA-266 2	
Allowable Stress, Ambient	165	N./mm ²
Allowable Stress, Operating	142	N./mm ²
Allowable Stress, Hydrotest	237.5	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	0	mm.
Yield Stress, Operating	213	N./mm ²
MDMT Curve	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element#	8/8	
Element From Node	80	
Element To Node	90	
Element Type	Flange	
Description	Body flange 1	
Distance "FROM" to "TO"	344	mm.
Flange Inside Diameter	711.2	mm.
Element Thickness	260	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	260	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	191	bars
Design Temperature Internal Pressure	200	°C
Design External Pressure	1.0342	bars
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.3200001	
Material Name	SA-266 2	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
Skirt 1	...	...	2000	...	2000
Skirt 2	...	...	3000	...	1000
Bottom D end	...	...	3000	...	...
Shell 1	...	...	6000	...	3000
Shell 2	...	...	8810	...	2810
Top D End	...	...	8810	...	...
body flange 2	...	...	10917.5	...	370
Body flange 1	...	...	6224.98	...	344

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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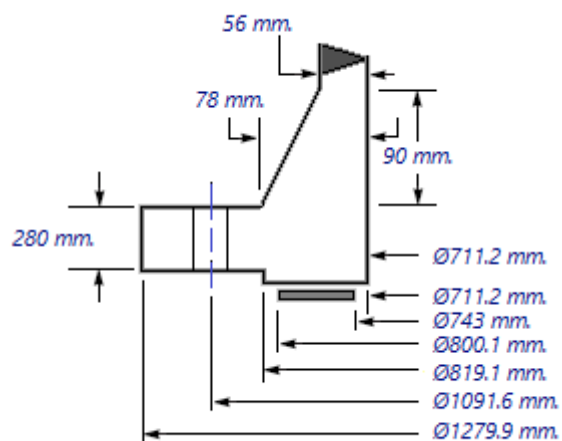


Flange Input Data Values

Description: New Flange :

body flange 2

Flange Type	Integral Weld Neck		
Design Pressure	P	191.00	bars
Design Temperature		200	°C
Internal Corrosion Allowance	ci	3.0000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	711.200	mm.
Flange Outside Diameter	A	1279.900	mm.
Flange Thickness	t	280.0000	mm.
Thickness of Hub at Small End	go	56.0000	mm.
Thickness of Hub at Large End	gl	78.0000	mm.
Length of Hub	h	90.0000	mm.
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	142.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	165.00	N./mm ²
Bolt Material		SA-36	
Bolt Allowable Stress At Temperature	Sb	142.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	165.00	N./mm ²
Diameter of Bolt Circle	C	1091.550	mm.
Nominal Bolt Diameter	a	100.0000	mm.
Type of Threads		TEMA Metric	
Number of Bolts		16	
Flange Face Outside Diameter	Fod	819.150	mm.
Flange Face Inside Diameter	Fid	711.200	mm.
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	800.100	mm.
Gasket Inside Diameter	Gi	742.950	mm.
Gasket Factor	m	5.5000	
Gasket Design Seating Stress	y	124.11	N./mm ²
Gasket Thickness	tg	3.1750	mm.
Raised face depth	Fd	0.0000	mm.
Flange Face Nubbin Width	w	31.7500	mm.
Axial Force on Flange		17132.38	Kgf
Bending Moment on Flange		44079.48	Kg-m.



Flange Bolt Area Problem Encountered for New Flange

The req'd bolt area (1158.06 cm²) is > than the actual bolt area (1064.245 cm²) for this flange design. Increase the number or size of bolts and attempt the analysis again.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	119.64	247.50	135.74	247.50
Radial Flange	24.25	165.00	27.52	165.00
Tangential Flange	103.84	165.00	117.82	165.00
Maximum Average	111.74	165.00	126.78	165.00
Bolting	154.52	165.00	27.26	165.00

Minimum Req'd Thickness, ambient, no Corrosion: 240.182 mm.

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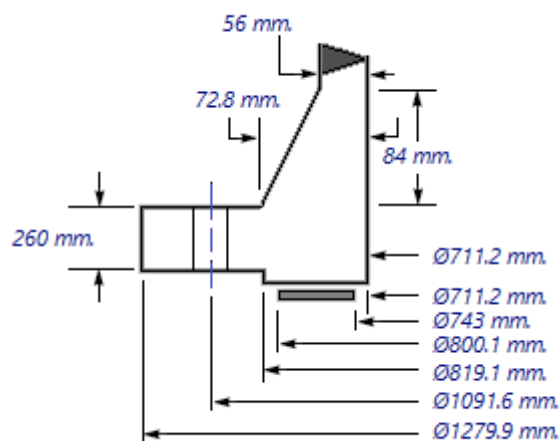


Flange Input Data Values

Description: New Flange :

Body flange 1

Flange Type	Integral Weld Neck		
Design Pressure	P	191.00	bars
Design Temperature		200	°C
Internal Corrosion Allowance	ci	3.0000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	711.200	mm.
Flange Outside Diameter	A	1279.900	mm.
Flange Thickness	t	260.0000	mm.
Thickness of Hub at Small End	go	56.0000	mm.
Thickness of Hub at Large End	gl	72.8000	mm.
Length of Hub	h	84.0000	mm.
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	142.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	165.00	N./mm ²
Bolt Material		SA-36	
Bolt Allowable Stress At Temperature	Sb	142.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	165.00	N./mm ²
Diameter of Bolt Circle	C	1091.550	mm.
Nominal Bolt Diameter	a	100.0000	mm.
Type of Threads		TEMA Metric	
Number of Bolts		16	
Flange Face Outside Diameter	Fod	819.150	mm.
Flange Face Inside Diameter	Fid	711.200	mm.
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	800.100	mm.
Gasket Inside Diameter	Gi	742.950	mm.
Gasket Factor	m	5.5000	
Gasket Design Seating Stress	y	124.11	N./mm ²
Gasket Thickness	tg	3.1750	mm.
Raised face depth	Fd	0.0000	mm.
Flange Face Nubbin Width	w	31.7500	mm.
Axial Force on Flange		17132.38	Kgf
Bending Moment on Flange		44079.48	Kg-m.



Flange Bolt Area Problem Encountered for New Flange

The req'd bolt area (1158.06 cm²) is > than the actual bolt area (1064.245 cm²) for this flange design. Increase the number or size of bolts and attempt the analysis again.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	152.32	247.50	171.27	247.50
Radial Flange	30.06	165.00	33.80	165.00
Tangential Flange	118.35	165.00	133.07	165.00
Maximum Average	135.33	165.00	152.17	165.00
Bolting	154.52	165.00	27.26	165.00

Minimum Req'd Thickness, ambient, no Corrosion: 248.310 mm.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
Skirt 1		...	20	1.5	2925	...	1
Skirt 2		...	20	1.5	2925	...	2
Bottom D end		191.24	104	3	2844	150	3
Shell 1		191.24	192	3	2750	150	4
Shell 2		191	192	3	2750	150	5
Top D End		191	104	3	2844	150	6
body flange 2		191	280	3	711.2	142	7
Body flange 1		191	260	3	711.2	142	8

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.W.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
Skirt 1		...	No Calc	No Calc	20	No Calc	1
Skirt 2		...	No Calc	No Calc	20	No Calc	2
Bottom D end		191	193.368	233.243	98	96.8009	3
Shell 1		191	192.547	228.724	192	190.385	4
Shell 2		191	192.782	228.724	192	190.14	5
Top D End		191	193.604	233.248	98	96.681	6
body flange 2		191	No Calc	No Calc	280	...	7
Body flange 1		191	No Calc	No Calc	260	...	8

Minimum 191 229

MAWP: 191 bars, limited by: Nozzle Reinforcement.

Warning: The computed MAWP is less than the design pressure.

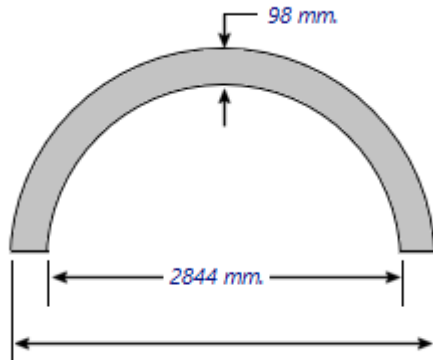
PV Elite could not compute the MAWP of one or more Flanges. Please check the reported MAWP by entering it in as the design pressure and performing an extra analysis. Please note that flanges computed by the EN 1591 and EN 13445 Annex G methods do not have a MAWP, MAP or required thickness calculation.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Bottom D end

Spherical Head Design Information:					
Design Pressure	P	191.235 bars	Int. Design Temperature		200 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		0.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	150.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	2844.000 mm.	Actual Stress		138.8 N./mm ²
Spec. Min. Thk t _e , or s _a		98.0000 mm.	Spec. Nominal Thk.		104.000 mm.
Surface Area		146315.0 cm ²	SG of Contents		1.000
Element Volume		6023311.5 cm ³	Weight of Contents		0.0 Kg _f
Empty Weight		11008.2 Kg _f	Operating Weight		11008.2 Kg _f
			Comp. All. Ope. Stress		112.1 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 35.550 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 17.775 mm.

Material treatment options: +N

Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:
 $= D/2 \cdot \exp(0.5P/(S \cdot E)) - 1 + c_i + c_o$
 $= 2850.0/2 \cdot (\exp(0.5 \cdot 191.235/(150.0 \cdot 1.0)) - 1) + 3.0 + 0.0$
 $= 93.801 + 3.0 + 0.0 = 96.8009 \text{ mm.}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D) - P_{liq}$
 $= 2 \cdot 150.0 \cdot 1.0 \cdot \ln((2 \cdot 95.0 + 2850.0)/2850.0) - 0.235$
 $= 193.603 - 0.235 = 193.3675 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D)$
 $= 2 \cdot 174.997 \cdot 1.0 \cdot \ln((2 \cdot 98.0 + 2844.0)/2844.0)$
 $= 233.2433 \text{ bars}$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D^2 / (E (D_o^2 - D^2))$$

$$= 191.235 \cdot 2850.0^2 / (1.0 (3040.0^2 - 2850.0^2))$$

$$= 138.808 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= 0.5 \cdot P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 0.5 \cdot 191.235 / (1.0 \cdot \ln((2 \cdot 95.0 + 2850.0) / 2850.0))$$

$$= 148.164 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.292 %
 For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$ 41.716 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159)$$

$$= \text{square root}(4 \cdot 146315.0 / 3.14159)$$

$$= 4316.180 \text{ mm.}$$

MDMT Calculations in the Spherical Portion:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : 13 °C
 User entered Minimum Design Temperature : -10 °C
 The Curve for this Material is : B
 From Figure 3.8M- with PWHT, the MDMT is : 21 °C
 Temperature reduction per Graph 3.13 : 8 °C
 The computed stress ratio is : 0.842

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (80.033 \cdot 1.0) / (98.0 - 3.0 - 0.0)$$

$$= 0.842$$

Warning: Impact Testing Is Mandatory!

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	191.235 bars
Governing Thk.	98.000 mm.	Min. Des. Mtl. Temp.	-10 °C
Stress/Thk. Ratio	0.842	Basic MDMT	21 °C
		Temperature Difference	8 °C
		Computed Min. Temp.	13 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 35.550 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 17.775 mm.

Material treatment options: +N

EXTERNAL PRESSURE RESULTS for: Bottom D end

Spherical Head From 30 to 40 Ext. Chart: CS-2 at 200 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 0.075 \cdot E_y(t / R_o)$$

$$= 0.075 \cdot 0.19258E+09(95.0/1520.0)$$

$$= 902.775 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / S_y$$

$$= 902.8 / 225.0$$

$$= 4.012$$

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 225.0/482.6$$

$$= 0.466$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 902.8/0.19258E+09$$

$$= 0.004687$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.466) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.166$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m_1$$

$$= 225.0(1 + 0.002) / \ln(1 + 0.002) 0.166192$$

$$= 633.384 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m_2 f(m_2 f_2 - R)$$

$$= 0.6(1.0 - 0.466)$$

$$= 0.320$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$$

$$= (482.6 \cdot \exp(0.32)) / (0.32 \cdot 0.32)$$

$$= 957.426 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R^{1.5} - 0.5 \cdot R^{2.5} - R^{3.5}$$

$$= 1.5 \cdot 0.466^{1.5} - 0.5 \cdot 0.466^{2.5} - 0.466^{3.5}$$

$$= 0.334$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.621	0.375E-05	0.296E-05	0.875E-05	-0.582E-05	0.983E+07	0.199E-01
2	120	-4.424	0.116E-06	0.116E-06	0.201E-05	-0.201E-05	0.133E+09	0.378E-02
3	180	-3.022	0.886E-06	0.876E-06	0.475E-05	-0.467E-05	0.423E+08	0.409E-03
4	211	-2.322	0.192E-05	0.182E-05	0.659E-05	-0.608E-05	0.209E+08	0.541E-02
5	196	-2.672	0.132E-05	0.129E-05	0.563E-05	-0.543E-05	0.299E+08	0.187E-02
6	188	-2.847	0.109E-05	0.107E-05	0.518E-05	-0.505E-05	0.356E+08	0.606E-03
7	184	-2.935	0.982E-06	0.969E-06	0.497E-05	-0.486E-05	0.388E+08	0.708E-04
8	182	-2.979	0.933E-06	0.922E-06	0.486E-05	-0.477E-05	0.405E+08	0.176E-03
9	183	-2.957	0.958E-06	0.945E-06	0.491E-05	-0.481E-05	0.397E+08	0.542E-04
10	184	-2.946	0.970E-06	0.957E-06	0.494E-05	-0.484E-05	0.392E+08	0.786E-05
11	184	-2.951	0.964E-06	0.951E-06	0.493E-05	-0.483E-05	0.395E+08	0.233E-04
12	184	-2.949	0.967E-06	0.954E-06	0.493E-05	-0.483E-05	0.394E+08	0.773E-05
13	184	-2.947	0.968E-06	0.956E-06	0.494E-05	-0.484E-05	0.393E+08	0.585E-07
14	184	-2.948	0.968E-06	0.955E-06	0.494E-05	-0.483E-05	0.393E+08	0.384E-05
15	184	-2.948	0.968E-06	0.955E-06	0.494E-05	-0.484E-05	0.393E+08	0.189E-05
16	184	-2.947	0.968E-06	0.955E-06	0.494E-05	-0.484E-05	0.393E+08	0.916E-06
17	184	-2.947	0.968E-06	0.955E-06	0.494E-05	-0.484E-05	0.393E+08	0.429E-06

Factor of Safety [FS]:

Since $Fic > 0.55 F_y$ and $Fic < F_y$

$$= 2.407 - 0.741 \cdot Fic / F_y$$

$$= 2.407 - 0.741 \cdot 184.23/225.0$$



= 1.800

Allowable Circumferential Compressive Stress [Fha]:

= F_{ic}/F_S
= 184.231/1.8
= 102.335 N./mm²

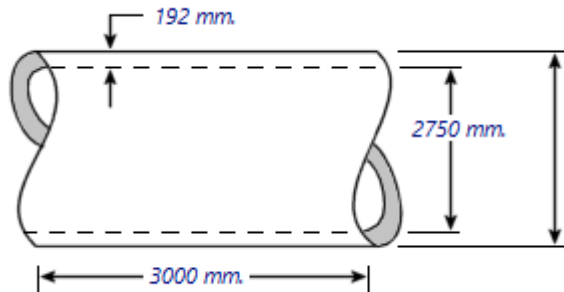
Allowable External Pressure at the given Thickness [Pa]:

= $2 \cdot F_{ha}[t/R_o]$
= $2 \cdot 102.335[95.0/1520.0]$
= 127.912 bars

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell 1

Cylindrical Shell Design Information:					
Design Pressure	P	191.235 bars	Int. Design Temperature	200 °C	
Material		SA-516 70	Ext. Design Temperature	200 °C	
Length		3000.000 mm.	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	150.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	2750.000 mm.	Actual Stress	139.4 N./mm ²	
Spec. Min. Thk t _e , or s _a		192.0000 mm.	Spec. Nominal Thk.	192.000 mm.	
Surface Area		295372.6 cm ²	SG of Contents	1.000	
Element Volume		17821918.0 cm ³	Weight of Contents	14248.8 Kg	
Empty Weight		41261.5 Kg	Operating Weight	55510.3 Kg	
			Comp. All. Ope. Stress	117.2 N./mm ²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2756.0 \cdot (\exp(191.235/(150.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 187.3851 + 3.0 + 0.0 = 190.3851 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 150.0 \cdot 1.0 \cdot \ln((2 \cdot 189.0 + 2756.0) / 2756.0) - 0.235$$

$$= 192.782 - 0.235 = 192.5465 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 174.997 \cdot 1.0 \cdot \ln((2 \cdot 192.0 + 2750.0) / 2750.0)$$

$$= 228.7238 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 191.235 \cdot 2756.0 / (1.0 (3134.0 - 2756.0))$$

$$= 139.4378 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 191.235 / (1.0 \cdot \ln((2 \cdot 189.0 + 2756.0) / 2756.0))$$

$$= 148.7954 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o) \quad 6.526 \%$$

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm..
 This component must be impact tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	191.235 bars
Governing Thk.	192.000 mm.	Min. Des. Mtl. Temp.	-10 °C
Stress/Thk. Ratio	-1.000	Basic MDMT	-17 °C
		Temperature Difference	0 °C
		Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

Material treatment options: +N

EXTERNAL PRESSURE RESULTS for: Shell 1

Cylindrical Shell From 40 to 50 Ext. Chart: CS-2 at 200 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 6758.0 / \sqrt{(1567.0 \cdot 189.0)}$$

$$= 12.418$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 1.5 \text{ and } M_x < 13, \text{ Ch} = 0.92 / (M_x - 0.579)$$

$$= 0.92 / (12.418 - 0.579)$$

$$= 0.0777$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot \text{Ch} \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.078 \cdot 0.19120\text{E}+09 (189.0 / 3134.0)$$

$$= 1433.745 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$

$$= 1433.7 / 225.0$$

$$= 6.372$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 225.0 / 482.6$$

$$= 0.466$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 1433.7 / 0.19120\text{E}+09$$

$$= 0.007498$$

Factor [m1]:

$$= (\ln(r) + (\text{epsp} - \text{epsys})) / \ln(\ln(1 + \text{epsp}) / \ln(1 + \text{epsys}))$$

$$= (\ln(0.466) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.166$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \text{epsys}) / \ln(1 + \text{epsys})^{m1}$$

$$= 225.0 (1 + 0.002) / \ln(1 + 0.002) \cdot 0.166192$$

$$= 633.384 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.466)$$

$$= 0.320$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.32)) / (0.32^{0.32})$$

$$= 957.426 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4661.5 - 0.5 \cdot 0.4662.5 - 0.4663.5$$

$$= 0.334$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.621	0.375E-05	0.296E-05	0.875E-05	-0.582E-05	0.983E+07	0.171E-01
2	120	-4.424	0.116E-06	0.116E-06	0.201E-05	-0.201E-05	0.132E+09	0.659E-02
3	180	-3.022	0.886E-06	0.876E-06	0.475E-05	-0.467E-05	0.422E+08	0.321E-02
4	211	-2.322	0.192E-05	0.182E-05	0.659E-05	-0.608E-05	0.209E+08	0.260E-02
5	196	-2.672	0.132E-05	0.129E-05	0.563E-05	-0.543E-05	0.299E+08	0.934E-03
6	203	-2.497	0.160E-05	0.154E-05	0.610E-05	-0.577E-05	0.250E+08	0.640E-03
7	199	-2.584	0.146E-05	0.141E-05	0.587E-05	-0.560E-05	0.273E+08	0.190E-03
8	201	-2.541	0.153E-05	0.148E-05	0.599E-05	-0.569E-05	0.262E+08	0.213E-03
9	200	-2.562	0.149E-05	0.145E-05	0.593E-05	-0.565E-05	0.267E+08	0.877E-05
10	200	-2.573	0.147E-05	0.143E-05	0.590E-05	-0.563E-05	0.270E+08	0.914E-04
11	200	-2.568	0.148E-05	0.144E-05	0.591E-05	-0.564E-05	0.269E+08	0.415E-04
12	200	-2.565	0.149E-05	0.144E-05	0.592E-05	-0.564E-05	0.268E+08	0.164E-04
13	200	-2.564	0.149E-05	0.144E-05	0.592E-05	-0.565E-05	0.268E+08	0.383E-05
14	200	-2.563	0.149E-05	0.145E-05	0.592E-05	-0.565E-05	0.268E+08	0.247E-05
15	200	-2.564	0.149E-05	0.144E-05	0.592E-05	-0.565E-05	0.268E+08	0.678E-06
16	200	-2.563	0.149E-05	0.145E-05	0.592E-05	-0.565E-05	0.268E+08	0.897E-06
17	200	-2.563	0.149E-05	0.145E-05	0.592E-05	-0.565E-05	0.268E+08	0.110E-06

Factor of Safety [FS]:
 Since $Fic > 0.55 F_y$ and $Fic < F_y$

$$= 2.407 - 0.741 \cdot Fic / F_y$$

$$= 2.407 - 0.741 \cdot 200.75 / 225.0$$

$$= 1.746$$

Allowable Circumferential Compressive Stress [Fha]:

$$= Fic / FS$$

$$= 200.749 / 1.746$$

$$= 114.986 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

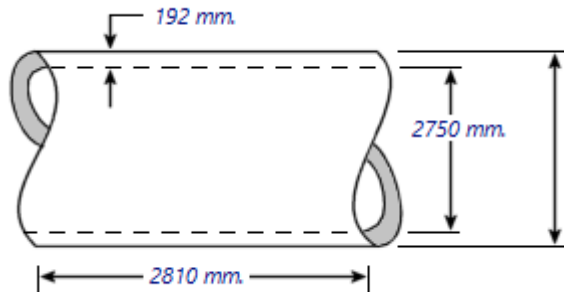
$$= 2 \cdot Fha (t / Do)$$

$$= 2 \cdot 114.986 (189.0 / 3134.0)$$

$$= 138.680 \text{ bars}$$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell 2

Cylindrical Shell Design Information:					
Design Pressure	P	191.000 bars	Int. Design Temperature		200 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		2810.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	150.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	2750.000 mm.	Actual Stress		139.3 N./mm ²
Spec. Min. Thk t _e or s _a		192.0000 mm.	Spec. Nominal Thk.		192.000 mm.
Surface Area		276665.7 cm ²	SG of Contents		0.000
Element Volume		16693198.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		38648.3 Kgf	Operating Weight		38648.3 Kgf
			Comp. All. Ope. Stress		117.2 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2756.0 \cdot (\exp(191.0/(150.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 187.1397 + 3.0 + 0.0 = 190.1397 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 150.0 \cdot 1.0 \cdot \ln((2 \cdot 189.0 + 2756.0) / 2756.0) - 0.0$$

$$= 192.782 - 0.0 = 192.7818 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 174.997 \cdot 1.0 \cdot \ln((2 \cdot 192.0 + 2750.0) / 2750.0)$$

$$= 228.7238 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 191.0 \cdot 2756.0 / (1.0 (3134.0 - 2756.0))$$

$$= 139.2663 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 191.0 / (1.0 \cdot \ln((2 \cdot 189.0 + 2756.0) / 2756.0))$$

$$= 148.6123 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o) \quad 6.526 \%$$

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm..
 This component must be impact tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	191.000 bars
Governing Thk.	192.000 mm.	Min. Des. Mtl. Temp.	-10 °C
Stress/Thk. Ratio	-1.000	Basic MDMT	-17 °C
		Temperature Difference	0 °C
		Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

Material treatment options: +N

EXTERNAL PRESSURE RESULTS for: Shell 2

Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 200 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 6758.0 / \sqrt{(1567.0 \cdot 189.0)}$$

$$= 12.418$$

Compute the Factor [Ch]:

Since $M_x > 1.5$ and $M_x < 13$, $Ch = 0.92 / (M_x - 0.579)$

$$= 0.92 / (12.418 - 0.579)$$

$$= 0.0777$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.078 \cdot 0.19120E+09 (189.0 / 3134.0)$$

$$= 1433.745 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 1433.7 / 225.0$$

$$= 6.372$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 225.0 / 482.6$$

$$= 0.466$$

Ratio [AE]:

$$= Fhe / E$$

$$= 1433.7 / 0.19120E+09$$

$$= 0.007498$$

Factor [m1]:

$$= (\ln(r) + (\text{epsp} - \text{epsys})) / \ln(\ln(1 + \text{epsp}) / \ln(1 + \text{epsys}))$$

$$= (\ln(0.466) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.166$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_s(1 + \text{epsys}) / \ln(1 + \text{epsys})^{m1}$$

$$= 225.0(1 + 0.002) / \ln(1 + 0.002)^{0.166192}$$

$$= 633.384 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.466)$$

$$= 0.320$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.32)) / (0.32^{0.32})$$

$$= 957.426 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4661.5 - 0.5 \cdot 0.4662.5 - 0.4663.5$$

$$= 0.334$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.621	0.375E-05	0.296E-05	0.875E-05	-0.582E-05	0.983E+07	0.171E-01
2	120	-4.424	0.116E-06	0.116E-06	0.201E-05	-0.201E-05	0.132E+09	0.659E-02
3	180	-3.022	0.886E-06	0.876E-06	0.475E-05	-0.467E-05	0.422E+08	0.321E-02
4	211	-2.322	0.192E-05	0.182E-05	0.659E-05	-0.608E-05	0.209E+08	0.260E-02
5	196	-2.672	0.132E-05	0.129E-05	0.563E-05	-0.543E-05	0.299E+08	0.934E-03
6	203	-2.497	0.160E-05	0.154E-05	0.610E-05	-0.577E-05	0.250E+08	0.640E-03
7	199	-2.584	0.146E-05	0.141E-05	0.587E-05	-0.560E-05	0.273E+08	0.190E-03
8	201	-2.541	0.153E-05	0.148E-05	0.599E-05	-0.569E-05	0.262E+08	0.213E-03
9	200	-2.562	0.149E-05	0.145E-05	0.593E-05	-0.565E-05	0.267E+08	0.877E-05
10	200	-2.573	0.147E-05	0.143E-05	0.590E-05	-0.563E-05	0.270E+08	0.914E-04
11	200	-2.568	0.148E-05	0.144E-05	0.591E-05	-0.564E-05	0.269E+08	0.415E-04
12	200	-2.565	0.149E-05	0.144E-05	0.592E-05	-0.564E-05	0.268E+08	0.164E-04
13	200	-2.564	0.149E-05	0.144E-05	0.592E-05	-0.565E-05	0.268E+08	0.383E-05
14	200	-2.563	0.149E-05	0.145E-05	0.592E-05	-0.565E-05	0.268E+08	0.247E-05
15	200	-2.564	0.149E-05	0.144E-05	0.592E-05	-0.565E-05	0.268E+08	0.678E-06
16	200	-2.563	0.149E-05	0.145E-05	0.592E-05	-0.565E-05	0.268E+08	0.897E-06
17	200	-2.563	0.149E-05	0.145E-05	0.592E-05	-0.565E-05	0.268E+08	0.110E-06

Factor of Safety [FS]:
 Since $Fic > 0.55 F_y$ and $Fic < F_y$

$$= 2.407 - 0.741 \cdot Fic / F_y$$

$$= 2.407 - 0.741 \cdot 200.75 / 225.0$$

$$= 1.746$$

Allowable Circumferential Compressive Stress [Fha]:

$$= Fic / FS$$

$$= 200.749 / 1.746$$

$$= 114.986 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

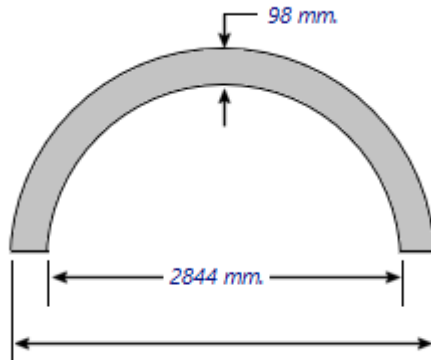
$$= 2 \cdot Fha(t / D_o)$$

$$= 2 \cdot 114.986(189.0 / 3134.0)$$

$$= 138.680 \text{ bars}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Top D End

Spherical Head Design Information:					
Design Pressure	P	191.000 bars	Int. Design Temperature		200 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		0.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	150.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	2844.000 mm.	Actual Stress		138.6 N./mm ²
Spec. Min. Thk t _e , or s _a		98.0000 mm.	Spec. Nominal Thk.		104.000 mm.
Surface Area		146315.0 cm ²	SG of Contents		0.000
Element Volume		6023311.5 cm ³	Weight of Contents		0.0 Kg _f
Empty Weight		11007.7 Kg _f	Operating Weight		11007.7 Kg _f
			Comp. All. Ope. Stress		112.1 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 35.550 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 17.775 mm.

Material treatment options: +N+IT

Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:
 $= D/2 \cdot (\exp(0.5P/(S \cdot E)) - 1) + c_i + c_o$
 $= 2850.0/2 \cdot (\exp(0.5 \cdot 191.0/(150.0 \cdot 1.0)) - 1) + 3.0 + 0.0$
 $= 93.681 + 3.0 + 0.0 = 96.681 \text{ mm.}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D) - P_{liq}$
 $= 2 \cdot 150.0 \cdot 1.0 \cdot \ln((2 \cdot 95.0 + 2850.0)/2850.0) - 0.0$
 $= 193.604 - 0.0 = 193.6045 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D)$
 $= 2 \cdot 175.0 \cdot 1.0 \cdot \ln((2 \cdot 98.0 + 2844.0)/2844.0)$
 $= 233.2475 \text{ bars}$



Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D^2 / (E (D_o^2 - D^2))$$

$$= 191.0 \cdot 2850.0^2 / (1.0 (3040.0^2 - 2850.0^2))$$

$$= 138.637 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= 0.5 \cdot P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 0.5 \cdot 191.0 / (1.0 \cdot \ln((2 \cdot 95.0 + 2850.0) / 2850.0))$$

$$= 147.982 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$	5.292 %
For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$	41.716 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159)$$

$$= \text{square root}(4 \cdot 146315.0 / 3.14159)$$

$$= 4316.180 \text{ mm.}$$

MDMT Calculations in the Spherical Portion:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 35.550 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 17.775 mm.

Material treatment options: +N+IT

EXTERNAL PRESSURE RESULTS for: Top D End

Spherical Head From 60 to 70 Ext. Chart: CS-2 at 200 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 0.075 \cdot E_y (t / R_o)$$

$$= 0.075 \cdot 0.19258E+09 (95.0 / 1520.0)$$

$$= 902.775 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / S_y$$

$$= 902.8 / 225.0$$

$$= 4.012$$

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 225.0 / 482.6$$

$$= 0.466$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 902.8 / 0.19258E+09$$

$$= 0.004687$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.466) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.166$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m_1$$

$$= 225.0 (1 + 0.002) / \ln(1 + 0.002) 0.166191$$

= 633.382 N./mm²

Curve Fitting Exponent for, Ferritic Steel [m2]:

= $m2f(m2f2 - R)$
 = $0.6(1.0 - 0.466)$
 = 0.320

Curve Fitting Constant [A2]:

= $(\sigma_{uts} \cdot \exp(m2)) / (m2 m2)$
 = $(482.6 \cdot \exp(0.32)) / (0.32 0.32)$
 = 957.424 N./mm²

Material Parameter for Stress-Strain Curve Model [K]:

= $1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$
 = $1.5 \cdot 0.4661.5 - 0.5 \cdot 0.4662.5 - 0.4663.5$
 = 0.334

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.621	0.375E-05	0.296E-05	0.875E-05	-0.582E-05	0.983E+07	0.199E-01
2	120	-4.424	0.116E-06	0.116E-06	0.201E-05	-0.201E-05	0.133E+09	0.378E-02
3	180	-3.023	0.886E-06	0.876E-06	0.475E-05	-0.467E-05	0.423E+08	0.409E-03
4	211	-2.322	0.192E-05	0.182E-05	0.659E-05	-0.608E-05	0.209E+08	0.541E-02
5	196	-2.672	0.132E-05	0.129E-05	0.563E-05	-0.543E-05	0.299E+08	0.187E-02
6	188	-2.847	0.109E-05	0.107E-05	0.518E-05	-0.505E-05	0.356E+08	0.606E-03
7	184	-2.935	0.982E-06	0.969E-06	0.497E-05	-0.486E-05	0.388E+08	0.705E-04
8	182	-2.979	0.933E-06	0.922E-06	0.486E-05	-0.477E-05	0.405E+08	0.176E-03
9	183	-2.957	0.958E-06	0.945E-06	0.491E-05	-0.481E-05	0.397E+08	0.544E-04
10	184	-2.946	0.970E-06	0.957E-06	0.494E-05	-0.484E-05	0.392E+08	0.764E-05
11	184	-2.951	0.964E-06	0.951E-06	0.493E-05	-0.483E-05	0.395E+08	0.235E-04
12	184	-2.949	0.967E-06	0.954E-06	0.493E-05	-0.483E-05	0.394E+08	0.795E-05
13	184	-2.947	0.968E-06	0.955E-06	0.494E-05	-0.484E-05	0.393E+08	0.159E-06
14	184	-2.947	0.969E-06	0.956E-06	0.494E-05	-0.484E-05	0.393E+08	0.374E-05
15	184	-2.947	0.969E-06	0.956E-06	0.494E-05	-0.484E-05	0.393E+08	0.179E-05
16	184	-2.947	0.969E-06	0.956E-06	0.494E-05	-0.484E-05	0.393E+08	0.815E-06
17	184	-2.947	0.968E-06	0.956E-06	0.494E-05	-0.484E-05	0.393E+08	0.328E-06

Factor of Safety [FS]:

Since $Fic > 0.55 Fy$ and $Fic < Fy$
 = $2.407 - 0.741 \cdot Fic / Fy$
 = $2.407 - 0.741 \cdot 184.24/225.0$
 = 1.800

Allowable Circumferential Compressive Stress [Fha]:

= Fic/FS
 = $184.238/1.8$
 = 102.341 N./mm²

Allowable External Pressure at the given Thickness [Pa]:

= $2 \cdot Fha[t/Ro]$
 = $2 \cdot 102.341[95.0/1520.0]$
 = 127.918 bars



Hydrostatic Test Pressure Results:

Test Pressure at high point in vessel per Equation 8.1
 $= \gamma_{St/s} \cdot MAWP(St/S)$
 $= 1.25 \cdot 190.6849(1.162)$
 $= 276.960 \text{ bars}$

Horizontal Test performed per: $\gamma_{St/s} \cdot MAWP(St/S)$

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Bottom D end	208.0	247.0	0.842	277.24
Shell 1	212.1	247.0	0.859	277.23
Shell 2	212.1	247.0	0.859	277.23
Top D End	208.0	247.0	0.842	277.24

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3_3"	Nozzle	165.00	142.00	1.162
N5A_2"	Nozzle	165.00	142.00	1.162
N5B_2"	Nozzle	165.00	142.00	1.162
N7A_2"	Nozzle	165.00	142.00	1.162
N8A_2"	Nozzle	165.00	142.00	1.162
N9A_2"	Nozzle	165.00	142.00	1.162
N7B_2"	Nozzle	165.00	142.00	1.162
N8B_2"	Nozzle	165.00	142.00	1.162
N9B_2"	Nozzle	165.00	142.00	1.162
N1_28"	Nozzle	165.00	142.00	1.162
N4_2"	Nozzle	165.00	142.00	1.162
M1_24"	Nozzle	165.00	142.00	1.162
N6_3"	Nozzle	165.00	142.00	1.162
N2_28"	Nozzle	165.00	142.00	1.162

Minimum

1.162

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Bottom D end	175.00	150.00	1.167
Shell 1	175.00	150.00	1.167
Shell 2	175.00	150.00	1.167
Top D End	175.00	150.00	1.167
body flange 2	165.00	142.00	1.162
Body flange 1	165.00	142.00	1.162

Minimum

1.162

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N3_3"	118.96	237.50	0.501
N5A_2"	83.52	237.50	0.352
N5B_2"	83.52	237.50	0.352
N7A_2"	83.52	237.50	0.352
N8A_2"	83.52	237.50	0.352
N9A_2"	83.52	237.50	0.352
N7B_2"	83.52	237.50	0.352
N8B_2"	83.52	237.50	0.352



N9B_2"	83.52	237.50	0.352
N1_28"	189.58	237.50	0.798
N4_2"	83.52	237.50	0.352
M1_24"	182.52	237.50	0.769
N6_3"	118.96	237.50	0.501
N2_28"	189.58	237.50	0.798

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	3000	...	...	No Calc	No Calc
20	30	3000	...	...	No Calc	No Calc
30	40	No Calc	3040	95	No Calc	No Calc
40	50	6758	3134	189	No Calc	No Calc
50	60	6758	3134	189	No Calc	No Calc
60	70	No Calc	3040	95	No Calc	No Calc
70	80	No Calc	...	277	No Calc	No Calc
80	90	No Calc	...	257	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	98	7.07028	1.03419	127.912
40	50	192	16.1987	1.03419	138.68
50	60	192	16.1987	1.03419	138.68
60	70	98	7.07028	1.03419	127.918
70	80	280	243.256	1.03419	No Calc
80	90	260	251.054	1.03419	No Calc

Minimum

128

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	3000	No Calc	No Calc	No Calc
20	30	3000	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	6758	25399974	No Calc	No Calc
50	60	6758	25399974	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	4206.27	...	3991.17	...	...
20	30	1434.15	...	1326.6	...	...
30	40	11008.2	6023311	10712.2	6061514	...
40	50	41261.5	17821918	40658.2	17899770	...
50	60	38648.3	16693198	38083.2	16766122	...
60	70	11007.7	6023311	10711.7	6061514	...
70	80	2044.24	148897	2024.94	149503	...
80	90	1894.29	138432	1876.34	138997	...
Total		111504	46849068.00	109384	47077416.00	0

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
30	Liqd	...	...	-1418.99	...	Liquid 30
30	Nozl	69.3816	...	-1473	...	N3_3"
30	Wght	3900	...	...	...	Btm Dish SF
40	Plat	3850.49	0.20320E-04	...	...	PLAT 1
40	Liqd	14248.8	...	1200	...	Liquid: 40
40	Nozl	59.4796	-1475	1249.97	...	N5A_2"
40	Nozl	59.4796	...	649.975	-1475	N5B_2"
40	Nozl	59.4796	...	2199.97	1475	N7A_2"
40	Nozl	59.4796	-1475	1999.97	...	N8A_2"
40	Nozl	59.4796	...	1249.97	1475	N9A_2"
40	Nozl	59.4796	...	649.975	1475	N7B_2"
40	Nozl	59.4796	-1475	449.975	...	N8B_2"
40	Nozl	59.4796	...	249.975	1475	N9B_2"
40	Wght	3000	...	2900	...	Lifting WT
40	Wght	1010	...	2905	...	Clip WT
40	Wght	10000	...	2910	...	Miscellaneous W
50	Plat	1857.12	...	1000	-1408.99	PLAT 2
50	Nozl	4905.51	-1600.01	224.975	...	N1_28"
50	Nozl	59.4796	...	599.975	1475	N4_2"
50	Nozl	4749.3	...	999.975	-1525	M1_24"
50	Wght	800	...	2500	-1400	Pipe Davit WT
50	Wght	50	...	1000	-1600	Manway davit
50	Wght	430	...	2400	...	Demister WT
50	Wght	30	...	2400	...	Demister suppor
50	Wght	3930	-2413	225.001	...	N1_OVH
60	Plat	4955.8	...	1800	...	PLAT 3
60	Nozl	82.732	950	1109.1	...	N6_3"
60	Nozl	3734.95	...	1715.75	...	N2_28"
60	Wght	3900	...	...	...	Top Dish SF
60	Wght	3948	2913	...	...	N2_OVH
60	Forc	...	...	1400	...	N2_FXFZ

Total Weight of Each Detail Type:

Platforms	10663.4
Liquid	14248.8
Nozzles	14077.2
Weights	30998.0
Sum of the Detail Weights	69987.4 kg.

Note:

Because some nozzles were attached to a head or flange, their weights include the attached head or flange.



Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	107566.1	107566.1	107566.1	107566.1	107566.1	107566.1
Nozzles	14077.2	14077.2	14077.2	14077.2	14077.2	14077.2
Wld Weights	23120.0	23120.0	23120.0	23120.0	23120.0	23120.0
Platforms	...	...	10663.4	10663.4	10663.4	10663.4
Empty Weights	...	...	...	7878.0	7878.0	...
Test Weights	...	7878.0	...	...	...	...
Ope Weights	...	...	...	...	...	7878.0
Ope. Liquid	...	...	...	...	...	14248.8
Test Liquid	...	46820.5	...	...	...	...
Totals	144763.3	199461.8	155426.7	163304.7	163304.7	177553.6

Field Installation Options:

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	144763.3 kg.
Shop Test Wt.	- Fabricated weight + water (full)	199461.8 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	155426.7 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	163304.7 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	163304.7 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	177553.6 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	97954.4 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	120556.1 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	98841.0 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	186296	183783
20	30	93148.2	91891.6
30	40	146315	127051
40	50	295373	259181
50	60	276666	242767
60	70	146315	127051
70	80	22603.9	8266.91
80	90	21608.8	7685.99

Total	1188325.6	1047678.2 cm ²
Total	1279.1	1127.7 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	4206.27	4206.27	4206.27	...	4206.27
20	30	1434.15	1434.15	1434.15	...	1434.15
30	40	14977.6	14977.6	20997.3	...	14977.6
40	50	59597.8	73846.7	77408.9	263.202	59597.8
50	60	55459.7	55459.7	72142.7	10971.8	55459.7
60	70	27629.2	27629.2	33648.8	...	27629.2

Empty Support Force + the Sum of the Y forces	182150.31	Kgf
Operating Support Force + the Sum of the Y forces	196399.16	Kgf
Hydro Support Force + the Sum of the Y forces	228683.62	Kgf

Cumulative Vessel Weight



From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	163305	177554	209838
20	30	159098	173347	205632
30	40	157664	171913	204198
40	50	142687	156935	183200
50	60	83088.8	83088.8	105791
60	70	27629.2	27629.2	33648.8

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	11235	11235	11235
20	30	11235	11235	11235
30	40	11235	11235	11235
40	50	11235	11235	11235
50	60	10971.8	10971.8	10971.8
60	70	...	...	...

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Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N3_3"	219.10	255.10	200	1500	GR 1.1	...	...	...	...
N6_3"	219.10	255.10	200	1500	GR 1.1	...	...	...	...
Min Rating	219.101	255.100, [for Core Elements]					...	...	...

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-2, 4.16.12 for more information.

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 7.56541 Hz.

The Natural Frequency for the Vessel (Ope...) is 7.40345 Hz.

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Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. Kgf	X,Z Moment and For Res Kg-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	18845.6	74871.8
70	80	...	...
80	90	...	...

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. Kgf	X,Z Moment and For Res Kg-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	18845.6	74871.8
70	80	...	...
80	90	...	...

User Input Forces and Moments:

From Node	Distance From	- - - Fx	Forces - - - Fy	- - - Fz	- - - Mx	Moments - - - My	- - - Mz
60	1400.00	18846.	-18846.		34286.		34286.

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	0
Base Elevation	300 mm.
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.167 \cdot 0.954) / (1 + 1.7 \cdot 3.4 \cdot 0.167)))$
 $= \min(0.85, 0.904)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	7.403 Hz
Natural Frequency of Vessel (Empty)	7.565 Hz
Natural Frequency of Vessel (Test)	7.120 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.741
Structure Height to Diameter ratio	3.472
Height to top of Structure	10329.999 mm.

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: D from Table 26.11-1
 $\alpha = 11.5$; $Zg = 589788$ mm.

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1000.0 + 300.0 = 1300.0$ mm.
 $= 4.265$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (4.265 ft.) < 15 ft.
 $= 2.41(15 / Zg)^{2/\alpha}$
 $= 2.41(15 / 1935.0)^{2/11.5}$
 $= 1.035$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[Kzt]$:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.035 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 111.85^2)$$

$$= 33.1 \text{ psf } [161.849] \text{ Kgs/m}^2$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.149 \cdot 0.85 \cdot 0.741 \cdot 84.256$$

$$= 1759.6 \text{ lbs. } [798.2] \text{ Kg}$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Skirt 1	1300.0	0.000	0.000	0.000	1.035	1.000	161.849
Skirt 2	2800.0	0.000	0.000	0.000	1.035	1.000	161.849
Bottom D end	3300.0	0.000	0.000	0.000	1.035	1.000	161.849
Shell 1	4800.0	0.000	0.000	0.000	1.044	1.000	163.225
Shell 2	7705.0	0.000	0.000	0.000	1.133	1.000	177.228
Top D End	9755.1	0.000	0.000	0.000	1.181	1.000	184.650

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, mm.
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, mm.
- f - Natural frequency of vibration (Hertz)
- f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
- L - Total height of structure mm.
- Lc - Total length of conical section(s) of vessel, mm.
- tb - Uncorroded plate thickness at bottom of vessel, mm.
- V30 - Design Wind Speed provided by user, Km/hr
- Vc - Critical wind velocity, Km/hr
- Vw - Maximum wind speed at top of structure, Km/hr
- W - Total corroded weight of structure, Kg
- Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kg
- Z - Maximum amplitude of vibration at top of vessel, mm.
- D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
- Vp - Vib. Chance, $\leq 0.320\text{E-}06$ (High); $0.320\text{E-}06 < 0.400\text{E-}06$ (Probable)
- P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: $0.0/8810.0 = 0.0$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[10.09/28.9^2] \cdot 10^4 = 120.813$ [Geometry Violation]

Compute the vibration possibility. If $Vp > 0.400\text{E-}06$ no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 179372 / (8810.0 \cdot 2871.488^2)$$

$$= 0.24692\text{E-}05$$

Since Vp is $> 0.400\text{E-}06$ no further vibration analysis is required !

Platform Load Calculations



ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf	Cf
PLAT 1	60000.00	3300.00	161.85	971.10	2.00
PLAT 2	60000.00	7300.00	175.28	1051.65	2.00
PLAT 3	184779.00	10909.97	187.13	3457.74	2.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf
Btm Dish SF	0.00	3299.97	161.85	0.00
Lifting WT	0.00	6200.00	169.97	0.00
Clip WT	0.00	6205.00	170.00	0.00
Miscellaneous	0.00	6210.00	170.02	0.00
Pipe Davit WT	0.00	8800.00	181.19	0.00
Manway davit	0.00	7300.00	175.28	0.00
Demister WT	0.00	8700.00	180.83	0.00
Demister supp	0.00	8700.00	180.83	0.00
N10VH	0.00	6525.00	171.54	0.00
Top Dish SF	0.00	9109.97	182.31	0.00
N20VH	0.00	9109.97	182.31	0.00

The Natural Frequency for the Vessel (Ope...) is 7.40345 Hz.

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1300	3913.8	78276	161.849	877.989
20	30	2800	3913.8	39138	161.849	438.995
30	40	3299.98	4012.8	1.2231	161.849	0.013719
40	50	4800	4136.88	124106	163.225	2138.27
50	60	7705	4136.88	116246	177.228	2523.79
60	70	9755.09	4012.8	47906.2	184.65	4866.84

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	1.500
Site Class	D
Component Elevation Ratio	z/h
Amplification Factor	Ap
Force Factor	0.000
Consider Vertical Acceleration	No
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR]
Component strength factor	[Rpo]
Overstrength factor	[Omega0]
Response Acceleration Factor	[SMS]
Response Acceleration Factor	[SM1]
Building Importance Factor	[Ie]
Site Latitude	0.000 deg
Site Longitude	0.000 deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.326 = 0.217$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.139 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(33.56960.75)$
 = 0.279 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/7.403)
 = 0.135

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.279$, 0.135) per 12.8.2
 = 0.135

Seismic Response Coefficient per equation 12.8-2 [Cs]:
 = $Sds / (R / I)$
 = $0.217 / (1.5 / 1.5)$
 = 0.217

Maximum value of C_s per equation 12.8-3 [Cs]:
 = $Sd1 / (T(R / I_e))$
 = $0.093 / (0.135(1.5 / 1.5))$
 = 0.686

Minimum value of C_s per equation 15.4-1 [Cs]:
 = $\max((0.044 \cdot SDS \cdot I_e), 0.030)$
 = $\max((0.044 \cdot 0.2173 \cdot 1.5), 0.030)$
 = 0.030

Total Base Shear [V]:
 = $C_s \cdot W$ (Equation 12.8-1):
 = $0.2173 \cdot 177553.5$
 = 38588.309 Kgf

Distribute the Base shear force to each element according to the equations
 $F_x = C_{vx} \cdot V$ (eqn. 12.8-11) and the vertical distribution factor
 $C_{vx} = W_x \cdot h_x^k / (\text{Sum of } W_i \cdot h_i^k)$ and k is an exponent which is related

to the period of Vibration.

In this case, the value of k was 1.0

The Natural Frequency for the Vessel (Ope...) is 7.40345 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf	Element Emp Load Kgf	Element Vertical Load Kgf
10	20	1000	4206.27	156.202	153.114	-18845.6
20	30	2500	1434.15	133.145	130.513	-18845.6
30	40	2999.98	14977.6	1668.59	1635.61	-18845.6
40	50	4500	73846.7	12340.5	9762.49	-18845.6
50	60	7405	55459.7	15250.7	14949.3	-18845.6
60	70	8809.98	27629.2	9039.21	8860.55	-18845.6

Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support mm.	Cumulative Shr Wind Cas Kg-f	Cumulative Shr Eqk Cas Kg-f	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1000	18845.6	18845.6	229281	229281
20	30	2500	18845.6	18845.6	192095	192095
30	40	2999.98	18845.6	18845.6	173585	173585
40	50	4500	18845.6	18845.6	173584	173584
50	60	7405	18845.6	18845.6	118687	118687
60	70	8809.98	18845.6	18845.6	69687.8	69687.8

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1000	10845.9	38588.3	76304.4	259430
20	30	2500	9967.91	38432.1	55490.1	182408
30	40	2999.98	9528.92	38299	45741.8	144043
40	50	4500	9528.91	36630.4	45741.5	144042
50	60	7405	7390.63	24289.9	20361.7	52659.6
60	70	8809.98	4866.84	9039.21	3139.77	5831.5

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear Kgf	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	18845.6	0.028131	0.10936	0.0001062
20	30	18845.6	0.16842	0.23899	0.00015227
30	40	18845.6	0.23899	0.23899	0.00015227
40	50	18845.6	0.47219	0.71388	0.0001636
50	60	18845.6	0.94663	1.18426	0.00017055
60	70	18845.6	1.18426	1.18426	0.00017055

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)
 Allowable deflection : 44.050 Actual deflection : 1.184 mm.

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kgf	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	10845.9	0.009589	0.036507	0.34678E-04
20	30	9967.91	0.055621	0.078068	0.48016E-04
30	40	9528.92	0.078069	0.07807	0.48016E-04
40	50	9528.91	0.15134	0.22657	0.50644E-04
50	60	7390.63	0.29818	0.37037	0.51489E-04
60	70	4866.84	0.37038	0.37038	0.51489E-04

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	520.156	3250.98
20	30	520.156	3250.98
30	40	533.314	3333.21
40	50	549.804	3436.28
50	60	549.804	3436.28
60	70	533.314	3333.21

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 44.050 Actual deflection : 0.370 mm.

Total Deflection in the Operating Condition + Applied Forces:
 Allowable deflection : 44.050 Actual deflection : 1.555 mm.

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm. ³	Section Modulus Corroded mm. ³
10	20	1850.4	1711.63	135323136	125237024
20	30	1850.4	1711.63	135323136	125237024
30	40	9057.72	8789.4	645434496	627548928
40	50	17745.7	17486.3	1230455680	1214768896
50	60	17745.7	17486.3	1230456064	1214769280
60	70	9057.72	8789.4	645434496	627548928

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Longitudinal Allowable Stresses:

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	82.5995	194.746	-87.483	-99.7437
20	30	104.999	246.996	-87.483	-99.7437
30	40	149.999	246.996	-112.085	-122.731
40	50	149.999	246.996	-117.203	-122.731
50	60	149.999	246.996	-117.203	-122.731
60	70	150	247	-112.085	-122.731

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have been computed in the new and cold condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	...	...	...
20	30	...	...	...
30	40	138.637	-0.85409	194.255
40	50	65.1643	-0.45626	92.7053
50	60	65.1644	-0.45626	92.7053
60	70	138.637	-0.85409	194.255

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	-9.35658	-10.173	...	0.87975	0.87975
20	30	-9.11558	-9.93197	...	0.87975	0.87975
30	40	-1.75915	-1.75915	...	0.17557	0.17557
40	50	-0.80023	-0.80023	...	0.090699	0.090699
50	60	-0.46599	-0.46599	...	0.088574	0.088574
60	70	-0.30827	-0.30827	...	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	...	5.97498	20.3146	...	...
20	30	...	4.34513	14.2834	...	...
30	40	...	0.7148	2.25094	...	...
40	50	...	0.36926	1.16283	...	...
50	60	...	0.16438	0.42511	...	...
60	70	...	0.049065	0.091128	...	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	19.0906
20	30	...	...	...	13.5442
30	40	...	...	...	2.15174
40	50	...	...	...	1.11158
50	60	...	...	...	0.41671
60	70	...	...	...	0.089327

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind N./mm ²	Longitudinal Stress Y Forces Seismic N./mm ²
------	----	--	---

10	20	-1.07976	-1.07976
20	30	-1.07976	-1.07976
30	40	-0.21027	-0.21027
40	50	-0.10569	-0.10569
50	60	-0.10569	-0.10569
60	70	-0.21027	-0.21027

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm ²	Earthquake For/Mom Corroded N./mm ²	Wind For/Mom No Corrosion N./mm ²	Earthquake For/Mom No Corrosion N./mm ²
10	20	17.9537	17.9537	16.6156	16.6156
20	30	15.0419	15.0419	13.9208	13.9208
30	40	2.71259	2.71259	2.63743	2.63743
40	50	1.40132	1.40132	1.38345	1.38345
50	60	0.95814	0.95814	0.94592	0.94592
60	70	1.089	1.089	1.05883	1.05883

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm ²
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...



Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Un-Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

For Division 2, the overall stress intensity is computed in accordance with the Section VIII Division II Example problem manual PTB-3, example 4.3.6, stress due to combined loads.
 The stresses are combined per equation 4.3.44 to determine the overall stress intensity.
 The unity check is from Section 4.4.12 equations 4.4.85 and 4.4.90.

An asterisk (*) in the final column denotes overstress.

Note: Performing Combined Stress Analysis per VIII-2 paragraph 4.3.10.2.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	35.24	82.60	-35.24	81.24	0.4267	0.4338	0.2916
20	30.46	105.00	-30.46	102.12	0.2901	0.2983	0.1891
30	5.57	150.00	-5.57	134.97	0.0371	0.0413	0.2205
40	2.77	150.00	-2.77	181.32	0.0184	0.0153	0.0103
50	1.78	150.00	-1.78	181.32	0.0119	0.0098	0.0067
60	1.66	150.00	-1.66	134.97	0.0110	0.0123	0.0126

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	48.36	82.60	-48.36	81.24	0.5855	0.5953	0.4458
20	39.66	105.00	-39.66	102.12	0.3777	0.3884	0.2750
30	7.01	150.00	-7.01	134.97	0.0467	0.0519	0.2886
40	3.51	150.00	-3.51	181.32	0.0234	0.0194	0.0144
50	2.04	150.00	-2.04	181.32	0.0136	0.0112	0.0081
60	1.70	150.00	-1.70	134.97	0.0113	0.0126	0.0144



Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	36.06	82.60	-36.06	81.24	0.4366	0.4439	0.2916
20	31.28	105.00	-31.28	102.12	0.2979	0.3063	0.1891
30	5.57	150.00	-5.57	134.97	0.0371	0.0413	0.2265
40	2.77	150.00	-2.77	181.32	0.0184	0.0153	0.0103
50	1.78	150.00	-1.78	181.32	0.0119	0.0098	0.0067
60	1.66	150.00	-1.66	134.97	0.0110	0.0123	0.0126

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	50.40	82.60	-50.40	81.24	0.6102	0.6204	0.4601
20	41.22	105.00	-41.22	102.12	0.3925	0.4036	0.2819
30	7.11	150.00	-7.11	134.97	0.0474	0.0527	0.3002
40	3.56	150.00	-3.56	181.32	0.0237	0.0196	0.0146
50	2.04	150.00	-2.04	181.32	0.0136	0.0113	0.0081
60	1.70	150.00	-1.70	134.97	0.0113	0.0126	0.0145

Analysis of Load Case 5 : NP+HW+HI

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	194.75	0.00	99.74	0.0000	-0.0000	0.0323
20	0.00	247.00	0.00	99.74	0.0000	-0.0000	0.0266
30	0.00	247.00	0.00	122.73	0.0000	-0.0000	...
40	0.00	247.00	0.00	122.73	0.0000	-0.0000	0.0017
50	0.00	247.00	0.00	122.73	0.0000	-0.0000	0.0019
60	0.00	247.00	0.00	122.73	0.0000	-0.0000	...

Analysis of Load Case 6 : NP+HW+HE

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	194.75	0.00	99.74	0.0000	-0.0000	0.0308
20	0.00	247.00	0.00	99.74	0.0000	-0.0000	0.0249
30	0.00	247.00	0.00	122.73	0.0000	-0.0000	...
40	0.00	247.00	0.00	122.73	0.0000	-0.0000	0.0015
50	0.00	247.00	0.00	122.73	0.0000	-0.0000	0.0015
60	0.00	247.00	0.00	122.73	0.0000	-0.0000	...

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	36.06	82.60	-36.06	81.24	0.4366	0.4439	0.2916
20	31.28	105.00	-31.28	102.12	0.2979	0.3063	0.1891
30	139.54	150.00		134.97	0.9303		0.2265
40	120.98	150.00		181.32	0.8065		0.0103
50	120.77	150.00		181.32	0.8051		0.0067
60	138.95	150.00		112.08	0.9263		...

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	50.40	82.60	-50.40	81.24	0.6102	0.6204	0.4601
20	41.22	105.00	-41.22	102.12	0.3925	0.4036	0.2819
30	140.33	150.00		134.97	0.9356		0.3002
40	121.03	150.00		181.32	0.8069		0.0146
50	120.78	150.00		181.32	0.8052		0.0081
60	138.97	150.00		112.08	0.9265		...

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
-----------	------------------	-----------------------	--------------------	-----------------------	-------------	-------------	-------------



10	36.06	82.60	-36.06	81.24	0.4366	0.4439	0.2916
20	31.28	105.00	-31.28	102.12	0.2979	0.3063	0.1891
30	6.43	150.00	-6.43	134.97	0.0428	0.0476	0.2265
40	3.22	150.00	-3.22	181.32	0.0215	0.0178	0.0159
50	2.24	150.00	-2.24	181.32	0.0149	0.0123	0.0126
60	2.51	150.00	-2.51	134.97	0.0167	0.0186	0.0092

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	50.40	82.60	-50.40	81.24	0.6102	0.6204	0.4601
20	41.22	105.00	-41.22	102.12	0.3925	0.4036	0.2819
30	7.96	150.00	-7.96	134.97	0.0531	0.0590	0.3002
40	4.02	150.00	-4.02	181.32	0.0268	0.0222	0.0198
50	2.50	150.00	-2.50	181.32	0.0167	0.0138	0.0136
60	2.55	150.00	-2.55	134.97	0.0170	0.0189	0.0111

Analysis of Load Case 11 : HP+HW+HI

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	194.75	0.00	99.74	0.0000	-0.0000	0.0323
20	0.00	247.00	0.00	99.74	0.0000	-0.0000	0.0266
30	201.48	247.00		122.73	0.8157		...
40	184.18	247.00		122.73	0.7457		0.0017
50	184.18	247.00		122.73	0.7457		0.0019
60	201.48	247.00		122.73	0.8157		...

Analysis of Load Case 12 : HP+HW+HE

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	194.75	0.00	99.74	0.0000	-0.0000	0.0308
20	0.00	247.00	0.00	99.74	0.0000	-0.0000	0.0249
30	201.48	247.00		122.73	0.8157		...
40	184.18	247.00		122.73	0.7457		0.0015
50	184.18	247.00		122.73	0.7457		0.0015
60	201.48	247.00		122.73	0.8157		...

Analysis of Load Case 13 : IP+WE+EW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	10.24	82.60	-10.24	81.24	0.1239	0.1260	0.1152
20	10.00	105.00	-10.00	102.12	0.0952	0.0979	0.0893
30	137.94	150.00		102.34	0.9196		0.0908
40	120.88	150.00		181.32	0.8059		0.0044
50	120.71	150.00		181.32	0.8048		0.0026
60	138.48	150.00		112.08	0.9232		...

Analysis of Load Case 14 : IP+WF+CW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	9.41	82.60	-9.41	81.24	0.1139	0.1158	0.1158
20	9.19	105.00	-9.19	102.12	0.0875	0.0900	0.0900
30	137.88	150.00		102.34	0.9192		0.0898
40	120.87	150.00		181.32	0.8058		0.0044
50	120.71	150.00		181.32	0.8047		0.0025
60	138.49	150.00		112.08	0.9233		...

Analysis of Load Case 15 : IP+VO+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	11.05	82.60	-11.05	81.24	0.1338	0.1360	0.1252
20	10.81	105.00	-10.81	102.12	0.1030	0.1059	0.0973
30	137.94	150.00		102.34	0.9196		0.0988



40	120.88	150.00	181.32	0.8059	0.0044
50	120.71	150.00	181.32	0.8048	0.0026
60	138.48	150.00	112.08	0.9232	...

Analysis of Load Case 16 : IP+VE+EW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	10.24	82.60	-10.24	81.24	0.1239	0.1260	0.1152
20	10.00	105.00	-10.00	102.12	0.0952	0.0979	0.0893
30	137.94	150.00		102.34	0.9196		0.0908
40	120.88	150.00		181.32	0.8059		0.0044
50	120.71	150.00		181.32	0.8048		0.0026
60	138.48	150.00		112.08	0.9232		...

Analysis of Load Case 17 : NP+VO+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	11.05	82.60	-11.05	81.24	0.1338	0.1360	0.1252
20	10.81	105.00	-10.81	102.12	0.1030	0.1059	0.0973
30	1.93	150.00	-1.93	102.34	0.0129	0.0189	0.0988
40	0.89	150.00	-0.89	181.32	0.0059	0.0049	0.0044
50	0.55	150.00	-0.55	181.32	0.0037	0.0031	0.0026
60	0.31	150.00	-0.31	102.34	0.0021	0.0030	0.0046

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	30.09	82.60	-30.09	81.24	0.3642	0.3703	0.2110
20	26.93	105.00	-26.93	102.12	0.2565	0.2637	0.1404
30	139.18	150.00		134.97	0.9279		0.1943
40	120.96	150.00		181.32	0.8064		0.0077
50	120.76	150.00		181.32	0.8051		0.0053
60	138.92	150.00		112.08	0.9262		...

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	30.09	82.60	-30.09	81.24	0.3642	0.3703	0.2110
20	26.93	105.00	-26.93	102.12	0.2565	0.2637	0.1404
30	5.71	150.00	-5.71	134.97	0.0381	0.0423	0.1943
40	2.85	150.00	-2.85	181.32	0.0190	0.0157	0.0138
50	2.07	150.00	-2.07	181.32	0.0138	0.0114	0.0116
60	2.46	150.00	-2.46	134.97	0.0164	0.0182	0.0080

Analysis of Load Case 20 : BL+IP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	11.05	82.60	-11.05	81.24	0.1338	0.1360	0.1252
20	10.81	105.00	-10.81	102.12	0.1030	0.1059	0.0973
30	137.94	150.00		102.34	0.9196		0.0988
40	120.88	150.00		181.32	0.8059		0.0044
50	120.71	150.00		181.32	0.8048		0.0026
60	138.48	150.00		112.08	0.9232		...

Absolute Maximum of the all of the Stress Ratio's 0.9356

Governing Element: Bottom D end
 Governing Load Case 8 : IP+OW+EQ+FS+BS

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt 1	...	-11.3	0.9	38.3	27.9	+ Tens
Skirt 1	...	-11.3	-0.9	-38.3	-50.4	- Comp



Skirt 2	...	-11.0	0.9	29.3	19.2	+ Tens
Skirt 2	...	-11.0	-0.9	-29.3	-41.2	- Comp
Bottom D end	138.6	-2.0	0.2	5.0	141.8	+ Tens
Bottom D end	138.6	-2.0	-0.2	-5.0	...	- Comp
Shell 1	65.2	-0.9	0.1	2.6	66.9	+ Tens
Shell 1	65.2	-0.9	-0.1	-2.6	...	- Comp
Shell 2	65.2	-0.6	0.1	1.4	66.1	+ Tens
Shell 2	65.2	-0.6	-0.1	-1.4	...	- Comp
Top D End	138.6	-0.5	...	1.2	139.3	+ Tens
Top D End	138.6	-0.5	...	-1.2	...	- Comp

Additional Results per Section 4.4, see 4.4.15

Nomenclature :

- L - Section Length mm.
- Fxa - Allowable comp mem stress due to compression with lamc ≤ 0.15 , N./mm²
- Fba - Allowable all comp mem stress due to bending moment, N./mm²
- Fva - Allowable shear stress when only shear is present, N./mm²
- Fha - Allowable hoop comp memb stress under ext pressure alone, N./mm²
- Fxha - Allowable axial comp w/hoop compression and lamc ≤ 0.15 , N./mm²
- Fhxa - Allowable hoop comp w/axial compression and lamc ≤ 0.15 , N./mm²
- Faha - Allowable axial comp w/hoop compression and lamc > 0.15 , N./mm²
- Fbha - Allowable axial comp from bending w/hoop compression, N./mm²
- Fhba - Allowable hoop comp w/long compression due to moment, N./mm²

Allowable Stress Results for Case 1 : NP+EW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.110	3000	81	85	62		6895	81	6895	6895	6895
20	0.124	3000	102	107	77		6895	102	6895	6895	6895
30						134	6895		6895	6895	6895
40	0.159	6758	181	181	141	114	6895	181	6895	6895	6895
50	0.159	6758	181	181	141	114	6895	181	6895	6895	6895
60						134	6895		6895	6895	6895
70											
80											

Allowable Stress Results for Case 2 : NP+EW+EE+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.110	3000	81	85	62		6895	81	6895	6895	6895
20	0.124	3000	102	107	77		6895	102	6895	6895	6895
30						134	6895		6895	6895	6895
40	0.159	6758	181	181	141	114	6895	181	6895	6895	6895
50	0.159	6758	181	181	141	114	6895	181	6895	6895	6895
60						134	6895		6895	6895	6895
70											
80											

Allowable Stress Results for Case 3 : NP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.110	3000	81	85	62		6895	81	6895	6895	6895
20	0.124	3000	102	107	77		6895	102	6895	6895	6895
30						134	6895		6895	6895	6895
40	0.159	6758	181	181	141	114	6895	181	6895	6895	6895
50	0.159	6758	181	181	141	114	6895	181	6895	6895	6895
60						134	6895		6895	6895	6895
70											
80											

Allowable Stress Results for Case 4 : NP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.110	3000	81	85	62		6895	81	6895	6895	6895
20	0.124	3000	102	107	77		6895	102	6895	6895	6895
30						134	6895		6895	6895	6895
40	0.159	6758	181	181	141	114	6895	181	6895	6895	6895

[illegible]

Allowable Stress Results for Case 5 : NP+HW+HI

[illegible]

Allowable Stress Results for Case 6 : NP+HW+HE

[illegible]

Allowable Stress Results for Case 7 : IP+OW+1.1WI+FW+BW

[illegible]

Allowable Stress Results for Case 8 : IP+OW+EQ+FS+BS

[illegible]

Allowable Stress Results for Case 9 : EP+OW+1.1WI+FW+BW

[illegible]

Allowable Stress Results for Case 10 : EP+OW+EQ+FS+BS

[illegible]



Allowable Stress Results for Case 11 : HP+HW+HI

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.109	3000	83	86	64		6895	83	6895	6895	6895
20	0.122	3000	104	109	80		6895	104	6895	6895	6895
30							6895		6895	6895	6895
40	0.155	6758	181	181	141	115	6895	181	6895	6895	6895
50	0.155	6758	181	181	141	115	6895	181	6895	6895	6895
60							6895		6895	6895	6895
70											
80											

Allowable Stress Results for Case 12 : HP+HW+HE

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.109	3000	83	86	64		6895	83	6895	6895	6895
20	0.122	3000	104	109	80		6895	104	6895	6895	6895
30							6895		6895	6895	6895
40	0.155	6758	181	181	141	115	6895	181	6895	6895	6895
50	0.155	6758	181	181	141	115	6895	181	6895	6895	6895
60							6895		6895	6895	6895
70											
80											

Allowable Stress Results for Case 13 : IP+WE+EW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.110	3000	81	85	62		6895	81	6895	6895	6895
20	0.124	3000	102	107	77		6895	102	6895	6895	6895
30							6895		6895	6895	6895
40	0.159	6758	181	181	141	114	6895	181	6895	6895	6895
50	0.159	6758	181	181	141	114	6895	181	6895	6895	6895
60							6895		6895	6895	6895
70											
80											

Allowable Stress Results for Case 14 : IP+WF+CW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.110	3000	81	85	62		6895	81	6895	6895	6895
20	0.124	3000	102	107	77		6895	102	6895	6895	6895
30							6895		6895	6895	6895
40	0.159	6758	181	181	141	114	6895	181	6895	6895	6895
50	0.159	6758	181	181	141	114	6895	181	6895	6895	6895
60							6895		6895	6895	6895
70											
80											

Allowable Stress Results for Case 15 : IP+VO+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.110	3000	81	85	62		6895	81	6895	6895	6895
20	0.124	3000	102	107	77		6895	102	6895	6895	6895
30							6895		6895	6895	6895
40	0.159	6758	181	181	141	114	6895	181	6895	6895	6895
50	0.159	6758	181	181	141	114	6895	181	6895	6895	6895
60							6895		6895	6895	6895
70											
80											

Allowable Stress Results for Case 16 : IP+VE+EW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.110	3000	81	85	62		6895	81	6895	6895	6895
20	0.124	3000	102	107	77		6895	102	6895	6895	6895
30							6895		6895	6895	6895
40	0.159	6758	181	181	141	114	6895	181	6895	6895	6895
50	0.159	6758	181	181	141	114	6895	181	6895	6895	6895
60							6895		6895	6895	6895
70											
80											

Allowable Stress Results for Case 17 : NP+VO+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.110	3000	81	85	62		6895	81	6895	6895	6895



20	0.124	3000	102	107	77		6895	102	6895	6895	6895
30						102	6895		6895	6895	6895
40	0.159	6758	181	181	141	114	6895	181	6895	6895	6895
50	0.159	6758	181	181	141	114	6895	181	6895	6895	6895
60						102	6895		6895	6895	6895
70											
80											

Allowable Stress Results for Case 18 : FS+BS+IP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.110	3000	81	85	62		6895	81	6895	6895	6895
20	0.124	3000	102	107	77		6895	102	6895	6895	6895
30						134	6895		6895	6895	6895
40	0.159	6758	181	181	141	114	6895	181	6895	6895	6895
50	0.159	6758	181	181	141	114	6895	181	6895	6895	6895
60							6895		6895	6895	6895
70											
80											

Allowable Stress Results for Case 19 : FS+BS+EP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.110	3000	81	85	62		6895	81	6895	6895	6895
20	0.124	3000	102	107	77		6895	102	6895	6895	6895
30						134	6895		6895	6895	6895
40	0.159	6758	181	181	141	114	199	135	198	121	74
50	0.159	6758	181	181	141	114	157	145	156	97	87
60						134	6895		6895	6895	6895
70											
80											

Allowable Stress Results for Case 20 : BL+IP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.110	3000	81	85	62		6895	81	6895	6895	6895
20	0.124	3000	102	107	77		6895	102	6895	6895	6895
30						102	6895		6895	6895	6895
40	0.159	6758	181	181	141	114	6895	181	6895	6895	6895
50	0.159	6758	181	181	141	114	6895	181	6895	6895	6895
60							6895		6895	6895	6895
70											
80											

Note:

Load cases involving Wind and Seismic are assumed to cause a net section bending moment. These load case components are: EQ, EE, WI, WE, VO, VE, VF, HE, HI, WF. Static bending moments due to the weights of nozzles etc. are considered to be local in nature and will not cause a net section bending moment. This is an important consideration when computing the unity check per section 4.4.12 equations 4.4.86, 4.4.87, 4.4.90, 4.4.91 and 4.4.92.

Skirt Support Analysis Per EN 13445-3 Section 16.12:

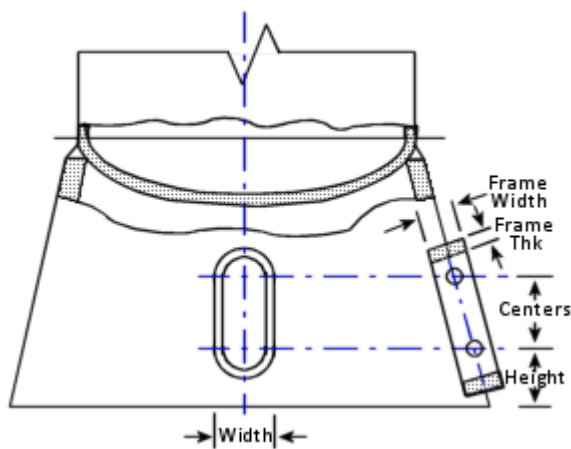
The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Skirt 2

Total vessel (empty) weight	Fg	157664.297 Kg
Head Weight below section 2-2	DeltaFg	14977.625 Kg
Weight of vessel contents (liquid)	Ff	14248.837 Kg
Vessel Gross Weight	Fg + Ff	171913.141 Kg
Skirt Mean Diameter	Dz	2945.000 mm.
Skirt Analysis Thickness (Corroded)	ez	18.500 mm.
Skirt Height	H	1000.000 mm.
Additional Axial Force at section 1-1	F1	-18845.611 Kg
Moment at section 1-1	M1	317627.969 Kg-m.
Applied Moment - Bottom of Skirt	Mb	374503.375 Kg-m.
Weight at Bottom of Skirt	W	173347.266 Kg

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Spherical
Head Mean Diameter at Skirt Top	Db	2945.000 mm.
Head Corroded Thickness	eb	95.000 mm.



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	18556112.000 cm ⁴
Moment of Inertia	Iyy =	18556112.000 cm ⁴
Section Modulus	W =	126017864.000 mm. ³
Section Metal Area	A =	1711.618 cm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:
 $= \sqrt{\frac{\min(Ixx, Iyy)}{Area}}$
 $= \sqrt{\frac{\min(18556112, 18556112)}{1711.618}}$
 $= 1041.215 \text{ mm.}$

Skirt Slenderness Ratio [rg/L]:
 $= rg / L$
 $= 1041.215/1000.0$
 $= 1.041$

Correct the Moment [M4]:
 top of skirt is the critical end
 Controlling Moment [M4] = 317627.969 Kg-m.

Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$= -M / Z - W / A$$

$$= -317627.969 / .50407E+09 - 173347.266 / 1711.61804$$

$$= -16.111 \text{ N./mm}^2$$

Upwind Side:

$$= +(M) / Z - W / A$$

$$= +317627.969 / .50407E+09 - 173347.266 / 1711.61804$$

$$= -3.753 \text{ N./mm}^2$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$= F1 - Fg - Ff + 4 \cdot M1 / Dz$$

$$= -18845.611 - 157664.297 - 14248.837 + 4 \cdot 317627.969 / 2945.0$$

$$= 240646.0 \text{ Kg}$$

Force at the joint, 16.12-2 [Fzq]:

$$= F1 - Fg - Ff - 4 \cdot M1 / Dz$$

$$= -18845.611 - 157664.297 - 14248.837 - 4 \cdot 317627.969 / 2945.0$$

$$= -622163.5 \text{ Kg}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$= (Fzp + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (240645.969 + 14977.625 + 14248.837) / (\pi \cdot 2945.0 \cdot 95.0) + 191.0 \cdot 2945.0 / (4 \cdot 95.0)$$

$$= 151.04 \text{ N./mm}^2$$

Stress [σ_{1qm}]:

$$= (Fzq + \Delta Fg + Ff) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (-0.6E+06 + 14977.625 + 14248.837) / (\pi \cdot 2945.0 \cdot 95.0) + 191.0 \cdot 2945.0 / (4 \cdot 95.0)$$

$$= 141.42 \text{ N./mm}^2$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$= (Ff + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (14248.837 + 14977.625) / (\pi \cdot 2945.0 \cdot 95.0) + 191.0 \cdot 2945.0 / (4 \cdot 95.0)$$

$$= 148.36 \text{ N./mm}^2$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$= Fzp / (\pi \cdot Dz \cdot ez)$$

$$= 240645.969 / (\pi \cdot 2945.0 \cdot 18.5)$$

$$= 13.79 \text{ N./mm}^2$$

Stress [σ_{3qm}]:

$$= Fzq / (\pi \cdot Dz \cdot ez)$$

$$= -622163.500 / (\pi \cdot 2945.0 \cdot 18.5)$$

$$= -35.65 \text{ N./mm}^2$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:

$$= 1 - 0.5(Db + eb - Dz + ez) / (r + eb)$$

$$= 1 - 0.5(2945.0 + 95.0 - 2945.0 + 18.5) / (1425.0 + 95.0)$$

$$= 0.9627$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.963) = 15.706 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(95.0^2 + 18.5^2 + 2 \cdot 95.0 \cdot 18.5 \cdot 0.963)}$$

$$= 56.46 \text{ mm.}$$

Bending moment at verification points p and q [Mp & Mq]:

$$\begin{aligned} M_p &= a \cdot F_{zp} = 56.46 \cdot 240646.0 = 13587.19 \text{ Kg-m.} \\ M_q &= a \cdot F_{zq} = 56.46 \cdot -622163.5 = -35128.17 \text{ Kg-m.} \end{aligned}$$

Correction factor [C]:

$$\begin{aligned} &= 0.63 - 0.057 \left(\frac{eb}{ez} \right)^2 \\ &= 0.63 - 0.057 \left(\frac{95.0}{18.5} \right)^2 \\ &= -0.873 \end{aligned}$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2) \\ &= -0.873 \cdot 6 \cdot 13587.187 / (\pi \cdot 2945.0 \cdot 95.0^2) \\ &= -8.36 \text{ N./mm}^2 \end{aligned}$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2) \\ &= -0.873 \cdot 6 \cdot -35128.168 / (\pi \cdot 2945.0 \cdot 95.0^2) \\ &= 21.61 \text{ N./mm}^2 \end{aligned}$$

Stress [$\sigma_{3pb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2) \\ &= -0.873 \cdot 6 \cdot 13587.187 / (\pi \cdot 2945.0 \cdot 18.5^2) \\ &= -220.43 \text{ N./mm}^2 \end{aligned}$$

Stress [$\sigma_{3qb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2) \\ &= -0.873 \cdot 6 \cdot -35128.168 / (\pi \cdot 2945.0 \cdot 18.5^2) \\ &= 569.90 \text{ N./mm}^2 \end{aligned}$$

Geometry variables, 16.12-32 to 36:

$$\begin{aligned} eb/D_b &= 3.74/115.945 = 0.0323 \\ y &= 125 \cdot eb/D_b = 125 \cdot 95.0/2945.0 = 4.0323 \\ \alpha &= 0 \end{aligned}$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= 0.0 \text{ N./mm}^2 \text{ (16.12-31)}$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$\begin{aligned} &= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)} \\ &= 151.045 - -8.359 + 0.0 \\ &= 159.40 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{1potot}]:

$$\begin{aligned} &= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)} \\ &= 151.045 + -8.359 - 0.0 \\ &= 142.69 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{1qitot}]:

$$\begin{aligned} &= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)} \\ &= 141.418 - 21.612 + 0.0 \\ &= 119.81 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{1qotot}]:

$$\begin{aligned} &= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)} \\ &= 141.418 + 21.612 - 0.0 \\ &= 163.03 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{2pitot}]:

$$\begin{aligned} &= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)} \\ &= 148.36 + -8.359 + 0.0 \\ &= 140.00 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{2potot}]:

$$\begin{aligned}
 &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\
 &= 148.36 - -8.359 - 0.0 \\
 &= 156.72 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{2qitot}]: \\
 &= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)} \\
 &= 148.36 + 21.612 + 0.0 \\
 &= 169.97 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{2qotot}]: \\
 &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\
 &= 148.36 - 21.612 - 0.0 \\
 &= 126.75 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3pitot}]: \\
 &= \sigma_{3pm} - \sigma_{3pb} \\
 &= 13.788 - -220.43 \\
 &= 234.22 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3potot}]: \\
 &= \sigma_{3pm} + \sigma_{3pb} \\
 &= 13.788 + -220.43 \\
 &= -206.64 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3qitot}]: \\
 &= \sigma_{3qm} - \sigma_{3qb} \\
 &= -35.647 - 569.897 \\
 &= -605.54 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3qotot}]: \\
 &= \sigma_{3qm} + \sigma_{3qb} \\
 &= -35.647 + 569.897 \\
 &= 534.25 \text{ N./mm}^2
 \end{aligned}$$

Allowable stresses:

$$\begin{aligned}
 \text{For } \sigma_{1pitot} \text{ and } \sigma_{1potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2 \\
 &= 150.0(3 - (1/1.5)) \cdot (151.045/150.0)^2 \\
 &= 348.60 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{1qitot} \text{ and } \sigma_{1qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2 \\
 &= 150.0(3 - (1/1.5)) \cdot (141.418/150.0)^2 \\
 &= 361.11 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{2pitot} \text{ and } \sigma_{2potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2 \\
 &= 150.0(3 - (1/1.5)) \cdot (148.36/150.0)^2 \\
 &= 352.17 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{2qitot} \text{ and } \sigma_{2qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2 \\
 &= 150.0(3 - (1/1.5)) \cdot (148.36/150.0)^2 \\
 &= 352.17 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{3pitot} \text{ \& } \sigma_{3potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{3pm} / f)^2 \\
 &= 150.0(3 - (1/1.5)) \cdot (13.788/150.0)^2 \\
 &= 449.15 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{3qitot} \text{ \& } \sigma_{3qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{3qm} / f)^2 \\
 &= 150.0(3 - (1/1.5)) \cdot (-35.647/150.0)^2 \\
 &= 444.35 \text{ N./mm}^2
 \end{aligned}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (N./mm²)



Stress	Actual	Allowed	Result	Equation
Stot1pi	159.40	348.60	Pass	46
Stot1po	142.69	348.60	Pass	47
Stot1qi	119.81	361.11	Pass	48
Stot1qo	163.03	361.11	Pass	49
Stot2pi	140.00	352.17	Pass	50
Stot2po	156.72	352.17	Pass	51
Stot2qi	169.97	352.17	Pass	52
Stot2qo	126.75	352.17	Pass	53
Stot3pi	234.22	449.15	Pass	54
Stot3po	-206.64	449.15	Pass	55
Stot3qi	-605.54	444.35	Fail	56
Stot3qo	534.25	444.35	Fail	57

WARNING - One of the Stresses is over the limit in an area without openings!

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Shop/Field Installation Options :

Platform(s) installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	7233.351 mm.
Center of Gravity of the Liquid	4200.000 mm.
Center of Gravity of the Nozzles	7555.398 mm.
Center of Gravity of the Added Weights (Operating)	6422.441 mm.
Center of Gravity of the Added Weights (Empty)	6422.441 mm.
Center of Gravity of Bare Shell New and Cold	5767.170 mm.
Center of Gravity of Bare Shell Corroded	5776.618 mm.
Vessel CG in the Operating Condition	5989.183 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	6132.625 mm.
Vessel CG in the Test Condition	6086.289 mm.

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	2964.9998	mm.
Skirt Thickness	STHK	20.0000	mm.
Skirt Internal Corrosion Allowance	SCA	0.7500	mm.
Skirt External Corrosion Allowance		0.7500	mm.
Skirt Material		SA-285 C	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	48.0000	mm.
Design Temperature of the Basing		38.00	°C
Basing Material		SA-285 C	
Basing Ope. All. Stress	BASOPE, f4	138.00	N./mm ²
Basing Yield Stress		207.00	N./mm ²
Inside Diameter of Basing	DI, D4	2850.9998	mm.
Outside Diameter of Basing	DOU, D5	3258.9998	mm.
Nominal Diameter of Bolts	BND, d(b0)	48.0000	mm.
Bolt Corrosion Allowance	BCA	1.5000	mm.
Bolt Material		SA-36	
Bolt Operating Allowable Stress	SA, fB	149.90	N./mm ²
Number of Bolts	RN, nb	16	
Diameter of Bolt Circle	DC, DBC	3119.0000	mm.
Bolt Allowable Shear Stress		99.920	N./mm ²
Ultimate Comp. Strength of Concrete	FPC	35.0	N./mm ²
Allowable Comp. Strength of Concrete	FC	15.8	N./mm ²
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	16.0000	mm.
Width of Gussets at Top Plate	TWDT	147.0000	mm.
Width of Gussets at Base Plate	BWDT	147.0000	mm.
Gusset Plate Elastic Modulus	E	201734288.0	KPa.
Gusset Plate Yield Stress	SY	228.0	N./mm ²
Height of Chair	h1	333.0000	mm.
Distance between Gussets	RG, b6	128.0000	mm.
Dist. from Bolt Center to Gusset (Rg/2)	CG	64.0000	mm.
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	42.0000	mm.
Radial Width of the Top Plate	TOPWTH, b5	147.0000	mm.
Anchor Bolt Hole Dia. in Top Plate	dia,H	64.0000	mm.
External Corrosion Allowance	CA	3.0000	mm.
Dead Weight of Vessel	DW	163304.7	Kgf
Operating Weight of Vessel	ROW	196399.2	Kgf
Earthquake Moment on Basing	EQMOM	261447.5	Kg-m.
Wind Moment on Basing	WIMOM	78321.8	Kg-m.
User Moment at the Base [Earthquake Case]		229280.7	Kg-m. (UEQMOM)
User Moment at the Base [WindLoad Case]		229280.7	Kg-m. (UWIMOM)
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 490728/3119.0) - 196399) / 16$$

= 27058.1 Kgf

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 307603/3119.0) - 163305) / 16$$

$$= 14448.5 \text{ Kgf}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/3119.0) - 196399) / 16$$

$$= -12274.9 \text{ Kgf}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in 1 Bolt (corroded) Abss : 11.5517 cm²
 Area Available in all the Bolts Abss * RN : 184.8264 cm²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.243	0.364	1.348	2.567	0.450	0.778	88608.7	285007.8	7570.8
4	0.303	0.273	1.518	2.435	0.437	0.781	91538.0	287937.1	8245.5
6	0.288	0.296	1.477	2.468	0.440	0.781	90791.5	287190.7	8068.9
8	0.300	0.301	1.508	2.443	0.438	0.781	91350.5	287749.6	8200.8
10	0.300	0.300	1.511	2.441	0.438	0.781	91397.3	287796.4	8211.9
11	0.300	0.300	1.509	2.442	0.438	0.781	91373.9	287773.1	8206.4

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 91373.9 / (1.886 \cdot 3119.0 \cdot 2.442)$$

$$= 124.742 \text{ N./mm}^2, \text{ Should be } \leq 149.9 \text{ N./mm}^2$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 16 \cdot 11.552 / (\pi \cdot 3119.0)$$

$$= 1.886 \text{ mm.}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 5.433[(2 \cdot 0.3 \cdot 3119.0 + 204.0) / (2 \cdot 0.3 \cdot 3119.0)]$$

$$= 6.026 \text{ N./mm}^2, \text{ Should be } \leq 15.8 \text{ N./mm}^2$$

Values for table 10.3, l = 147.0, b = 306.207, l/b = 0.480067

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(1539.896, 4408.616) = 4408.616 \text{ Kgf}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 4408.616 / 136.6)} + 3.0$$

$$= 46.575 \text{ mm.}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 147.0, b = 128.0, b/l = 0.870748

As b/l (0.871) is less than 1, inverting b/l = 1.148.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2l \sin(\pi \cdot a/l)) / (\pi \cdot g))) + 1] - [(0.7 - g2)P / (4\pi)]$$

$$= 17657.01 / (4 \cdot \pi) [1.3(\ln((2 \cdot 147.0 \cdot \sin(\pi \cdot 77.0/147.0)) / (\pi \cdot 40.0))) + 1] - [(0.7 - 0.116) \cdot 17657.01 / (4 \cdot \pi)]$$

$$= 2132.1897 \text{ Kgf}$$

Required Thickness of Continuous Top Ring [Tc]:



$$= \sqrt{(6 \cdot \text{Abs}(\text{Mo}) / \text{Fb}) + \text{Ca}}$$

$$= \sqrt{(6 \cdot \text{Abs}(2132.19)/152.0) + 3.0}$$

$$= 31.7299 \text{ mm.}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot \text{lavg}) + \text{Ca}$$

$$= 17657.01 / (124.11 \cdot 147.0) + 3.0$$

$$= 12.525 \text{ (not less than } 9.525 + 3.0 \text{) mm.}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 3119.0 / 16 = 612.414 \text{ mm.}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76 (P \cdot a / (m \cdot (h + tba) \cdot 1.5 \cdot \text{Sktope}))^{2/3} \cdot r^{1/3} + \text{Ca}$$

$$= 1.76 (17657 \cdot 77.0 / (612.41 \cdot 291.0 \cdot 177))^{2/3} \cdot 1482.51^{1/3} + \text{Ca}$$

$$= 11.303 + 1.5 = 12.803 \text{ mm.}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 57433.9 / (11.552 \cdot 16)$$

$$= 30.5 \text{ N./mm}^2. \text{ Must be less than } 99.9 \text{ N./mm}^2.$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	46.5750	mm.
Actual Basing Thickness as entered by user	48.0000	mm.
Required Thickness of Chair Cap	31.7299	mm.
Actual Top Ring Thickness as entered by user	42.0000	mm.
Required Gusset thickness, + CA	12.5250	mm.
Actual Gusset Thickness as entered by user	16.0000	mm.
Required Thickness of Skirt for Local Stress	12.8031	mm.
Given Thickness of Skirt	20.0000	mm.
Required Gusset Height to meet local stress	109.7664	mm.

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - \text{CA})$$

$$= 50.401 \cdot (20.0 - 1.5)$$

$$= 95.08 \text{ Kg/mm.}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 95 / [(0.4 \cdot 177) \cdot 2 \cdot 0.707]$$

$$= 9.314 \text{ mm.}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 17657.0 / (170.0 + 2 \cdot (243.0 + 42.0))$$

$$= 23.861 \text{ Kg/mm.}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 17657.0 \cdot 77.0 / (170.0 \cdot (285.0) + 0.6667 \cdot (285.0)^2)$$

$$= 13.251 \text{ Kg/mm.}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{23.86^2 + 13.25^2}$$

$$= 27.294 \text{ Kg/mm.}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 27.29 / [(0.4 \cdot 177) \cdot 2 \cdot 0.707]$$



= 2.674 mm.

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$= \text{Bolt Load} / (2 \cdot \text{TopWth})$$

$$= 17657.0 / (2 \cdot 147.0)$$

$$= 60.058 \text{ Kg/mm.}$$

Weld Load [Wh]:

$$= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth})$$

$$= 17657.0 \cdot 77.0 / (2 \cdot 285.0 \cdot 147.0)$$

$$= 16.226 \text{ Kg/mm.}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{60.06^2 + 16.23^2}$$

$$= 62.211 \text{ Kg/mm.}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 62.21 / [(0.4 \cdot 177) \cdot 2 \cdot 0.707]$$

$$= 6.094 \text{ mm.}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	9.3138	mm.
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm.
Required Top Plate to Skirt Weld Size	6.3500	mm.
Required Gusset to Top Plate Double Fillet Weld Size	6.0942	mm.

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N33"

From: 30

Pressure for Reinforcement Calculations	P	190.920	bars
Temperature for Internal Pressure	Temp	200	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	150.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Hemispherical Head	D	2844.00	mm.
Head Finished (Minimum) Thickness	t	98.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

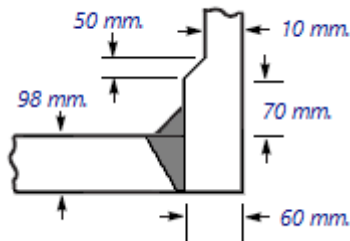
Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	142.00	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm.
Height of Beveled Transition	L'	50.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		60.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	98.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		1500	
Flange Grade		GR 1.1	
Piping Exit Angle [North Clockwise reference]		90.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		2000.0	mm.
Centerline Distance to Tangent		1850.0	mm.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N33"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	3.000	in.
Actual Thickness Used in Calculation	0.394	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 82.2 \cdot (\exp(190.92/(142.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 5.9151 + 3.0 + 0.0 = 8.9151 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 142.0 \cdot 1.0 \cdot \ln((2 \cdot 7.0 + 82.2) / 82.2) - 0.0$$

$$= 223.314 - 0.0 = 223.3144 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	8.020 mm
Min. required shell thickness, internal p + c	96.674 mm.
Min. required shell thickness, external p + c	10.070 mm.

Nozzle neck minimum required thickness	: 8.020 mm.
--	-------------

Minimum nozzle neck thickness 10.008 mm., is greater than required thickness 8.020 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	150.9,	Allowable :	142.0 N./mm²	Failed
Expansion :	0.0,	Allowable :	241.7 N./mm ²	Passed
Occasional :	51.7,	Allowable :	188.9 N./mm ²	Passed
Shear :	79.4,	Allowable :	99.4 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[142.0/150.0, 1]$$

$$= 0.947$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 60.0 - 3.0$$

$$= 57.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 10.0 - 3.0$$

$$= 7.000 \text{ mm.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2844.0/2 + 3.0$$

$$= 1425.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:



$$\begin{aligned}
 &= \min(\sqrt{(\text{Reff} \cdot t)}, 2 \cdot Rn) \\
 &= \min(\sqrt{(1425.0 \cdot 95.0)}, 2 \cdot 41.1) \\
 &= 82.200 \text{ mm.}
 \end{aligned}$$

Intermediate value [Cn]:

$$\begin{aligned}
 &= \min((t + te)/tn)^{0.35}, 1) \\
 &= \min((95.0 + 0.0)/57.0)^{0.35}, 1) \\
 &= \min(1.196, 1) \\
 &= 1.000
 \end{aligned}$$

Intermediate value [Cp]:

$$= 9999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned}
 &= \min(t + te + Fp \cdot \sqrt{(Rn \cdot tn)}, Lpr1 + t) \\
 &= \min(95.0 + 0.0 + 1.0 \cdot \sqrt{(41.1 \cdot 57.0)}, 200.0 + 95.0) \\
 &= 143.401 \text{ mm.}
 \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned}
 &= t + ((A5 \cdot frp) / LR) \\
 &= 95.0 + ((0.0 \cdot 1.0) / 82.2) \\
 &= 95.000 \text{ mm.}
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned}
 &= t \cdot LR \\
 &= 95.0 \cdot 82.2 \\
 &= 78.090 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned}
 &= tn \cdot LH \\
 &= 57.0 \cdot 143.401 \\
 &= 81.739 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned}
 &= 0.5 \cdot \text{Leg}^2 \\
 &= 0.5 \cdot 10.0^2 \\
 &= 0.500 \text{ cm}^2
 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned}
 &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\
 &= 78.09 + 0.947(81.739 + 0.0) + 0.5 + 0.0 + 0.0 \\
 &= 155.969 \text{ cm}^2
 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned}
 &= tn / \ln[1 + tn/Rn] \\
 &= 57.0 / \ln[1 + 57.0/41.1] \\
 &= 65.519 \text{ mm.}
 \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned}
 &= \text{teff} / \ln[1 + \text{teff}/\text{Reff}] \\
 &= 95.0 / \ln[1 + 95.0/1425.0] \\
 &= 1471.990 \text{ mm.}
 \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned}
 &= 1.5 \cdot S \cdot E \\
 &= 1.5 \cdot 150.0 \cdot 1.0 \\
 &= 225.0 \text{ N./mm}^2
 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned}
 &= P \cdot Rxn \cdot LH \\
 &= 190.92 \cdot 65.519 \cdot 143.401 \\
 &= 18292.3 \text{ Kgf}
 \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned}
 &= P \cdot Rxs \cdot (LR + tn)/2 \\
 &= 190.92 \cdot 1471.99 \cdot (82.2 + 57.0)/2 \\
 &= 199463.1 \text{ Kgf}
 \end{aligned}$$



Discontinuity Force from Internal Pressure [FY]:

$$= (P \cdot Rxs \cdot Rnc) / 2$$

$$= (190.92 \cdot 1471.99 \cdot 41.1) / 2$$

$$= 58893.2 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (199463.1 + 58893.2 + 18292.3) / 190.92$$

$$= 1421.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / (2 \cdot teff))$$

$$= 224.998 / (2 \cdot 1420.954 / 155.969 - 1471.99 / (2 \cdot 95.0))$$

$$= 214.8 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t/Rxs]$$

$$= 2 \cdot 150.0[95.0/1471.99]$$

$$= 193.6 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(214.811, 193.602)$$

$$= 193.602 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (18292.279 + 199463.094 + 58893.195) / 155.969$$

$$= 173.947 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / (2 \cdot teff)$$

$$= 190.92 \cdot 1471.99 / (2 \cdot 95.0)$$

$$= 147.9 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 173.947 - 147.921, 147.921)$$

$$= 200.0 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	225.00	N./mm ²
Local Primary Membrane Stress	PL	199.97	N./mm ²
Maximum Allowable Working Pressure	Pmax	193.60	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (41.1 + 57.0) / 41.1$$

$$= 2.387 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn)$$

$$= \pi / 2 \cdot (41.1 + 57.0)$$

$$= 154.095 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 57.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(58893 \cdot 2.39, 1.5 \cdot 142.0(81.739 + 0.0), \pi / 4 \cdot 190.9 \cdot 41.1^2 \cdot 2.39^2)$$

$$= 14715.539 \text{ Kg}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$



= 14715.539/(154.095(0.49•7.071 + 0.6•95.0 + 0.49•0.0))
= 15.489 < or = to 149.999 Weld Stress Passed

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

= $t_n / \ln[1 + t_n/R_n]$
= 57.0/ln[1 + 57.0/41.1]
= 65.519 mm.

Shell Radius for Force Calculation [Rxs]:

= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
= 95.0/ln[1 + 95.0/1425.0]
= 1471.990 mm.

Allowable Local Primary Membrane Stress [Sallow]:

= 1.5 • S
= 1.5 • 150.0
= 225.0 N./mm²

Force acting on the Nozzle [fN]:

= P • Rxn • LH
= 1.034 • 65.519 • 143.401
= 99.1 Kgf

Force acting on the Shell [fS]:

= P • Rxs • (LR + t_n)/2
= 1.034 • 1471.99 • (82.2 + 57.0)/2
= 1080.5 Kgf

Discontinuity Force from External Pressure [fY]:

= (P • Rxs • R_{nc})/2
= (1.034 • 1471.99 • 41.1)/2
= 319.0 Kgf

Area Resisting External Pressure [Ap]:

= (fS + fY + fN)/P
= (1080.5 + 319.0 + 99.1)/1.034
= 1421.0 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= Sallow / (2 • Ap/AT - Rxs/(2 • t_{eff}))
= 224.998/(2 • 1420.954/155.969 - 1471.99/(2•95.0))
= 214.8 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:

= 2 • S[t/Rxs]
= 2 • 150.0[95.0/1471.99]
= 193.6 bars

Maximum Allowable Working Pressure [Pmax]:

= min(Pmax1, Pmax2)
= min(214.811, 193.602)
= 193.602 bars

Average Primary Membrane Stress [σAvg]:

= (fN + fS + fY) / AT
= (99.087 + 1080.466 + 319.017)/155.969
= 0.942 N./mm²

General Primary Membrane Stress [σCirc]:

= P • Rxs / (2 • t_{eff})
= 1.034 • 1471.99/(2 • 95.0)
= 0.8 N./mm²

Maximum Local Primary Membrane Stress [PL]:

= max(2 • σAvg - σCirc, σCirc)
= max(2 • 0.942 - 0.801, 0.801)
= 1.1 N./mm²



Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	225.00	N./mm ²
Local Primary Membrane Stress	PL	1.08	N./mm ²
Maximum Allowable External Pressure	Pmax	193.60	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -15 °C
 User entered Minimum Design Temperature : -10 °C
 The Curve for this Material is : A
 From Figure 3.8M- with PWHT, the MDMT is : 1 °C
 Temperature reduction per Graph 3.13 : 15 °C
 The computed stress ratio is : 0.720

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (5.042 \cdot 1.0) / (10.0 - 3.0 - 0.0)$$

$$= 0.720$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : -10 °C
 The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.088

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (5.042 \cdot 1.0) / (60.0 - 3.0 - 0.0)$$

$$= 0.088$$

Gov. MDMT of the nozzle to shell joint welded assembly : -15 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction : -32 °C

The Drop for this Nozzle is : 3.3879 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 301.3879 mm.

Input Echo, WRC107/537 Item 1, Description: N33" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	2844.000	mm.
Vessel Thickness	Tv	98.000	mm.
Design Temperature	T1	200.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	175.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	150.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow

Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm.
Nozzle Diameter	Dn	76.200	mm.
Nozzle Thickness	Tn	60.000	mm.

Nozzle Material	SA-105		
Nozzle UNS Number	K03504		
Nozzle Cold S.I. Allowable	SNmc	165.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	142.00	N./mm ²

Design Internal Pressure	Dp	190.920	bars
Include Pressure Thrust		No	

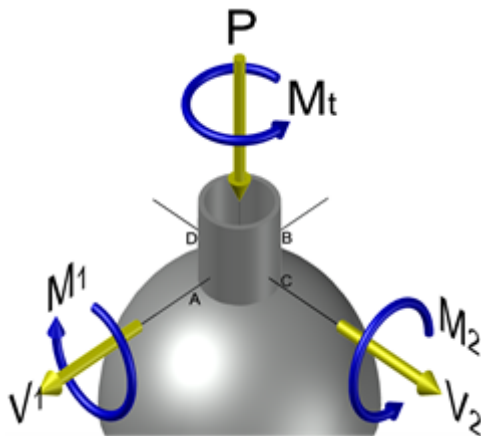
External Forces and Moments in WRC 107/537 Convention:

Radial Load	(SUS)	P	-1835.6	Kgf
Longitudinal Shear	(SUS)	(V1) V1	1622.5	Kgf
Circumferential Shear	(SUS)	(Vc) V2	1622.5	Kgf
Circumferential Moment	(SUS)	(Mc) M1	319.2	Kg-m.
Longitudinal Moment	(SUS)	(Ml) M2	319.2	Kg-m.
Torsional Moment	(SUS)	Mt	412.9	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 196.2 + 2 \cdot 1.65 \cdot \sqrt{(1472.5 (98.0 - 3.0))}$$

$$= 1430.451 \text{ mm.}$$

WRC 107 Stress Calculation for SUStained loads:

Radial Load	P	-1835.6	Kgf
Circumferential Shear	(VC) V2	1622.5	Kgf
Longitudinal Shear	(VL) V1	1622.5	Kgf
Circumferential Moment	(MC) M1	319.2	Kg-m.
Longitudinal Moment	(ML) M2	319.2	Kg-m.
Torsional Moment	MT	412.9	Kg-m.

Dimensionless Param: U = 0.26 TAU = 5.00 (1.22) RHO = 1.67

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 2	0.11625	(A,B,C,D)
M(x) / P	SP 2	0.10940	(A,B,C,D)



$N(x) * T * \sqrt{Rm * T} / MC$ SM 2 0.30989 (A,B,C,D)
 $M(x) * \sqrt{Rm * T} / MC$ SM 2 0.36618 (A,B,C,D)
 $N(x) * T * \sqrt{Rm * T} / ML$ SM 2 0.30989 (A,B,C,D)
 $M(x) * \sqrt{Rm * T} / ML$ SM 2 0.36618 (A,B,C,D)

 $N(y) * T / P$ SP 2 0.21514 (A,B,C,D)
 $M(y) / P$ SP 2 0.07901 (A,B,C,D)
 $N(y) * T * \sqrt{Rm * T} / MC$ SM 2 0.06202 (A,B,C,D)
 $M(y) * \sqrt{Rm * T} / MC$ SM 2 0.36669 (A,B,C,D)
 $N(y) * T * \sqrt{Rm * T} / ML$ SM 2 0.06202 (A,B,C,D)
 $M(y) * \sqrt{Rm * T} / ML$ SM 2 0.36669 (A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Rad.	Bend. P	1.3	-1.3	1.3	-1.3	1.3	-1.3	1.3	-1.3
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-2.0	2.0	2.0	-2.0
Rad.	Memb. ML	-0.3	-0.3	0.3	0.3	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-2.0	2.0	2.0	-2.0	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-0.8	0.7	3.9	-2.8	-0.8	0.7	3.9	-2.8
Tang.	Memb. P	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
Tang.	Bend. P	0.9	-0.9	0.9	-0.9	0.9	-0.9	0.9	-0.9
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-2.0	2.0	2.0	-2.0
Tang.	Memb. ML	-0.1	-0.1	0.1	0.1	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-2.0	2.0	2.0	-2.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-0.7	1.5	3.5	-2.5	-0.7	1.5	3.5	-2.5
Shear VC		0.5	0.5	-0.5	-0.5	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Shear MT		0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7
Tot. Shear		1.2	1.2	0.2	0.2	0.2	0.2	1.2	1.2
Str. Int.		2.5	2.6	3.9	2.9	0.9	1.5	4.9	3.9

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	147.9	147.9	147.9	147.9	147.9	147.9	147.9	147.9
Rad.	Pl (SUS)	-0.1	-0.1	0.5	0.5	-0.1	-0.1	0.5	0.5
Rad.	Q (SUS)	-0.7	0.7	3.3	-3.3	-0.7	0.7	3.3	-3.3
Long.	Pm (SUS)	147.9	147.9	147.9	147.9	147.9	147.9	147.9	147.9
Long.	Pl (SUS)	0.4	0.4	0.5	0.5	0.4	0.4	0.5	0.5
Long.	Q (SUS)	-1.1	1.1	3.0	-3.0	-1.1	1.1	3.0	-3.0
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.5	0.5	-0.5	-0.5	-0.5	-0.5	0.5	0.5
Shear	Q (SUS)	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7
Pm (SUS)		147.9	147.9	147.9	147.9	147.9	147.9	147.9	147.9
Pm+Pl (SUS)		148.7	148.7	149.0	149.0	148.7	148.7	149.0	149.0



Pm+Pl+Q (Total)	148.4	150.3	151.8	145.5	147.3	149.4	152.9	146.5
-----------------	-------	-------	-------	-------	-------	-------	-------	-------

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	147.92	150.00	Passed
Pm+Pl (SUS)	148.97	225.00	Passed
Pm+Pl+Q (TOTAL)	152.85	450.02	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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Input, Nozzle Desc: N5A2"

From: 40

Pressure for Reinforcement Calculations	P	190.798	bars
Temperature for Internal Pressure	Temp	200	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	150.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2750.00	mm.
Design Length of Section	L	6758.0000	mm.
Shell Finished (Minimum) Thickness	t	192.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1249.97	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

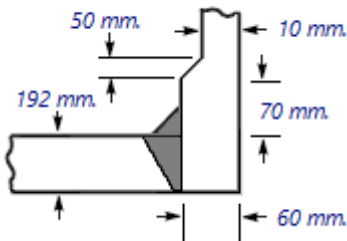
Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	142.00	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm.
Height of Beveled Transition	L'	50.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		60.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	192.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N5A2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.394	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 56.8 \cdot (\exp(190.798/(142.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 4.0845 + 3.0 + 0.0 = 7.0845 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 142.0 \cdot 1.0 \cdot \ln((2 \cdot 7.0 + 56.8) / 56.8) - 0.0$$

$$= 312.837 - 0.0 = 312.8375 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	7.520 mm
Min. required shell thickness, internal p + c	190.198 mm.
Min. required shell thickness, external p + c	19.199 mm.

Nozzle neck minimum required thickness	: 7.520 mm.
--	-------------

Minimum nozzle neck thickness 10.008 mm., is greater than required thickness 7.520 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[142.0/150.0, 1]$$

$$= 0.947$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 60.0 - 3.0$$

$$= 57.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 10.0 - 3.0$$

$$= 7.000 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2756.0/189.0$$

$$= 14.582 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2750.0/2 + 3.0$$

$$= 1378.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1378.0 \cdot 189.0}, 2 \cdot 28.4)$$

$$= 56.800 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 189.0, 0.0] + \sqrt{(28.4 \cdot 57.0)}$$

$$= 40.234 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 200.0$$

$$= 200.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(189.0 + 0.0)$$

$$= 1512.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[40.234, 200.0, 1512.0] + 189.0$$

$$= 229.234 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 189.0 + ((0.0 \cdot 1.0) / 56.8)$$

$$= 189.000 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 28.4 + 57.0) / (\sqrt{(2756.0 + 189.0) \cdot 189.0}))$$

$$= 0.153$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 189.0 \cdot 56.8 \cdot \max((0.153/5)^{0.85}, 1)$$

$$= 107.352 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 57.0 \cdot 229.234$$

$$= 130.664 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 107.352 + 0.947(130.664 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 231.547 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 57.0 / \ln[1 + 57.0/28.4]$$

$$= 51.773 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 189.0 / \ln[1 + 189.0/1378.0]$$

$$= 1470.476 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 150.0 \cdot 1.0$$

$$= 225.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 190.798 \cdot 51.773 \cdot 229.234$$

$$= 23091.6 \text{ Kg}$$



Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + tn)$$

$$= 190.798 \cdot 1470.476(56.8 + 57.0)$$

$$= 325589.1 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 190.798 \cdot 1470.476 \cdot 28.4$$

$$= 81254.2 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN)/P$$

$$= (325589.1 + 81254.2 + 23091.6)/190.798$$

$$= 2209.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$

$$= 224.998 / (2 \cdot 2209.699/231.547 - 1470.476/189.0)$$

$$= 199.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 150.0[189.0/1470.476]$$

$$= 192.8 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(198.994, 192.782)$$

$$= 192.782 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (23091.58 + 325589.094 + 81254.211)/231.547$$

$$= 182.093 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 190.798 \cdot 1470.476/189.0$$

$$= 148.5 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 182.093 - 148.455, 148.455)$$

$$= 215.7 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	225.00	N./mm ²
Local Primary Membrane Stress	PL	215.73	N./mm ²
Maximum Allowable Working Pressure	Pmax	192.78	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (28.4 + 57.0)/28.4$$

$$= 3.007 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn)$$

$$= \pi/2 \cdot (28.4 + 57.0)$$

$$= 134.146 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 57.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$



$$= \min(81254 \cdot 3.01, 1.5 \cdot 142.0(130.664 + 0.0), \pi/4 \cdot 190.8 \cdot 28.4^2 \cdot 3.01^2)$$

$$= 11144.873 \text{ Kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 11144.873 / (134.146(0.49 \cdot 7.071 + 0.6 \cdot 189.0 + 0.49 \cdot 0.0))$$

$$= 6.972 < \text{or} = \text{to } 149.999 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 57.0 / \ln[1 + 57.0/28.4]$$

$$= 51.773 \text{ mm.}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 189.0 / \ln[1 + 189.0/1378.0]$$

$$= 1470.476 \text{ mm.}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 150.0$$

$$= 225.0 \text{ N./mm}^2$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.034 \cdot 51.773 \cdot 229.234$$

$$= 125.2 \text{ Kgf}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs}(L_R + t_n)$$

$$= 1.034 \cdot 1470.476(56.8 + 57.0)$$

$$= 1764.8 \text{ Kgf}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.034 \cdot 1470.476 \cdot 28.4$$

$$= 440.4 \text{ Kgf}$$

Area Resisting External Pressure [A_p]:

$$= (f_S + f_Y + f_N) / P$$

$$= (1764.8 + 440.4 + 125.2) / 1.034$$

$$= 2209.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$

$$= 224.998 / (2 \cdot 2209.699 / 231.547 - 1470.476 / 189.0)$$

$$= 199.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$= S[t / R_{xs}]$$

$$= 150.0[189.0 / 1470.476]$$

$$= 192.8 \text{ bars}$$

Maximum Allowable Working Pressure [P_{max}]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(198.994, 192.782)$$

$$= 192.782 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / AT$$

$$= (125.164 + 1764.807 + 440.426) / 231.547$$

$$= 0.987 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 1.034 \cdot 1470.476 / 189.0$$

$$= 0.8 \text{ N./mm}^2$$



Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 0.987 - 0.805, 0.805)$$

$$= 1.2 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	225.00	N./mm ²
Local Primary Membrane Stress	PL	1.17	N./mm ²
Maximum Allowable External Pressure	Pmax	192.78	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-36 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A
From Figure 3.8M- with PWHT, the MDMT is	:	1 °C
Temperature reduction per Graph 3.13	:	36 °C
The computed stress ratio is	:	0.497

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (3.482 \cdot 1.0) / (10.0 - 3.0 - 0.0)$$

$$= 0.497$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A

As the Stress Ratio < 0.240, the MDMT is	:	-104 °C
The computed stress ratio is	:	0.061

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (3.482 \cdot 1.0) / (60.0 - 3.0 - 0.0)$$

$$= 0.061$$

Gov. MDMT of the nozzle to shell joint welded assembly : -36 °C

The Drop for this Nozzle is : 2.6546 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 394.6546 mm.



Input, Nozzle Desc: N5B2"

From: 40

Pressure for Reinforcement Calculations	P	190.856	bars
Temperature for Internal Pressure	Temp	200	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	150.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2750.00	mm.
Design Length of Section	L	6758.0000	mm.
Shell Finished (Minimum) Thickness	t	192.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		649.97	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

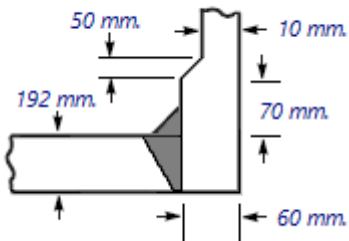
Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	142.00	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm.
Height of Beveled Transition	L'	50.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		60.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	192.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N5B2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.394	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 56.8 \cdot (\exp(190.856/(142.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 4.0858 + 3.0 + 0.0 = 7.0858 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 142.0 \cdot 1.0 \cdot \ln((2 \cdot 7.0 + 56.8) / 56.8) - 0.0$$

$$= 312.837 - 0.0 = 312.8375 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	7.520 mm
Min. required shell thickness, internal p + c	190.259 mm.
Min. required shell thickness, external p + c	19.199 mm.

Nozzle neck minimum required thickness	: 7.520 mm.
--	-------------

Minimum nozzle neck thickness 10.008 mm., is greater than required thickness 7.520 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[142.0/150.0, 1]$$

$$= 0.947$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 60.0 - 3.0$$

$$= 57.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 10.0 - 3.0$$

$$= 7.000 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2756.0/189.0$$

$$= 14.582 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2750.0/2 + 3.0$$

$$= 1378.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{R_{eff} \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1378.0 \cdot 189.0}, 2 \cdot 28.4)$$

$$= 56.800 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$



$$= \min[1.5 \cdot 189.0, 0.0] + \sqrt{(28.4 \cdot 57.0)}$$

$$= 40.234 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 200.0$$

$$= 200.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(189.0 + 0.0)$$

$$= 1512.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[40.234, 200.0, 1512.0] + 189.0$$

$$= 229.234 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 189.0 + ((0.0 \cdot 1.0) / 56.8)$$

$$= 189.000 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 28.4 + 57.0) / (\sqrt{(2756.0 + 189.0) \cdot 189.0}))$$

$$= 0.153$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 189.0 \cdot 56.8 \cdot \max((0.153/5)^{0.85}, 1)$$

$$= 107.352 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 57.0 \cdot 229.234$$

$$= 130.664 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 107.352 + 0.947(130.664 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 231.547 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 57.0 / \ln[1 + 57.0/28.4]$$

$$= 51.773 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 189.0 / \ln[1 + 189.0/1378.0]$$

$$= 1470.476 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 150.0 \cdot 1.0$$

$$= 225.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 190.856 \cdot 51.773 \cdot 229.234$$

$$= 23098.7 \text{ Kg}$$



Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + tn)$$

$$= 190.856 \cdot 1470.476(56.8 + 57.0)$$

$$= 325689.4 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 190.856 \cdot 1470.476 \cdot 28.4$$

$$= 81279.3 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN)/P$$

$$= (325689.4 + 81279.3 + 23098.7)/190.856$$

$$= 2209.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$

$$= 224.998 / (2 \cdot 2209.699/231.547 - 1470.476/189.0)$$

$$= 199.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 150.0[189.0/1470.476]$$

$$= 192.8 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(198.994, 192.782)$$

$$= 192.782 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (23098.699 + 325689.438 + 81279.258)/231.547$$

$$= 182.149 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 190.856 \cdot 1470.476/189.0$$

$$= 148.5 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 182.149 - 148.501, 148.501)$$

$$= 215.8 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	225.00	N./mm ²
Local Primary Membrane Stress	PL	215.80	N./mm ²
Maximum Allowable Working Pressure	Pmax	192.78	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (28.4 + 57.0)/28.4$$

$$= 3.007 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn)$$

$$= \pi/2 \cdot (28.4 + 57.0)$$

$$= 134.146 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 57.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$



$$= \min(81279 \cdot 3.01, 1.5 \cdot 142.0(130.664 + 0.0), \pi/4 \cdot 190.9 \cdot 28.4^2 \cdot 3.01^2)$$

$$= 11148.310 \text{ Kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 11148.31 / (134.146(0.49 \cdot 7.071 + 0.6 \cdot 189.0 + 0.49 \cdot 0.0))$$

$$= 6.974 < \text{or} = \text{to } 149.999 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 57.0 / \ln[1 + 57.0/28.4]$$

$$= 51.773 \text{ mm.}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 189.0 / \ln[1 + 189.0/1378.0]$$

$$= 1470.476 \text{ mm.}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 150.0$$

$$= 225.0 \text{ N./mm}^2$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.034 \cdot 51.773 \cdot 229.234$$

$$= 125.2 \text{ Kgf}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs}(L_R + t_n)$$

$$= 1.034 \cdot 1470.476(56.8 + 57.0)$$

$$= 1764.8 \text{ Kgf}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.034 \cdot 1470.476 \cdot 28.4$$

$$= 440.4 \text{ Kgf}$$

Area Resisting External Pressure [A_p]:

$$= (f_S + f_Y + f_N) / P$$

$$= (1764.8 + 440.4 + 125.2) / 1.034$$

$$= 2209.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$

$$= 224.998 / (2 \cdot 2209.699 / 231.547 - 1470.476 / 189.0)$$

$$= 199.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$= S[t / R_{xs}]$$

$$= 150.0[189.0 / 1470.476]$$

$$= 192.8 \text{ bars}$$

Maximum Allowable Working Pressure [P_{max}]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(198.994, 192.782)$$

$$= 192.782 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / AT$$

$$= (125.164 + 1764.807 + 440.426) / 231.547$$

$$= 0.987 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 1.034 \cdot 1470.476 / 189.0$$

$$= 0.8 \text{ N./mm}^2$$



Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 0.987 - 0.805, 0.805)$$

$$= 1.2 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	225.00	N./mm ²
Local Primary Membrane Stress	PL	1.17	N./mm ²
Maximum Allowable External Pressure	Pmax	192.78	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-36 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A
From Figure 3.8M- with PWHT, the MDMT is	:	1 °C
Temperature reduction per Graph 3.13	:	36 °C
The computed stress ratio is	:	0.498

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (3.483 \cdot 1.0) / (10.0 - 3.0 - 0.0)$$

$$= 0.498$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A

As the Stress Ratio < 0.240, the MDMT is	:	-104 °C
The computed stress ratio is	:	0.061

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (3.483 \cdot 1.0) / (60.0 - 3.0 - 0.0)$$

$$= 0.061$$

Gov. MDMT of the nozzle to shell joint welded assembly : -36 °C

The Drop for this Nozzle is : 2.6546 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 394.6546 mm.



Input, Nozzle Desc: N7A2"

From: 40

Pressure for Reinforcement Calculations	P	190.705	bars
Temperature for Internal Pressure	Temp	200	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	150.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2750.00	mm.
Design Length of Section	L	6758.0000	mm.
Shell Finished (Minimum) Thickness	t	192.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2199.97	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

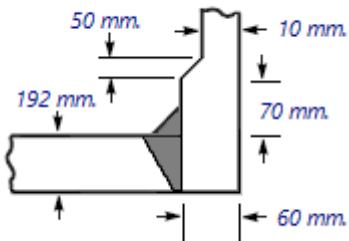
Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	142.00	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm.
Height of Beveled Transition	L'	50.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		60.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	192.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N7A2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.394	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 56.8 \cdot (\exp(190.705/(142.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 4.0824 + 3.0 + 0.0 = 7.0824 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 142.0 \cdot 1.0 \cdot \ln((2 \cdot 7.0 + 56.8) / 56.8) - 0.0$$

$$= 312.837 - 0.0 = 312.8375 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 7.520 mm
 Min. required shell thickness, internal p + c : 190.100 mm.
 Min. required shell thickness, external p + c : 19.199 mm.

Nozzle neck minimum required thickness : 7.520 mm.

Minimum nozzle neck thickness 10.008 mm., is greater than required thickness 7.520 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[142.0/150.0, 1]$$

$$= 0.947$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 60.0 - 3.0$$

$$= 57.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 10.0 - 3.0$$

$$= 7.000 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2756.0/189.0$$

$$= 14.582 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2750.0/2 + 3.0$$

$$= 1378.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1378.0 \cdot 189.0}, 2 \cdot 28.4)$$

$$= 56.800 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$



$$= \min[1.5 \cdot 189.0, 0.0] + \sqrt{(28.4 \cdot 57.0)}$$

$$= 40.234 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 200.0$$

$$= 200.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(189.0 + 0.0)$$

$$= 1512.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[40.234, 200.0, 1512.0] + 189.0$$

$$= 229.234 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 189.0 + ((0.0 \cdot 1.0) / 56.8)$$

$$= 189.000 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 28.4 + 57.0) / (\sqrt{(2756.0 + 189.0) \cdot 189.0}))$$

$$= 0.153$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 189.0 \cdot 56.8 \cdot \max((0.153/5)^{0.85}, 1)$$

$$= 107.352 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 57.0 \cdot 229.234$$

$$= 130.664 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 107.352 + 0.947(130.664 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 231.547 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 57.0 / \ln[1 + 57.0/28.4]$$

$$= 51.773 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 189.0 / \ln[1 + 189.0/1378.0]$$

$$= 1470.476 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 150.0 \cdot 1.0$$

$$= 225.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 190.705 \cdot 51.773 \cdot 229.234$$

$$= 23080.3 \text{ Kg}$$



Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + tn)$$

$$= 190.705 \cdot 1470.476(56.8 + 57.0)$$

$$= 325430.2 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 190.705 \cdot 1470.476 \cdot 28.4$$

$$= 81214.6 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN)/P$$

$$= (325430.2 + 81214.6 + 23080.3)/190.705$$

$$= 2209.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$

$$= 224.998 / (2 \cdot 2209.699/231.547 - 1470.476/189.0)$$

$$= 199.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 150.0[189.0/1470.476]$$

$$= 192.8 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(198.994, 192.782)$$

$$= 192.782 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (23080.311 + 325430.188 + 81214.555)/231.547$$

$$= 182.004 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 190.705 \cdot 1470.476/189.0$$

$$= 148.4 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 182.004 - 148.382, 148.382)$$

$$= 215.6 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	225.00	N./mm ²
Local Primary Membrane Stress	PL	215.63	N./mm ²
Maximum Allowable Working Pressure	Pmax	192.78	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (28.4 + 57.0)/28.4$$

$$= 3.007 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn)$$

$$= \pi/2 \cdot (28.4 + 57.0)$$

$$= 134.146 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 57.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$



$$= \min(81215 \cdot 3.01, 1.5 \cdot 142.0(130.664 + 0.0), \pi/4 \cdot 190.7 \cdot 28.4^2 \cdot 3.01^2)$$

$$= 11139.435 \text{ Kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 11139.435 / (134.146(0.49 \cdot 7.071 + 0.6 \cdot 189.0 + 0.49 \cdot 0.0))$$

$$= 6.968 < \text{or} = \text{to } 149.999 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 57.0 / \ln[1 + 57.0/28.4]$$

$$= 51.773 \text{ mm.}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 189.0 / \ln[1 + 189.0/1378.0]$$

$$= 1470.476 \text{ mm.}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 150.0$$

$$= 225.0 \text{ N./mm}^2$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.034 \cdot 51.773 \cdot 229.234$$

$$= 125.2 \text{ Kgf}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs}(L_R + t_n)$$

$$= 1.034 \cdot 1470.476(56.8 + 57.0)$$

$$= 1764.8 \text{ Kgf}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.034 \cdot 1470.476 \cdot 28.4$$

$$= 440.4 \text{ Kgf}$$

Area Resisting External Pressure [A_p]:

$$= (f_S + f_Y + f_N) / P$$

$$= (1764.8 + 440.4 + 125.2) / 1.034$$

$$= 2209.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$

$$= 224.998 / (2 \cdot 2209.699 / 231.547 - 1470.476 / 189.0)$$

$$= 199.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$= S[t / R_{xs}]$$

$$= 150.0[189.0 / 1470.476]$$

$$= 192.8 \text{ bars}$$

Maximum Allowable Working Pressure [P_{max}]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(198.994, 192.782)$$

$$= 192.782 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / AT$$

$$= (125.164 + 1764.807 + 440.426) / 231.547$$

$$= 0.987 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 1.034 \cdot 1470.476 / 189.0$$

$$= 0.8 \text{ N./mm}^2$$



Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 0.987 - 0.805, 0.805)$$

$$= 1.2 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	225.00	N./mm ²
Local Primary Membrane Stress	PL	1.17	N./mm ²
Maximum Allowable External Pressure	Pmax	192.78	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-36 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A
From Figure 3.8M- with PWHT, the MDMT is	:	1 °C
Temperature reduction per Graph 3.13	:	36 °C
The computed stress ratio is	:	0.497

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (3.48 \cdot 1.0) / (10.0 - 3.0 - 0.0)$$

$$= 0.497$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A

As the Stress Ratio < 0.240, the MDMT is	:	-104 °C
The computed stress ratio is	:	0.061

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (3.48 \cdot 1.0) / (60.0 - 3.0 - 0.0)$$

$$= 0.061$$

Gov. MDMT of the nozzle to shell joint welded assembly : -36 °C

The Drop for this Nozzle is : 2.6546 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 394.6546 mm.



Input, Nozzle Desc: N8A2"

From: 40

Pressure for Reinforcement Calculations	P	190.724	bars
Temperature for Internal Pressure	Temp	200	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	150.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2750.00	mm.
Design Length of Section	L	6758.0000	mm.
Shell Finished (Minimum) Thickness	t	192.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1999.97	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

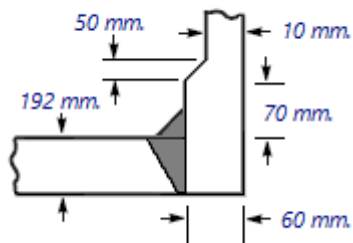
Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	142.00	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm.
Height of Beveled Transition	L'	50.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		60.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	192.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N8A2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.394	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 56.8 \cdot (\exp(190.724/(142.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 4.0828 + 3.0 + 0.0 = 7.0828 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 142.0 \cdot 1.0 \cdot \ln((2 \cdot 7.0 + 56.8) / 56.8) - 0.0$$

$$= 312.837 - 0.0 = 312.8375 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	7.520 mm
Min. required shell thickness, internal p + c :	190.121 mm.
Min. required shell thickness, external p + c :	19.199 mm.

Nozzle neck minimum required thickness	: 7.520 mm.
--	-------------

Minimum nozzle neck thickness 10.008 mm., is greater than required thickness 7.520 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[142.0/150.0, 1]$$

$$= 0.947$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 60.0 - 3.0$$

$$= 57.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 10.0 - 3.0$$

$$= 7.000 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2756.0/189.0$$

$$= 14.582 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2750.0/2 + 3.0$$

$$= 1378.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{R_{eff} \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1378.0 \cdot 189.0}, 2 \cdot 28.4)$$

$$= 56.800 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 189.0, 0.0] + \sqrt{(28.4 \cdot 57.0)}$$

$$= 40.234 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 200.0$$

$$= 200.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(189.0 + 0.0)$$

$$= 1512.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[40.234, 200.0, 1512.0] + 189.0$$

$$= 229.234 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 189.0 + ((0.0 \cdot 1.0) / 56.8)$$

$$= 189.000 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 28.4 + 57.0) / (\sqrt{(2756.0 + 189.0) \cdot 189.0}))$$

$$= 0.153$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 189.0 \cdot 56.8 \cdot \max((0.153/5)^{0.85}, 1)$$

$$= 107.352 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 57.0 \cdot 229.234$$

$$= 130.664 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 107.352 + 0.947(130.664 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 231.547 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 57.0 / \ln[1 + 57.0/28.4]$$

$$= 51.773 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 189.0 / \ln[1 + 189.0/1378.0]$$

$$= 1470.476 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 150.0 \cdot 1.0$$

$$= 225.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 190.724 \cdot 51.773 \cdot 229.234$$

$$= 23082.7 \text{ Kg}$$



Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + tn)$$

$$= 190.724 \cdot 1470.476(56.8 + 57.0)$$

$$= 325463.7 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 190.724 \cdot 1470.476 \cdot 28.4$$

$$= 81222.9 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN)/P$$

$$= (325463.7 + 81222.9 + 23082.7)/190.724$$

$$= 2209.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$

$$= 224.998 / (2 \cdot 2209.699/231.547 - 1470.476/189.0)$$

$$= 199.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 150.0[189.0/1470.476]$$

$$= 192.8 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(198.994, 192.782)$$

$$= 192.782 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (23082.686 + 325463.656 + 81222.906)/231.547$$

$$= 182.023 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 190.724 \cdot 1470.476/189.0$$

$$= 148.4 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 182.023 - 148.398, 148.398)$$

$$= 215.6 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	225.00	N./mm ²
Local Primary Membrane Stress	PL	215.65	N./mm ²
Maximum Allowable Working Pressure	Pmax	192.78	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (28.4 + 57.0)/28.4$$

$$= 3.007 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn)$$

$$= \pi/2 \cdot (28.4 + 57.0)$$

$$= 134.146 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 57.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$



$$= \min(81223 \cdot 3.01, 1.5 \cdot 142.0(130.664 + 0.0), \pi/4 \cdot 190.7 \cdot 28.4^2 \cdot 3.01^2)$$
$$= 11140.580 \text{ Kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$
$$= 11140.58 / (134.146(0.49 \cdot 7.071 + 0.6 \cdot 189.0 + 0.49 \cdot 0.0))$$
$$= 6.969 < \text{or} = \text{to } 149.999 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 57.0 / \ln[1 + 57.0/28.4]$$
$$= 51.773 \text{ mm.}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 189.0 / \ln[1 + 189.0/1378.0]$$
$$= 1470.476 \text{ mm.}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$= 1.5 \cdot S$$
$$= 1.5 \cdot 150.0$$
$$= 225.0 \text{ N./mm}^2$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H$$
$$= 1.034 \cdot 51.773 \cdot 229.234$$
$$= 125.2 \text{ Kgf}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs}(L_R + t_n)$$
$$= 1.034 \cdot 1470.476(56.8 + 57.0)$$
$$= 1764.8 \text{ Kgf}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc}$$
$$= 1.034 \cdot 1470.476 \cdot 28.4$$
$$= 440.4 \text{ Kgf}$$

Area Resisting External Pressure [A_p]:

$$= (f_S + f_Y + f_N) / P$$
$$= (1764.8 + 440.4 + 125.2) / 1.034$$
$$= 2209.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$
$$= 224.998 / (2 \cdot 2209.699 / 231.547 - 1470.476 / 189.0)$$
$$= 199.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$= S[t / R_{xs}]$$
$$= 150.0[189.0 / 1470.476]$$
$$= 192.8 \text{ bars}$$

Maximum Allowable Working Pressure [P_{max}]:

$$= \min(P_{max1}, P_{max2})$$
$$= \min(198.994, 192.782)$$
$$= 192.782 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / AT$$
$$= (125.164 + 1764.807 + 440.426) / 231.547$$
$$= 0.987 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$
$$= 1.034 \cdot 1470.476 / 189.0$$
$$= 0.8 \text{ N./mm}^2$$



Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}}) \\ &= \max(2 \cdot 0.987 - 0.805, 0.805) \\ &= 1.2 \text{ N./mm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	225.00	N./mm ²
Local Primary Membrane Stress	PL	1.17	N./mm ²
Maximum Allowable External Pressure	Pmax	192.78	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-36 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A
From Figure 3.8M- with PWHT, the MDMT is	:	1 °C
Temperature reduction per Graph 3.13	:	36 °C
The computed stress ratio is	:	0.497

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (3.48 \cdot 1.0) / (10.0 - 3.0 - 0.0) \\ &= 0.497 \end{aligned}$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A

As the Stress Ratio < 0.240, the MDMT is	:	-104 °C
The computed stress ratio is	:	0.061

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (3.48 \cdot 1.0) / (60.0 - 3.0 - 0.0) \\ &= 0.061 \end{aligned}$$

Gov. MDMT of the nozzle to shell joint welded assembly : -36 °C

The Drop for this Nozzle is : 2.6546 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 394.6546 mm.

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Input, Nozzle Desc: N9A2"

From: 40

Pressure for Reinforcement Calculations	P	190.798	bars
Temperature for Internal Pressure	Temp	200	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	150.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2750.00	mm.
Design Length of Section	L	6758.0000	mm.
Shell Finished (Minimum) Thickness	t	192.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1249.97	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

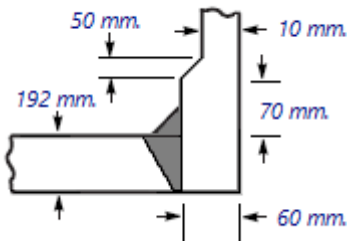
Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	142.00	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm.
Height of Beveled Transition	L'	50.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		60.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	192.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N9A2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.394	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 56.8 \cdot (\exp(190.798/(142.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 4.0845 + 3.0 + 0.0 = 7.0845 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 142.0 \cdot 1.0 \cdot \ln((2 \cdot 7.0 + 56.8) / 56.8) - 0.0$$

$$= 312.837 - 0.0 = 312.8375 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	7.520 mm
Min. required shell thickness, internal p + c :	190.198 mm.
Min. required shell thickness, external p + c :	19.199 mm.

Nozzle neck minimum required thickness	: 7.520 mm.
--	-------------

Minimum nozzle neck thickness 10.008 mm., is greater than required thickness 7.520 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[142.0/150.0, 1]$$

$$= 0.947$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 60.0 - 3.0$$

$$= 57.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 10.0 - 3.0$$

$$= 7.000 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2756.0/189.0$$

$$= 14.582 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2750.0/2 + 3.0$$

$$= 1378.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1378.0 \cdot 189.0}, 2 \cdot 28.4)$$

$$= 56.800 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 189.0, 0.0] + \sqrt{(28.4 \cdot 57.0)}$$

$$= 40.234 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 200.0$$

$$= 200.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(189.0 + 0.0)$$

$$= 1512.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[40.234, 200.0, 1512.0] + 189.0$$

$$= 229.234 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 189.0 + ((0.0 \cdot 1.0) / 56.8)$$

$$= 189.000 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 28.4 + 57.0) / (\sqrt{(2756.0 + 189.0) \cdot 189.0}))$$

$$= 0.153$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 189.0 \cdot 56.8 \cdot \max((0.153/5)^{0.85}, 1)$$

$$= 107.352 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 57.0 \cdot 229.234$$

$$= 130.664 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 107.352 + 0.947(130.664 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 231.547 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 57.0 / \ln[1 + 57.0/28.4]$$

$$= 51.773 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 189.0 / \ln[1 + 189.0/1378.0]$$

$$= 1470.476 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 150.0 \cdot 1.0$$

$$= 225.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 190.798 \cdot 51.773 \cdot 229.234$$

$$= 23091.6 \text{ Kg}$$



Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + tn)$$

$$= 190.798 \cdot 1470.476(56.8 + 57.0)$$

$$= 325589.1 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 190.798 \cdot 1470.476 \cdot 28.4$$

$$= 81254.2 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN)/P$$

$$= (325589.1 + 81254.2 + 23091.6)/190.798$$

$$= 2209.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$

$$= 224.998 / (2 \cdot 2209.699/231.547 - 1470.476/189.0)$$

$$= 199.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 150.0[189.0/1470.476]$$

$$= 192.8 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(198.994, 192.782)$$

$$= 192.782 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (23091.58 + 325589.094 + 81254.211)/231.547$$

$$= 182.093 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 190.798 \cdot 1470.476/189.0$$

$$= 148.5 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 182.093 - 148.455, 148.455)$$

$$= 215.7 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	225.00	N./mm ²
Local Primary Membrane Stress	PL	215.73	N./mm ²
Maximum Allowable Working Pressure	Pmax	192.78	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (28.4 + 57.0)/28.4$$

$$= 3.007 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn)$$

$$= \pi/2 \cdot (28.4 + 57.0)$$

$$= 134.146 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 57.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$



$$= \min(81254 \cdot 3.01, 1.5 \cdot 142.0(130.664 + 0.0), \pi/4 \cdot 190.8 \cdot 28.4^2 \cdot 3.01^2)$$
$$= 11144.873 \text{ Kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$
$$= 11144.873 / (134.146(0.49 \cdot 7.071 + 0.6 \cdot 189.0 + 0.49 \cdot 0.0))$$
$$= 6.972 < \text{or} = \text{to } 149.999 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 57.0 / \ln[1 + 57.0/28.4]$$
$$= 51.773 \text{ mm.}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 189.0 / \ln[1 + 189.0/1378.0]$$
$$= 1470.476 \text{ mm.}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$= 1.5 \cdot S$$
$$= 1.5 \cdot 150.0$$
$$= 225.0 \text{ N./mm}^2$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H$$
$$= 1.034 \cdot 51.773 \cdot 229.234$$
$$= 125.2 \text{ Kgf}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs}(L_R + t_n)$$
$$= 1.034 \cdot 1470.476(56.8 + 57.0)$$
$$= 1764.8 \text{ Kgf}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc}$$
$$= 1.034 \cdot 1470.476 \cdot 28.4$$
$$= 440.4 \text{ Kgf}$$

Area Resisting External Pressure [A_p]:

$$= (f_S + f_Y + f_N) / P$$
$$= (1764.8 + 440.4 + 125.2) / 1.034$$
$$= 2209.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$
$$= 224.998 / (2 \cdot 2209.699 / 231.547 - 1470.476 / 189.0)$$
$$= 199.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$= S / [t / R_{xs}]$$
$$= 150.0 [189.0 / 1470.476]$$
$$= 192.8 \text{ bars}$$

Maximum Allowable Working Pressure [P_{max}]:

$$= \min(P_{max1}, P_{max2})$$
$$= \min(198.994, 192.782)$$
$$= 192.782 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / AT$$
$$= (125.164 + 1764.807 + 440.426) / 231.547$$
$$= 0.987 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$
$$= 1.034 \cdot 1470.476 / 189.0$$
$$= 0.8 \text{ N./mm}^2$$



Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 0.987 - 0.805, 0.805)$$

$$= 1.2 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	225.00	N./mm ²
Local Primary Membrane Stress	PL	1.17	N./mm ²
Maximum Allowable External Pressure	Pmax	192.78	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-36 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A
From Figure 3.8M- with PWHT, the MDMT is	:	1 °C
Temperature reduction per Graph 3.13	:	36 °C
The computed stress ratio is	:	0.497

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (3.482 \cdot 1.0) / (10.0 - 3.0 - 0.0)$$

$$= 0.497$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A

As the Stress Ratio < 0.240, the MDMT is	:	-104 °C
The computed stress ratio is	:	0.061

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (3.482 \cdot 1.0) / (60.0 - 3.0 - 0.0)$$

$$= 0.061$$

Gov. MDMT of the nozzle to shell joint welded assembly : -36 °C

The Drop for this Nozzle is : 2.6546 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 394.6546 mm.

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Input, Nozzle Desc: N7B2"

From: 40

Pressure for Reinforcement Calculations	P	190.856	bars
Temperature for Internal Pressure	Temp	200	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	150.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2750.00	mm.
Design Length of Section	L	6758.0000	mm.
Shell Finished (Minimum) Thickness	t	192.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		649.97	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

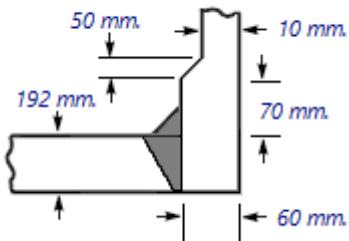
Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	142.00	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm.
Height of Beveled Transition	L'	50.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		60.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	192.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N7B2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.394	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 56.8 \cdot (\exp(190.856/(142.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 4.0858 + 3.0 + 0.0 = 7.0858 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 142.0 \cdot 1.0 \cdot \ln((2 \cdot 7.0 + 56.8) / 56.8) - 0.0$$

$$= 312.837 - 0.0 = 312.8375 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	7.520 mm
Min. required shell thickness, internal p + c :	190.259 mm.
Min. required shell thickness, external p + c :	19.199 mm.

Nozzle neck minimum required thickness	: 7.520 mm.
--	-------------

Minimum nozzle neck thickness 10.008 mm., is greater than required thickness 7.520 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[142.0/150.0, 1]$$

$$= 0.947$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 60.0 - 3.0$$

$$= 57.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 10.0 - 3.0$$

$$= 7.000 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2756.0/189.0$$

$$= 14.582 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2750.0/2 + 3.0$$

$$= 1378.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1378.0 \cdot 189.0}, 2 \cdot 28.4)$$

$$= 56.800 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$



$$= \min[1.5 \cdot 189.0, 0.0] + \sqrt{(28.4 \cdot 57.0)}$$

$$= 40.234 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 200.0$$

$$= 200.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(189.0 + 0.0)$$

$$= 1512.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[40.234, 200.0, 1512.0] + 189.0$$

$$= 229.234 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 189.0 + ((0.0 \cdot 1.0) / 56.8)$$

$$= 189.000 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 28.4 + 57.0) / (\sqrt{(2756.0 + 189.0) \cdot 189.0}))$$

$$= 0.153$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 189.0 \cdot 56.8 \cdot \max((0.153/5)^{0.85}, 1)$$

$$= 107.352 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 57.0 \cdot 229.234$$

$$= 130.664 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 107.352 + 0.947(130.664 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 231.547 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 57.0 / \ln[1 + 57.0/28.4]$$

$$= 51.773 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 189.0 / \ln[1 + 189.0/1378.0]$$

$$= 1470.476 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 150.0 \cdot 1.0$$

$$= 225.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 190.856 \cdot 51.773 \cdot 229.234$$

$$= 23098.7 \text{ Kg}$$



Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + tn)$$

$$= 190.856 \cdot 1470.476(56.8 + 57.0)$$

$$= 325689.4 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 190.856 \cdot 1470.476 \cdot 28.4$$

$$= 81279.3 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN)/P$$

$$= (325689.4 + 81279.3 + 23098.7)/190.856$$

$$= 2209.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$

$$= 224.998 / (2 \cdot 2209.699/231.547 - 1470.476/189.0)$$

$$= 199.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 150.0[189.0/1470.476]$$

$$= 192.8 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(198.994, 192.782)$$

$$= 192.782 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (23098.699 + 325689.438 + 81279.258)/231.547$$

$$= 182.149 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 190.856 \cdot 1470.476/189.0$$

$$= 148.5 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 182.149 - 148.501, 148.501)$$

$$= 215.8 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	225.00	N./mm ²
Local Primary Membrane Stress	PL	215.80	N./mm ²
Maximum Allowable Working Pressure	Pmax	192.78	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (28.4 + 57.0)/28.4$$

$$= 3.007 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn)$$

$$= \pi/2 \cdot (28.4 + 57.0)$$

$$= 134.146 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 57.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$



$$= \min(81279 \cdot 3.01, 1.5 \cdot 142.0(130.664 + 0.0), \pi/4 \cdot 190.9 \cdot 28.4^2 \cdot 3.01^2)$$

$$= 11148.310 \text{ Kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 11148.31 / (134.146(0.49 \cdot 7.071 + 0.6 \cdot 189.0 + 0.49 \cdot 0.0))$$

$$= 6.974 < \text{or} = \text{to } 149.999 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 57.0 / \ln[1 + 57.0/28.4]$$

$$= 51.773 \text{ mm.}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 189.0 / \ln[1 + 189.0/1378.0]$$

$$= 1470.476 \text{ mm.}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 150.0$$

$$= 225.0 \text{ N./mm}^2$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.034 \cdot 51.773 \cdot 229.234$$

$$= 125.2 \text{ Kgf}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs}(L_R + t_n)$$

$$= 1.034 \cdot 1470.476(56.8 + 57.0)$$

$$= 1764.8 \text{ Kgf}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.034 \cdot 1470.476 \cdot 28.4$$

$$= 440.4 \text{ Kgf}$$

Area Resisting External Pressure [A_p]:

$$= (f_S + f_Y + f_N) / P$$

$$= (1764.8 + 440.4 + 125.2) / 1.034$$

$$= 2209.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$

$$= 224.998 / (2 \cdot 2209.699 / 231.547 - 1470.476 / 189.0)$$

$$= 199.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$= S[t / R_{xs}]$$

$$= 150.0[189.0 / 1470.476]$$

$$= 192.8 \text{ bars}$$

Maximum Allowable Working Pressure [P_{max}]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(198.994, 192.782)$$

$$= 192.782 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / AT$$

$$= (125.164 + 1764.807 + 440.426) / 231.547$$

$$= 0.987 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 1.034 \cdot 1470.476 / 189.0$$

$$= 0.8 \text{ N./mm}^2$$



Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 0.987 - 0.805, 0.805)$$

$$= 1.2 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	225.00	N./mm ²
Local Primary Membrane Stress	PL	1.17	N./mm ²
Maximum Allowable External Pressure	Pmax	192.78	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-36 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A
From Figure 3.8M- with PWHT, the MDMT is	:	1 °C
Temperature reduction per Graph 3.13	:	36 °C
The computed stress ratio is	:	0.498

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (3.483 \cdot 1.0) / (10.0 - 3.0 - 0.0)$$

$$= 0.498$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A

As the Stress Ratio < 0.240, the MDMT is	:	-104 °C
The computed stress ratio is	:	0.061

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (3.483 \cdot 1.0) / (60.0 - 3.0 - 0.0)$$

$$= 0.061$$

Gov. MDMT of the nozzle to shell joint welded assembly : -36 °C

The Drop for this Nozzle is : 2.6546 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 394.6546 mm.



Input, Nozzle Desc: N8B2"

From: 40

Pressure for Reinforcement Calculations	P	190.876	bars
Temperature for Internal Pressure	Temp	200	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	150.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2750.00	mm.
Design Length of Section	L	6758.0000	mm.
Shell Finished (Minimum) Thickness	t	192.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		449.98	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

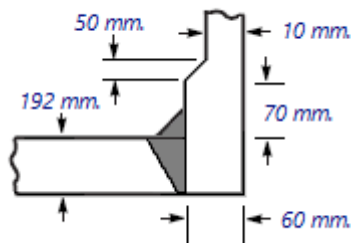
Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	142.00	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm.
Height of Beveled Transition	L'	50.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		60.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	192.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N8B2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.394	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 56.8 \cdot (\exp(190.876/(142.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 4.0863 + 3.0 + 0.0 = 7.0863 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 142.0 \cdot 1.0 \cdot \ln((2 \cdot 7.0 + 56.8) / 56.8) - 0.0$$

$$= 312.837 - 0.0 = 312.8375 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	7.520 mm
Min. required shell thickness, internal p + c	190.280 mm.
Min. required shell thickness, external p + c	19.199 mm.

Nozzle neck minimum required thickness	: 7.520 mm.
--	-------------

Minimum nozzle neck thickness 10.008 mm., is greater than required thickness 7.520 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[142.0/150.0, 1]$$

$$= 0.947$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 60.0 - 3.0$$

$$= 57.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 10.0 - 3.0$$

$$= 7.000 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2756.0/189.0$$

$$= 14.582 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2750.0/2 + 3.0$$

$$= 1378.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{R_{eff} \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1378.0 \cdot 189.0}, 2 \cdot 28.4)$$

$$= 56.800 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$



$$= \min[1.5 \cdot 189.0, 0.0] + \sqrt{(28.4 \cdot 57.0)}$$

$$= 40.234 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 200.0$$

$$= 200.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(189.0 + 0.0)$$

$$= 1512.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[40.234, 200.0, 1512.0] + 189.0$$

$$= 229.234 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 189.0 + ((0.0 \cdot 1.0) / 56.8)$$

$$= 189.000 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 28.4 + 57.0) / (\sqrt{(2756.0 + 189.0) \cdot 189.0}))$$

$$= 0.153$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 189.0 \cdot 56.8 \cdot \max((0.153/5)^{0.85}, 1)$$

$$= 107.352 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 57.0 \cdot 229.234$$

$$= 130.664 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 107.352 + 0.947(130.664 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 231.547 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 57.0 / \ln[1 + 57.0/28.4]$$

$$= 51.773 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 189.0 / \ln[1 + 189.0/1378.0]$$

$$= 1470.476 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 150.0 \cdot 1.0$$

$$= 225.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 190.876 \cdot 51.773 \cdot 229.234$$

$$= 23101.1 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + tn)$$

$$= 190.876 \cdot 1470.476(56.8 + 57.0)$$

$$= 325722.9 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 190.876 \cdot 1470.476 \cdot 28.4$$

$$= 81287.6 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN)/P$$

$$= (325722.9 + 81287.6 + 23101.1)/190.876$$

$$= 2209.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$

$$= 224.998 / (2 \cdot 2209.699/231.547 - 1470.476/189.0)$$

$$= 199.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 150.0[189.0/1470.476]$$

$$= 192.8 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(198.994, 192.782)$$

$$= 192.782 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (23101.072 + 325722.938 + 81287.609)/231.547$$

$$= 182.168 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 190.876 \cdot 1470.476/189.0$$

$$= 148.5 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 182.168 - 148.516, 148.516)$$

$$= 215.8 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	225.00	N./mm ²
Local Primary Membrane Stress	PL	215.82	N./mm ²
Maximum Allowable Working Pressure	Pmax	192.78	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (28.4 + 57.0)/28.4$$

$$= 3.007 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn)$$

$$= \pi/2 \cdot (28.4 + 57.0)$$

$$= 134.146 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 57.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$



$$= \min(81288 \cdot 3.01, 1.5 \cdot 142.0(130.664 + 0.0), \pi/4 \cdot 190.9 \cdot 28.4^2 \cdot 3.01^2)$$

$$= 11149.455 \text{ Kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 11149.455 / (134.146(0.49 \cdot 7.071 + 0.6 \cdot 189.0 + 0.49 \cdot 0.0))$$

$$= 6.975 < \text{or} = \text{to } 149.999 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 57.0 / \ln[1 + 57.0/28.4]$$

$$= 51.773 \text{ mm.}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 189.0 / \ln[1 + 189.0/1378.0]$$

$$= 1470.476 \text{ mm.}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 150.0$$

$$= 225.0 \text{ N./mm}^2$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.034 \cdot 51.773 \cdot 229.234$$

$$= 125.2 \text{ Kgf}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs}(L_R + t_n)$$

$$= 1.034 \cdot 1470.476(56.8 + 57.0)$$

$$= 1764.8 \text{ Kgf}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.034 \cdot 1470.476 \cdot 28.4$$

$$= 440.4 \text{ Kgf}$$

Area Resisting External Pressure [A_p]:

$$= (f_S + f_Y + f_N) / P$$

$$= (1764.8 + 440.4 + 125.2) / 1.034$$

$$= 2209.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$

$$= 224.998 / (2 \cdot 2209.699 / 231.547 - 1470.476 / 189.0)$$

$$= 199.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$= S[t / R_{xs}]$$

$$= 150.0[189.0 / 1470.476]$$

$$= 192.8 \text{ bars}$$

Maximum Allowable Working Pressure [P_{max}]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(198.994, 192.782)$$

$$= 192.782 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / AT$$

$$= (125.164 + 1764.807 + 440.426) / 231.547$$

$$= 0.987 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 1.034 \cdot 1470.476 / 189.0$$

$$= 0.8 \text{ N./mm}^2$$



Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 0.987 - 0.805, 0.805)$$

$$= 1.2 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	225.00	N./mm ²
Local Primary Membrane Stress	PL	1.17	N./mm ²
Maximum Allowable External Pressure	Pmax	192.78	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-36 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A
From Figure 3.8M- with PWHT, the MDMT is	:	1 °C
Temperature reduction per Graph 3.13	:	36 °C
The computed stress ratio is	:	0.498

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (3.483 \cdot 1.0) / (10.0 - 3.0 - 0.0)$$

$$= 0.498$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A

As the Stress Ratio < 0.240, the MDMT is	:	-104 °C
The computed stress ratio is	:	0.061

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (3.483 \cdot 1.0) / (60.0 - 3.0 - 0.0)$$

$$= 0.061$$

Gov. MDMT of the nozzle to shell joint welded assembly : -36 °C

The Drop for this Nozzle is : 2.6546 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 394.6546 mm.



Input, Nozzle Desc: N9B2"

From: 40

Pressure for Reinforcement Calculations	P	190.896	bars
Temperature for Internal Pressure	Temp	200	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	150.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2750.00	mm.
Design Length of Section	L	6758.0000	mm.
Shell Finished (Minimum) Thickness	t	192.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		249.98	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

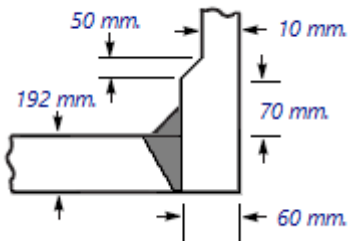
Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	142.00	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm.
Height of Beveled Transition	L'	50.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		60.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	192.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N9B2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.394	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 56.8 \cdot (\exp(190.896/(142.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 4.0867 + 3.0 + 0.0 = 7.0867 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 142.0 \cdot 1.0 \cdot \ln((2 \cdot 7.0 + 56.8) / 56.8) - 0.0$$

$$= 312.837 - 0.0 = 312.8375 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	7.520 mm
Min. required shell thickness, internal p + c	190.300 mm.
Min. required shell thickness, external p + c	19.199 mm.

Nozzle neck minimum required thickness	: 7.520 mm.
--	-------------

Minimum nozzle neck thickness 10.008 mm., is greater than required thickness 7.520 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[142.0/150.0, 1]$$

$$= 0.947$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 60.0 - 3.0$$

$$= 57.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 10.0 - 3.0$$

$$= 7.000 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2756.0/189.0$$

$$= 14.582 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2750.0/2 + 3.0$$

$$= 1378.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1378.0 \cdot 189.0}, 2 \cdot 28.4)$$

$$= 56.800 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 189.0, 0.0] + \sqrt{(28.4 \cdot 57.0)}$$

$$= 40.234 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 200.0$$

$$= 200.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(189.0 + 0.0)$$

$$= 1512.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[40.234, 200.0, 1512.0] + 189.0$$

$$= 229.234 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 189.0 + ((0.0 \cdot 1.0) / 56.8)$$

$$= 189.000 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 28.4 + 57.0) / (\sqrt{(2756.0 + 189.0) \cdot 189.0}))$$

$$= 0.153$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 189.0 \cdot 56.8 \cdot \max((0.153/5)^{0.85}, 1)$$

$$= 107.352 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 57.0 \cdot 229.234$$

$$= 130.664 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 107.352 + 0.947(130.664 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 231.547 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 57.0 / \ln[1 + 57.0/28.4]$$

$$= 51.773 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 189.0 / \ln[1 + 189.0/1378.0]$$

$$= 1470.476 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 150.0 \cdot 1.0$$

$$= 225.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 190.896 \cdot 51.773 \cdot 229.234$$

$$= 23103.4 \text{ Kg}$$



Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + tn)$$

$$= 190.896 \cdot 1470.476(56.8 + 57.0)$$

$$= 325756.3 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 190.896 \cdot 1470.476 \cdot 28.4$$

$$= 81296.0 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN)/P$$

$$= (325756.3 + 81296.0 + 23103.4)/190.896$$

$$= 2209.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$

$$= 224.998 / (2 \cdot 2209.699/231.547 - 1470.476/189.0)$$

$$= 199.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 150.0[189.0/1470.476]$$

$$= 192.8 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(198.994, 192.782)$$

$$= 192.782 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (23103.443 + 325756.344 + 81295.953)/231.547$$

$$= 182.186 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 190.896 \cdot 1470.476/189.0$$

$$= 148.5 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 182.186 - 148.531, 148.531)$$

$$= 215.8 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	225.00	N./mm ²
Local Primary Membrane Stress	PL	215.84	N./mm ²
Maximum Allowable Working Pressure	Pmax	192.78	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (28.4 + 57.0)/28.4$$

$$= 3.007 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn)$$

$$= \pi/2 \cdot (28.4 + 57.0)$$

$$= 134.146 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 57.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$



$$= \min(81296 \cdot 3.01, 1.5 \cdot 142.0(130.664 + 0.0), \pi/4 \cdot 190.9 \cdot 28.4^2 \cdot 3.01^2)$$
$$= 11150.600 \text{ Kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$
$$= 11150.6 / (134.146(0.49 \cdot 7.071 + 0.6 \cdot 189.0 + 0.49 \cdot 0.0))$$
$$= 6.975 < \text{or} = \text{to } 149.999 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 57.0 / \ln[1 + 57.0/28.4]$$
$$= 51.773 \text{ mm.}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 189.0 / \ln[1 + 189.0/1378.0]$$
$$= 1470.476 \text{ mm.}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$= 1.5 \cdot S$$
$$= 1.5 \cdot 150.0$$
$$= 225.0 \text{ N./mm}^2$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H$$
$$= 1.034 \cdot 51.773 \cdot 229.234$$
$$= 125.2 \text{ Kgf}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs}(L_R + t_n)$$
$$= 1.034 \cdot 1470.476(56.8 + 57.0)$$
$$= 1764.8 \text{ Kgf}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc}$$
$$= 1.034 \cdot 1470.476 \cdot 28.4$$
$$= 440.4 \text{ Kgf}$$

Area Resisting External Pressure [A_p]:

$$= (f_S + f_Y + f_N) / P$$
$$= (1764.8 + 440.4 + 125.2) / 1.034$$
$$= 2209.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$
$$= 224.998 / (2 \cdot 2209.699 / 231.547 - 1470.476 / 189.0)$$
$$= 199.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$= S[t / R_{xs}]$$
$$= 150.0[189.0 / 1470.476]$$
$$= 192.8 \text{ bars}$$

Maximum Allowable Working Pressure [P_{max}]:

$$= \min(P_{max1}, P_{max2})$$
$$= \min(198.994, 192.782)$$
$$= 192.782 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / AT$$
$$= (125.164 + 1764.807 + 440.426) / 231.547$$
$$= 0.987 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$
$$= 1.034 \cdot 1470.476 / 189.0$$
$$= 0.8 \text{ N./mm}^2$$



Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 0.987 - 0.805, 0.805)$$

$$= 1.2 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	225.00	N./mm ²
Local Primary Membrane Stress	PL	1.17	N./mm ²
Maximum Allowable External Pressure	Pmax	192.78	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-36 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A
From Figure 3.8M- with PWHT, the MDMT is	:	1 °C
Temperature reduction per Graph 3.13	:	36 °C
The computed stress ratio is	:	0.498

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (3.484 \cdot 1.0) / (10.0 - 3.0 - 0.0)$$

$$= 0.498$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A

As the Stress Ratio < 0.240, the MDMT is	:	-104 °C
The computed stress ratio is	:	0.061

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (3.484 \cdot 1.0) / (60.0 - 3.0 - 0.0)$$

$$= 0.061$$

Gov. MDMT of the nozzle to shell joint welded assembly : -36 °C

The Drop for this Nozzle is : 2.6546 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 394.6546 mm.



Input, Nozzle Desc: N128"

From: 50

Pressure for Reinforcement Calculations	P	190.685	bars
Temperature for Internal Pressure	Temp	200	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	150.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2750.00	mm.
Design Length of Section	L	6758.0000	mm.
Shell Finished (Minimum) Thickness	t	192.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		3224.98	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

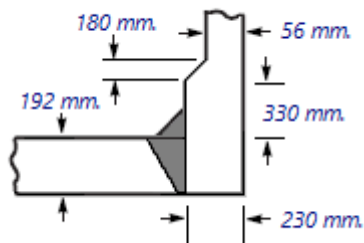
Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	142.00	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		28.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	56.0000	mm.
Hub Height of Integral Nozzle	h	330.0000	mm.
Height of Beveled Transition	L'	180.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		230.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	450.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	192.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N128"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	28.000	in.
Actual Thickness Used in Calculation	2.205	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 717.2 \cdot (\exp(190.685/(142.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 51.5413 + 3.0 + 0.0 = 54.5412 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 142.0 \cdot 1.0 \cdot \ln((2 \cdot 53.0 + 717.2) / 717.2) - 0.0$$

$$= 195.726 - 0.0 = 195.7261 \text{ bars}$$

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 192.18 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 1095.94 mm.

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.340 mm
 Min. required shell thickness, internal p + c : 190.079 mm.
 Min. required shell thickness, external p + c : 19.199 mm.

Nozzle neck minimum required thickness : 11.340 mm.

Minimum nozzle neck thickness 56.007 mm., is greater than required thickness 11.340 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 75.4	Allowable	: 142.0 N./mm ²	Passed
Expansion	: 0.0	Allowable	: 308.4 N./mm ²	Passed
Occasional	: 60.1	Allowable	: 188.9 N./mm ²	Passed
Shear	: 11.8	Allowable	: 99.4 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[142.0/150.0, 1]$$

$$= 0.947$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 230.0 - 3.0$$

$$= 227.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 56.0 - 3.0$$

$$= 53.000 \text{ mm.}$$



Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2756.0/189.0$$

$$= 14.582 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2750.0/2 + 3.0$$

$$= 1378.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{(Reff \cdot t)}, 2 \cdot R_n)$$

$$= \min(\sqrt{(1378.0 \cdot 189.0)}, 2 \cdot 358.6)$$

$$= 510.335 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 189.0, 0.0] + \sqrt{(358.6 \cdot 227.0)}$$

$$= 285.311 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 450.0$$

$$= 450.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(189.0 + 0.0)$$

$$= 1512.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[285.311, 450.0, 1512.0] + 189.0$$

$$= 474.311 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 189.0 + ((0.0 \cdot 1.0) / 510.335)$$

$$= 189.000 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + teff)} \cdot teff))$$

$$= \min(12, (2 \cdot 358.6 + 227.0) / (\sqrt{(2756.0 + 189.0)} \cdot 189.0))$$

$$= 1.266$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 189.0 \cdot 510.335 \cdot \max((1.266/5)^{0.85}, 1)$$

$$= 964.534 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 227.0 \cdot 474.311$$

$$= 1076.685 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$

$$= 964.534 + 0.947(1076.685 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 1984.296 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$



$$= 227.0 / \ln[1 + 227.0/358.6]$$

$$= 462.860 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 189.0 / \ln[1 + 189.0/1378.0]$$

$$= 1470.476 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 150.0 \cdot 1.0$$

$$= 225.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 190.685 \cdot 462.86 \cdot 474.311$$

$$= 426899.0 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 190.685 \cdot 1470.476 (510.335 + 227.0)$$

$$= 2108316.8 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 190.685 \cdot 1470.476 \cdot 358.6$$

$$= 1025371.2 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (2108317 + 1025371 + 426899.0) / 190.685$$

$$= 18310.9 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 224.998 / (2 \cdot 18310.859 / 1984.296 - 1470.476 / 189.0)$$

$$= 210.7 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 150.0 [189.0 / 1470.476]$$

$$= 192.8 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(210.749, 192.782)$$

$$= 192.782 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (426899.031 + 2108317 + 1025371) / 1984.296$$

$$= 175.972 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 190.685 \cdot 1470.476 / 189.0$$

$$= 148.4 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 175.972 - 148.367, 148.367)$$

$$= 203.6 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	225.00	N./mm ²
Local Primary Membrane Stress	PL	203.58	N./mm ²
Maximum Allowable Working Pressure	Pmax	192.78	bars

Strength of Nozzle Attachment Welds per 4.5.14



Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (358.6 + 227.0) / 358.6$$

$$= 1.633 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (358.6 + 227.0)$$

$$= 919.858 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 227.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(1025371 \cdot 1.63, 1.5 \cdot 142.0(1076.685 + 0.0), \pi/4 \cdot 190.7 \cdot 358.6^2 \cdot 1.63^2)$$

$$= 523727.000 \text{ Kgf}$$

Weld Stress Value [τ]:

$$= fwelds / (L_t(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 523727 / (919.858(0.49 \cdot 7.071 + 0.6 \cdot 189.0 + 0.49 \cdot 0.0))$$

$$= 47.778 < \text{ or } = \text{ to } 149.999 \text{ Weld Stress Passed}$$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 227.0 / \ln[1 + 227.0/358.6]$$

$$= 462.860 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 189.0 / \ln[1 + 189.0/1378.0]$$

$$= 1470.476 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 150.0$$

$$= 225.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.034 \cdot 462.86 \cdot 474.311$$

$$= 2315.3 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs}(L_R + t_n)$$

$$= 1.034 \cdot 1470.476(510.335 + 227.0)$$

$$= 11434.6 \text{ Kgf}$$

Discontinuity Force from External Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.034 \cdot 1470.476 \cdot 358.6$$

$$= 5561.2 \text{ Kgf}$$

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (11434.6 + 5561.2 + 2315.3) / 1.034$$

$$= 18310.9 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 224.998 / (2 \cdot 18310.861 / 1984.296 - 1470.476 / 189.0)$$

$$= 210.7 \text{ bars}$$



Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 150.0[189.0/1470.476]$$

$$= 192.8 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(210.749, 192.782)$$

$$= 192.782 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$

$$= (2315.31 + 11434.571 + 5561.158) / 1984.296$$

$$= 0.954 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / teff$$

$$= 1.034 \cdot 1470.476 / 189.0$$

$$= 0.8 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 0.954 - 0.805, 0.805)$$

$$= 1.1 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	225.00	N./mm ²
Local Primary Membrane Stress	PL	1.10	N./mm ²
Maximum Allowable External Pressure	Pmax	192.78	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm..
 This component must be impact tested.

The Drop for this Nozzle is : 130.9350 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 772.9351 mm.

Input Echo, WRC297 Item 1, Description: N128" :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	3.0000	mm.	
Shell External Corrosion Allowance	0.0000	mm.	
Shell Diameter	2750.000	mm.	
Shell Thickness	192.0000	mm.	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	175.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	150.00	N./mm ²
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm.	
Nozzle Diameter	711.200	mm.	
Nozzle Thickness	230.0000	mm.	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-105		
Nozzle Cold S.I. Allowable	SNmc	165.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	142.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:



These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	190.68	bars
Radial Load	P	-17132.38	Kgf
Circumferential Shear	Vc	12849.26	Kgf
Longitudinal Shear	Vl	17132.38	Kgf
Circumferential Moment	Mc	23976.08	Kg-m.
Longitudinal Moment	Ml	31168.92	Kg-m.
Torsional Moment	Mt	35964.05	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_v/T_v (16.323) is < 20
 check the Limitations of WRC bulletin 297.
 Warning - The ratio, D_n/T_n (5.092) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$
 $= 1171.2 + 2 \cdot 1.65 \cdot \sqrt{(1472.5(192.0 - 3.0))}$
 $= 2912.095 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2945.000 mm.
Nozzle Outside Diameter	(d) =	1171.200 mm.
Vessel Thickness used	(T) =	189.000 mm.
Nozzle Thickness used	(t) =	227.000 mm.
T / t	=	0.833
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.570$$

Nr/P	=	0.093
Mr/P	=	0.148
M0/P	=	0.047
N0/P	=	0.109
MrD/Mc	=	0.272
NrDL/Mc	=	0.129
M0d/Mc	=	0.094
N0DL/Mc	=	0.054
MrD/Ml	=	0.141
NrDL/Ml	=	0.083
M0D/Ml	=	0.050
N0DL/Ml	=	0.135

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	1	-1	1	-1
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	0	0	0	0
Outplane Bending (Ml)	-2	2	2	-2
Normal Pressure Stress	130	149	130	149
Outplane Stress Summary	128	149	135	147

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------



Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	4	-4	4	-4
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-6	6	6	-6
Inplane Pressure Stress	65	65	65	65

Inplane Stress Summary	62	66	76	55
------------------------	----	----	----	----

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	1	1	0	0
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside

Two * Max Shear Stress	128	149	135	147
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	1	-1	1	-1
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-3	3	3	-3
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	65	65	65	65

Outplane Stress Summary	63	67	70	61
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	4	-4	4	-4
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-9	9	9	-9
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	130	149	130	149

Inplane Stress Summary	124	153	144	136
------------------------	-----	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	0	0	1	1
----------------------	---	---	---	---

Vessel Stresses (N./mm²):



CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	124	153	144	137

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	27	50	27	50
Hoop Stress Summary	26	49	28	51

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-3	3	3	-3
Axial Pressure Stress	13	13	13	13
Axial Stress Summary	10	12	21	11

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	1	1	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	27	49	28	51

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	27	50	27	50
Hoop Stress Summary	27	50	28	51

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to	Cu Left	Cl Left	Du Right	Dl Right
--	------------	------------	-------------	-------------



pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-4	4	4	-4
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	13	13	13	13
Axial Stress Summary	9	15	22	8

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	27	50	28	51

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	130	149	130	149	130	149	130	149
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	-1	1	3	-3	-5	5	13	-13
Long. Pm (SUS)	65	65	65	65	65	65	65	65
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	-2	2	10	-10	-2	2	4	-4
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	130.0	149.0	130.0	149.0	130.0	149.0	130.0	149.0
Pm+Pl (SUS)	130.0	149.0	130.0	149.0	130.0	149.0	130.0	149.0
Pm+Pl+Q (Total)	129.0	150.0	133.0	146.0	125.0	154.0	143.0	136.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	149.00	150.00	Passed
Pm+Pl (SUS)	149.00	225.00	Passed
Pm+Pl+Q (TOTAL)	154.00	450.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):



Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	27	50	27	50	27	50	27	50
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	13	13	13	13	13	13	13	13
Long. Pl (SUS)	-2	-2	2	2	-1	-1	1	1
Long. Q (SUS)	-2	2	4	-4	-3	3	5	-5
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	27.0	50.0	27.0	50.0	27.0	50.0	27.0	50.0
Pm+Pl (SUS)	27.0	50.0	27.0	50.0	27.0	50.0	27.0	50.0
Pm+Pl+Q (Total)	27.0	50.0	27.0	50.0	27.0	50.0	27.0	50.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	50.00	142.00	Passed
Pm+Pl (SUS)	50.00	213.00	Passed
Pm+Pl+Q (TOTAL)	50.00	426.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N42"

From: 50

Pressure for Reinforcement Calculations	P	190.685	bars
Temperature for Internal Pressure	Temp	200	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	150.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2750.00	mm.
Design Length of Section	L	6758.0000	mm.
Shell Finished (Minimum) Thickness	t	192.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		3599.97	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

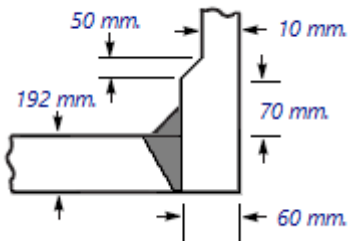
Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	142.00	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm.
Height of Beveled Transition	L'	50.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		60.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	91.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N42"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.394	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 56.8 \cdot (\exp(190.685/(142.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 4.0819 + 3.0 + 0.0 = 7.0819 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 142.0 \cdot 1.0 \cdot \ln((2 \cdot 7.0 + 56.8) / 56.8) - 0.0$$

$$= 312.837 - 0.0 = 312.8375 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 7.520 mm
 Min. required shell thickness, internal p + c : 190.079 mm.
 Min. required shell thickness, external p + c : 19.199 mm.

Nozzle neck minimum required thickness : 7.520 mm.

Minimum nozzle neck thickness 10.008 mm., is greater than required thickness 7.520 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 122.3,	Allowable	: 142.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 261.5 N./mm ²	Passed
Occasional	: 34.4,	Allowable	: 188.9 N./mm ²	Passed
Shear	: 74.7,	Allowable	: 99.4 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[142.0/150.0, 1]$$

$$= 0.947$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 60.0 - 3.0$$

$$= 57.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 10.0 - 3.0$$

$$= 7.000 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2756.0/189.0$$

$$= 14.582 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:



$$\begin{aligned}
 &= Di/2 + \text{corrosion allowance} \\
 &= 2750.0/2 + 3.0 \\
 &= 1378.000 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Effective Length of Vessel Wall [LR]:} \\
 &= \min(\sqrt{Reff \cdot t}, 2 \cdot Rn) \\
 &= \min(\sqrt{1378.0 \cdot 189.0}, 2 \cdot 28.4) \\
 &= 56.800 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Thickness Limit Candidate [LH1]:} \\
 &= \min[1.5t, te] + \sqrt{Rn \cdot tn} \\
 &= \min[1.5 \cdot 189.0, 0.0] + \sqrt{28.4 \cdot 57.0} \\
 &= 40.234 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Thickness Limit Candidate [LH2]:} \\
 &= Lpr1 \\
 &= 200.0 \\
 &= 200.000 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Thickness Limit Candidate [LH3]:} \\
 &= 8(t + te) \\
 &= 8(189.0 + 0.0) \\
 &= 1512.000 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Effective Nozzle Wall Length Outside the Vessel [LH]:} \\
 &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\
 &= \min[40.234, 200.0, 1512.0] + 189.0 \\
 &= 229.234 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Effective Vessel Thickness [teff]:} \\
 &= t + ((A5 \cdot frp) / LR) \\
 &= 189.0 + ((0.0 \cdot 1.0) / 56.8) \\
 &= 189.000 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Parameter } [\lambda]: \\
 &= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff})) \\
 &= \min(12, (2 \cdot 28.4 + 57.0) / (\sqrt{(2756.0 + 189.0) \cdot 189.0})) \\
 &= 0.153
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

$$\begin{aligned}
 &\text{Area Contributed by the Vessel Wall [A1]:} \\
 &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\
 &= 189.0 \cdot 56.8 \cdot \max((0.153/5)^{0.85}, 1) \\
 &= 107.352 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Area Contributed by the Nozzle Outside the Vessel Wall [A2]:} \\
 &= tn \cdot LH \\
 &= 57.0 \cdot 229.234 \\
 &= 130.664 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Area Contributed by the Outside Fillet Weld [A41]:} \\
 &= 0.5 \cdot Leg^2 \\
 &= 0.5 \cdot 10.0^2 \\
 &= 0.500 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{The total area contributed by A1 through A43 [AT]:} \\
 &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\
 &= 107.352 + 0.947(130.664 + 0.0) + 0.5 + 0.0 + 0.0 \\
 &= 231.547 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Nozzle Radius for Force Calculation [Rxn]:} \\
 &= tn / \ln[1 + tn/Rn] \\
 &= 57.0 / \ln[1 + 57.0/28.4] \\
 &= 51.773 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Shell Radius for Force Calculation [Rxs]:} \\
 &= teff / \ln[1 + teff/Reff] \\
 &= 189.0 / \ln[1 + 189.0/1378.0] \\
 &= 1470.476 \text{ mm.}
 \end{aligned}$$



Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 150.0 \cdot 1.0$$

$$= 225.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 190.685 \cdot 51.773 \cdot 229.234$$

$$= 23077.9 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 190.685 \cdot 1470.476 (56.8 + 57.0)$$

$$= 325396.7 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 190.685 \cdot 1470.476 \cdot 28.4$$

$$= 81206.2 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (325396.7 + 81206.2 + 23077.9) / 190.685$$

$$= 2209.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 224.998 / (2 \cdot 2209.699 / 231.547 - 1470.476 / 189.0)$$

$$= 199.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / t / R_{xs}$$

$$= 150.0 / 189.0 / 1470.476$$

$$= 192.8 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(198.994, 192.782)$$

$$= 192.782 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (23077.938 + 325396.688 + 81206.195) / 231.547$$

$$= 181.985 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 190.685 \cdot 1470.476 / 189.0$$

$$= 148.4 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 181.985 - 148.367, 148.367)$$

$$= 215.6 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	225.00	N./mm ²
Local Primary Membrane Stress	PL	215.60	N./mm ²
Maximum Allowable Working Pressure	Pmax	192.78	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (28.4 + 57.0) / 28.4$$

$$= 3.007 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (28.4 + 57.0)$$

$$= 134.146 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:
 = 7.071, 0.000, 0.000, mm.

Required Weld Throat Dimensions :
 tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 57.000, 6.000)$ Passed

Weld Load Value [fwelds]:
 $= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
 $= \min(81206 \cdot 3.01, 1.5 \cdot 142.0(130.664 + 0.0), \pi/4 \cdot 190.7 \cdot 28.4^2 \cdot 3.01^2)$
 $= 11138.289 \text{ Kgf}$

Weld Stress Value [τ]:
 $= \text{fwelds} / (L_t(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
 $= 11138.289 / (134.146(0.49 \cdot 7.071 + 0.6 \cdot 91.0 + 0.49 \cdot 0.0))$
 $= 14.024 < \text{or} = \text{to } 149.999 \text{ Weld Stress Passed}$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:
 $= t_n / \ln[1 + t_n/R_n]$
 $= 57.0 / \ln[1 + 57.0/28.4]$
 $= 51.773 \text{ mm.}$

Shell Radius for Force Calculation [Rxs]:
 $= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
 $= 189.0 / \ln[1 + 189.0/1378.0]$
 $= 1470.476 \text{ mm.}$

Allowable Local Primary Membrane Stress [Sallow]:
 $= 1.5 \cdot S$
 $= 1.5 \cdot 150.0$
 $= 225.0 \text{ N./mm}^2$

Force acting on the Nozzle [fN]:
 $= P \cdot R_{xn} \cdot L_H$
 $= 1.034 \cdot 51.773 \cdot 229.234$
 $= 125.2 \text{ Kgf}$

Force acting on the Shell [fS]:
 $= P \cdot R_{xs}(L_R + t_n)$
 $= 1.034 \cdot 1470.476(56.8 + 57.0)$
 $= 1764.8 \text{ Kgf}$

Discontinuity Force from External Pressure [fY]:
 $= P \cdot R_{xs} \cdot R_{nc}$
 $= 1.034 \cdot 1470.476 \cdot 28.4$
 $= 440.4 \text{ Kgf}$

Area Resisting External Pressure [Ap]:
 $= (f_S + f_Y + f_N)/P$
 $= (1764.8 + 440.4 + 125.2)/1.034$
 $= 2209.7 \text{ cm}^2$

Maximum Allowable Working Pressure Candidate [Pmax1]:
 $= S_{allow} / (2 \cdot A_p / A_T - R_{xs}/t_{eff})$
 $= 224.998 / (2 \cdot 2209.699/231.547 - 1470.476/189.0)$
 $= 199.0 \text{ bars}$

Maximum Allowable Working Pressure Candidate [Pmax2]:
 $= S/[t/R_{xs}]$
 $= 150.0[189.0/1470.476]$
 $= 192.8 \text{ bars}$

Maximum Allowable Working Pressure [Pmax]:
 $= \min(P_{max1}, P_{max2})$
 $= \min(198.994, 192.782)$
 $= 192.782 \text{ bars}$

Average Primary Membrane Stress [σ_{Avg}]:
 $= (fN + fS + fY) / AT$
 $= (125.164 + 1764.807 + 440.426) / 231.547$
 $= 0.987 \text{ N./mm}^2$

General Primary Membrane Stress [σ_{Circ}]:
 $= P \cdot Rxs / teff$
 $= 1.034 \cdot 1470.476 / 189.0$
 $= 0.8 \text{ N./mm}^2$

Maximum Local Primary Membrane Stress [PL]:
 $= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
 $= \max(2 \cdot 0.987 - 0.805, 0.805)$
 $= 1.2 \text{ N./mm}^2$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	225.00	N./mm ²
Local Primary Membrane Stress	PL	1.17	N./mm ²
Maximum Allowable External Pressure	Pmax	192.78	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -36 °C
 User entered Minimum Design Temperature : -10 °C
 The Curve for this Material is : A
 From Figure 3.8M- with PWHT, the MDMT is : 1 °C
 Temperature reduction per Graph 3.13 : 36 °C
 The computed stress ratio is : 0.497

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (3.48 \cdot 1.0) / (10.0 - 3.0 - 0.0)$
 $= 0.497$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : -10 °C
 The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.061

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (3.48 \cdot 1.0) / (60.0 - 3.0 - 0.0)$
 $= 0.061$

Gov. MDMT of the nozzle to shell joint welded assembly : -36 °C

The Drop for this Nozzle is : 2.6546 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 394.6546 mm.

Input Echo, WRC297 Item 1, Description: N42" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2750.000	mm.
Shell Thickness	192.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	175.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	150.00 N./mm ²



Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	50.800	mm.
Nozzle Thickness	60.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-105	
Nozzle Cold S.I. Allowable	SNmc	165.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	142.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	190.68	bars
Radial Load	P	-1223.74	Kgf
Circumferential Shear	Vc	917.81	Kgf
Longitudinal Shear	Vl	1223.74	Kgf
Circumferential Moment	Mc	122.33	Kg-m.
Longitudinal Moment	Ml	159.03	Kg-m.
Torsional Moment	Mt	183.49	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_v/T_v (16.323) is < 20
 check the limitations of WRC bulletin 297.

Warning - The ratio, D_n/T_v (.890) is < 5
 check the limitations of WRC bulletin 297.

Warning - The ratio, D_n/T_n (2.847) is < 10
 check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 170.8 + 2 \cdot 1.65 \cdot \sqrt{1472.5(192.0 - 3.0)}$$

$$= 1911.695 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2945.000	mm.
Nozzle Outside Diameter	(d) =	170.800	mm.
Vessel Thickness used	(T) =	189.000	mm.
Nozzle Thickness used	(t) =	57.000	mm.
T / t	=	3.316	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.229$$

Nr/P	=	0.246
Mr/P	=	0.211
M0/P	=	0.238
N0/P	=	0.205
MrD/Mc	=	0.179
NrDL/Mc	=	0.181
M0d/Mc	=	0.235
N0DL/Mc	=	-0.129
MrD/Ml	=	0.178
NrDL/Ml	=	0.186
M0D/Ml	=	0.231
N0DL/Ml	=	-0.124

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside



Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	0	0	0	0
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	130	149	130	149

Outplane Stress Summary	130	149	130	148
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	0	0	0	0
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	65	65	65	65

Inplane Stress Summary	65	64	65	64
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	0	0	0	0
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	130	149	130	148
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	0	0	0	0
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	65	65	65	65

Outplane Stress Summary	65	64	65	64
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	0	0	0	0
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0



Inplane Pressure Stress	130	149	130	149
-------------------------	-----	-----	-----	-----

Inplane Stress Summary	130	149	130	148
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	0	0	0	0
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	130	149	130	148
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	28	5	28

Hoop Stress Summary	5	28	5	28
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	0	0	10	3
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	1	1	0	0
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	6	29	10	28
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Stresses in the hoop direction)	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	28	5	28
Hoop Stress Summary	5	28	5	28

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-1	1	1	-1
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	1	0	8	2

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	5	28	9	28

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	130	149	130	149	130	149	130	149
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	65	65	65	65	65	65	65	65
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	0	0	0	0	0	0	0	0
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	130.0	149.0	130.0	149.0	130.0	149.0	130.0	149.0
Pm+Pl (SUS)	130.0	149.0	130.0	149.0	130.0	149.0	130.0	149.0
Pm+Pl+Q (Total)	130.0	149.0	130.0	149.0	130.0	149.0	130.0	149.0



Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	149.00	150.00	Passed
Pm+Pl (SUS)	149.00	225.00	Passed
Pm+Pl+Q (TOTAL)	149.00	450.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	5	28	5	28	5	28	5	28
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-3	-3	3	3	-2	-2	2	2
Long. Q (SUS)	0	0	2	-2	0	0	2	-2
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	5.0	28.0	5.0	28.0	5.0	28.0	5.0	28.0
Pm+Pl (SUS)	6.0	29.0	5.0	28.0	5.0	28.0	5.0	28.0
Pm+Pl+Q (Total)	6.0	29.0	7.0	28.0	5.0	28.0	6.0	28.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	28.00	142.00	Passed
Pm+Pl (SUS)	29.00	213.00	Passed
Pm+Pl+Q (TOTAL)	29.00	426.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: M124"

From: 50

Pressure for Reinforcement Calculations	P	190.685	bars
Temperature for Internal Pressure	Temp	200	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	150.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2750.00	mm.
Design Length of Section	L	6758.0000	mm.
Shell Finished (Minimum) Thickness	t	192.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		3999.97	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

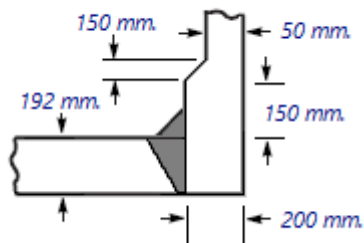
Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	142.00	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	50.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	150.0000	mm.
Height of Beveled Transition	L'	150.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		200.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	192.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: M124"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	1.969	in.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 615.6 \cdot (\exp(190.685/(142.0 \cdot 1.0)) - 1) + 3.0 + 0.0$
 $= 44.2398 + 3.0 + 0.0 = 47.2398 \text{ mm. Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 142.0 \cdot 1.0 \cdot \ln((2 \cdot 47.0 + 615.6) / 615.6) - 0.0$
 $= 201.774 - 0.0 = 201.7742 \text{ bars}$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[142.0/150.0, 1]$
 $= 0.947$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 200.0 - 3.0$
 $= 197.000 \text{ mm.}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 50.0 - 3.0$
 $= 47.000 \text{ mm.}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 2756.0/189.0$
 $= 14.582 \text{ must be less than } 400.$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 2750.0/2 + 3.0$
 $= 1378.000 \text{ mm.}$

Effective Length of Vessel Wall [LR]:
 $= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$
 $= \min(\sqrt{1378.0 \cdot 189.0}, 2 \cdot 307.8)$
 $= 510.335 \text{ mm.}$

Thickness Limit Candidate [LH1]:
 $= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$
 $= \min[1.5 \cdot 189.0, 0.0] + \sqrt{307.8 \cdot 197.0}$
 $= 246.245 \text{ mm.}$



Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 300.0$$

$$= 300.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(189.0 + 0.0)$$

$$= 1512.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[246.245, 300.0, 1512.0] + 189.0$$

$$= 435.245 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 189.0 + ((0.0 \cdot 1.0) / 510.335)$$

$$= 189.000 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 307.8 + 197.0) / (\sqrt{(2756.0 + 189.0) \cdot 189.0}))$$

$$= 1.089$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 189.0 \cdot 510.335 \cdot \max((1.089/5)^{0.85}, 1)$$

$$= 964.534 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$

$$= 667.83 + 143.287$$

$$= 811.117 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$

$$= 964.534 + 0.947(811.117 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 1732.892 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 197.0 / \ln[1 + 197.0/307.8]$$

$$= 398.211 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 189.0 / \ln[1 + 189.0/1378.0]$$

$$= 1470.476 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 150.0 \cdot 1.0$$

$$= 225.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 190.685 \cdot 398.211 \cdot 435.245$$

$$= 337023.6 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs}(LR + t_n)$$



$$= 190.685 \cdot 1470.476 (510.335 + 197.0)$$

$$= 2022535.5 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 190.685 \cdot 1470.476 \cdot 307.8$$

$$= 880115.2 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN)/P$$

$$= (2022536 + 880115 + 337023.6)/190.685$$

$$= 16660.5 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$

$$= 224.998 / (2 \cdot 16660.518/1732.892 - 1470.476/189.0)$$

$$= 196.5 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 150.0[189.0/1470.476]$$

$$= 192.8 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(196.523, 192.782)$$

$$= 192.782 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (337023.625 + 2022536 + 880115)/1732.892$$

$$= 183.341 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 190.685 \cdot 1470.476/189.0$$

$$= 148.4 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 183.341 - 148.367, 148.367)$$

$$= 218.3 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	225.00	N./mm ²
Local Primary Membrane Stress	PL	218.31	N./mm ²
Maximum Allowable Working Pressure	Pmax	192.78	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (307.8 + 197.0)/307.8$$

$$= 1.640 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn)$$

$$= \pi/2 \cdot (307.8 + 197.0)$$

$$= 792.938 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 197.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(880115 \cdot 1.64, 1.5 \cdot 142.0(811.117 + 0.0), \pi/4 \cdot 190.7 \cdot 307.8^2 \cdot 1.64^2)$$

$$= 389171.906 \text{ Kgf}$$



Weld Stress Value [τ]:
= $f_{welds} / (L_t(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
= $389171.906 / (792.938(0.49 \cdot 7.071 + 0.6 \cdot 189.0 + 0.49 \cdot 0.0))$
= 41.186 < or = to 149.999 Weld Stress Passed

Please check Code paragraph 4.5.17.1 regarding openings in compression. A
Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:
= $t_n / \ln[1 + t_n/R_n]$
= $197.0 / \ln[1 + 197.0/307.8]$
= 398.211 mm.

Shell Radius for Force Calculation [R_{xs}]:
= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
= $189.0 / \ln[1 + 189.0/1378.0]$
= 1470.476 mm.

Allowable Local Primary Membrane Stress [S_{allow}]:
= $1.5 \cdot S$
= $1.5 \cdot 150.0$
= 225.0 N./mm²

Force acting on the Nozzle [f_N]:
= $P \cdot R_{xn} \cdot L_H$
= $1.034 \cdot 398.211 \cdot 435.245$
= 1827.9 Kgf

Force acting on the Shell [f_S]:
= $P \cdot R_{xs} (L_R + t_n)$
= $1.034 \cdot 1470.476 (510.335 + 197.0)$
= 10969.3 Kgf

Discontinuity Force from External Pressure [f_Y]:
= $P \cdot R_{xs} \cdot R_{nc}$
= $1.034 \cdot 1470.476 \cdot 307.8$
= 4773.4 Kgf

Area Resisting External Pressure [A_p]:
= $(f_S + f_Y + f_N) / P$
= $(10969.3 + 4773.4 + 1827.9) / 1.034$
= 16660.5 cm²

Maximum Allowable Working Pressure Candidate [P_{max1}]:
= $S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$
= $224.998 / (2 \cdot 16660.52 / 1732.892 - 1470.476 / 189.0)$
= 196.5 bars

Maximum Allowable Working Pressure Candidate [P_{max2}]:
= $S / (t / R_{xs})$
= $150.0 [189.0 / 1470.476]$
= 192.8 bars

Maximum Allowable Working Pressure [P_{max}]:
= $\min(P_{max1}, P_{max2})$
= $\min(196.523, 192.782)$
= 192.782 bars

Average Primary Membrane Stress [σ_{Avg}]:
= $(f_N + f_S + f_Y) / A_T$
= $(1827.866 + 10969.332 + 4773.353) / 1732.892$
= 0.994 N./mm²

General Primary Membrane Stress [σ_{Circ}]:
= $P \cdot R_{xs} / t_{eff}$
= $1.034 \cdot 1470.476 / 189.0$
= 0.8 N./mm²



Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 0.994 - 0.805, 0.805)$$

$$= 1.2 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	225.00	N./mm ²
Local Primary Membrane Stress	PL	1.18	N./mm ²
Maximum Allowable External Pressure	Pmax	192.78	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: 19 °C
User entered Minimum Design Temperature	: -10 °C
The Curve for this Material is	: A
From Figure 3.8M- with PWHT, the MDMT is	: 30 °C
Temperature reduction per Graph 3.13	: 11 °C
The computed stress ratio is	: 0.802

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (37.711 \cdot 1.0) / (50.0 - 3.0 - 0.0)$$

$$= 0.802$$

Warning: Impact Testing Is Mandatory!

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm.. This component must be impact tested.

Gov. MDMT of the nozzle to shell joint welded assembly : 19 °C

The Drop for this Nozzle is : 96.0153 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 588.0153 mm.

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Input, Nozzle Desc: N63"

From: 60

Pressure for Reinforcement Calculations	P	190.685	bars
Temperature for Internal Pressure	Temp	200	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	150.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Hemispherical Head	D	2844.00	mm.
Head Finished (Minimum) Thickness	t	98.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	950.0001	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

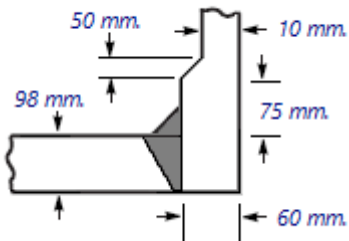
Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	142.00	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	75.0000	mm.
Height of Beveled Transition	L'	50.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		60.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		98.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N63"

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Actual Inside Diameter Used in Calculation	3.000	in.
Actual Thickness Used in Calculation	0.394	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 82.2 \cdot (\exp(190.685/(142.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 5.9073 + 3.0 + 0.0 = 8.9073 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 142.0 \cdot 1.0 \cdot \ln((2 \cdot 7.0 + 82.2) / 82.2) - 0.0$$

$$= 223.314 - 0.0 = 223.3144 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	8.020 mm
Min. required shell thickness, internal p + c :	96.554 mm.
Min. required shell thickness, external p + c :	10.070 mm.

Nozzle neck minimum required thickness : 8.020 mm.

Minimum nozzle neck thickness 10.008 mm., is greater than required thickness 8.020 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[142.0/150.0, 1]$$

$$= 0.947$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 60.0 - 3.0$$

$$= 57.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 10.0 - 3.0$$

$$= 7.000 \text{ mm.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2844.0/2 + 3.0$$

$$= 1425.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1425.0 \cdot 95.0}, 2 \cdot 41.1)$$

$$= 82.200 \text{ mm.}$$

Intermediate value [Cn]:

$$= \min(((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min(((95.0 + 0.0)/57.0)^{0.35}, 1)$$

$$= \min(1.196, 1)$$

$$= 1.000$$

Intermediate value [Cp]:

= 9999999.000

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t)$$

$$= \min(95.0 + 0.0 + 1.0 \cdot \sqrt{(41.1 \cdot 57.0)}, 200.0 + 95.0)$$

$$= 143.401 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A_5 \cdot f_{rp}) / LR)$$

$$= 95.0 + ((0.0 \cdot 1.0) / 82.2)$$

$$= 95.000 \text{ mm.}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$

$$= 95.0 \cdot 82.2$$

$$= 78.090 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 57.0 \cdot 143.401$$

$$= 81.739 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A_1 + f_{rn}(A_2 + A_3) + A_{41} + A_{42} + A_{43}$$

$$= 78.09 + 0.947(81.739 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 155.969 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 57.0 / \ln[1 + 57.0/41.1]$$

$$= 65.519 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 95.0 / \ln[1 + 95.0/1425.0]$$

$$= 1471.990 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 150.0 \cdot 1.0$$

$$= 225.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 190.685 \cdot 65.519 \cdot 143.401$$

$$= 18269.7 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (LR + t_n)/2$$

$$= 190.685 \cdot 1471.99 \cdot (82.2 + 57.0)/2$$

$$= 199217.3 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc})/2$$

$$= (190.685 \cdot 1471.99 \cdot 55.141)/2$$

$$= 78916.1 \text{ Kgf}$$

Note: Rnc in the above formula is divided by sin(theta).

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N)/P$$

$$= (199217.3 + 78916.1 + 18269.7)/190.685$$

$$= 1524.3 \text{ cm}^2$$



Maximum Allowable Working Pressure Candidate [Pmax1]:
 = Sallow / (2 • Ap/AT - Rxs/(2 • teff))
 = 225.0 / (2 • 1524.299/155.969 - 1471.99/(2•95.0))
 = 190.7 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:
 = 2 • S[t/Rxs]
 = 2 • 150.0[95.0/1471.99]
 = 193.6 bars

Maximum Allowable Working Pressure [Pmax]:
 = min(Pmax1, Pmax2)
 = min(190.685, 193.604)
 = 190.685 bars

Average Primary Membrane Stress [σAvg]:
 = (fN + fS + fY) / AT
 = (18269.74 + 199217.328 + 78916.148)/155.969
 = 186.369 N./mm²

General Primary Membrane Stress [σCirc]:
 = P • Rxs / (2 • teff)
 = 190.685 • 1471.99 / (2 • 95.0)
 = 147.7 N./mm²

Maximum Local Primary Membrane Stress [PL]:
 = max(2 • σAvg - σCirc, σCirc)
 = max(2 • 186.369 - 147.738, 147.738)
 = 225.0 N./mm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	225.00	N./mm ²
Local Primary Membrane Stress	PL	225.00	N./mm ²
Maximum Allowable Working Pressure	Pmax	190.68	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:
 = (Rnc + tn) / Rnc
 = (55.141 + 57.0)/55.141
 = 2.034 For set-in Nozzles

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [LT]:
 = $\pi/2 \cdot \sqrt{((Rnc+tn)^2 + (Rn+tn)^2)/2}$
 = $\pi/2 \cdot \sqrt{((55.141+57.0)^2 + (41.1+57.0)^2)/2}$
 = 165.491 mm.

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:
 = 7.071, 0.000, 0.000, mm.

Required Weld Throat Dimensions :
 tc, L41T = 7.071, must be >= min(0.7 • 57.000, 6.000) Passed

Weld Load Value [fwelds]:
 = min(fy • ky, 1.5 • Sn(A2 + A3), $\pi/4 \cdot P \cdot Rn^2 \cdot ky^2$)
 = min(78916 • 2.03, 1.5 • 142.0(81.739+0.0), $\pi/4 \cdot 190.7 \cdot 41.1^2 \cdot 2.03^2$)
 = 10669.952 Kgf

Weld Stress Value [τ]:
 = fwelds/(LT(0.49•L41T + 0.6•tw1 + 0.49•L43T))
 = 10669.952/(165.491(0.49•7.071 + 0.6•95.0 + 0.49•0.0))
 = 10.457 < or = to 150.000 Weld Stress Passed

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:
 = tn / ln[1 + tn/Rn]
 = 57.0/ln[1 + 57.0/41.1]
 = 65.519 mm.



Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 95.0 / \ln[1 + 95.0/1425.0]$$

$$= 1471.990 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 150.0$$

$$= 225.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.034 \cdot 65.519 \cdot 143.401$$

$$= 99.1 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (L_R + t_n)/2$$

$$= 1.034 \cdot 1471.99 \cdot (82.2 + 57.0)/2$$

$$= 1080.5 \text{ Kgf}$$

Discontinuity Force from External Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc})/2$$

$$= (1.034 \cdot 1471.99 \cdot 55.141)/2$$

$$= 428.0 \text{ Kgf}$$

Note: Rnc in the above formula is divided by sin(theta).

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N)/P$$

$$= (1080.5 + 428.0 + 99.1)/1.034$$

$$= 1524.3 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / (2 \cdot t_{eff}))$$

$$= 225.0 / (2 \cdot 1524.299 / 155.969 - 1471.99 / (2 \cdot 95.0))$$

$$= 190.7 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S / (t / R_{xs})$$

$$= 2 \cdot 150.0 / [95.0 / 1471.99]$$

$$= 193.6 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(190.685, 193.604)$$

$$= 190.685 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (99.087 + 1080.466 + 428.006) / 155.969$$

$$= 1.011 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / (2 \cdot t_{eff})$$

$$= 1.034 \cdot 1471.99 / (2 \cdot 95.0)$$

$$= 0.8 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 1.011 - 0.801, 0.801)$$

$$= 1.2 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	225.00	N./mm ²
Local Primary Membrane Stress	PL	1.22	N./mm ²
Maximum Allowable External Pressure	Pmax	190.68	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:



Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -15 °C
User entered Minimum Design Temperature : -10 °C
The Curve for this Material is : A
From Figure 3.8M- with PWHT, the MDMT is : 1 °C
Temperature reduction per Graph 3.13 : 16 °C
The computed stress ratio is : 0.719

The computed thickness ratio [sr]:

$$= (tr \cdot E*) / (tnom - ci - cext)$$
$$= (5.036 \cdot 1.0) / (10.0 - 3.0 - 0.0)$$
$$= 0.719$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -10 °C
The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.088

The computed thickness ratio [sr]:

$$= (tr \cdot E*) / (tnom - ci - cext)$$
$$= (5.036 \cdot 1.0) / (60.0 - 3.0 - 0.0)$$
$$= 0.088$$

Gov. MDMT of the nozzle to shell joint welded assembly : -15 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction : -32 °C

The Drop for this Nozzle is : 97.0783 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 425.5239 mm.

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Input, Nozzle Desc: N228"

From: 60

Pressure for Reinforcement Calculations	P	190.685	bars
Temperature for Internal Pressure	Temp	200	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	150.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Hemispherical Head	D	2844.00	mm.
Head Finished (Minimum) Thickness	t	98.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

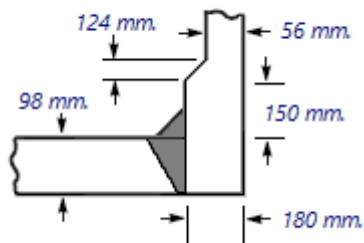
Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	142.00	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		28.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	56.0000	mm.
Hub Height of Integral Nozzle	h	150.0000	mm.
Height of Beveled Transition	L'	124.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		180.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	685.5000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	98.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N228"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	28.000	in.
Actual Thickness Used in Calculation	2.205	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 717.2 \cdot (\exp(190.685/(142.0 \cdot 1.0)) - 1) + 3.0 + 0.0$
 $= 51.5413 + 3.0 + 0.0 = 54.5412 \text{ mm. Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 142.0 \cdot 1.0 \cdot \ln((2 \cdot 53.0 + 717.2) / 717.2) - 0.0$
 $= 195.726 - 0.0 = 195.7261 \text{ bars}$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.340 mm
 Min. required shell thickness, internal p + c : 96.554 mm.
 Min. required shell thickness, external p + c : 10.070 mm.

Nozzle neck minimum required thickness : 11.340 mm.

Minimum nozzle neck thickness 56.007 mm., is greater than required thickness 11.340 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 75.4,	Allowable	: 142.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 308.4 N./mm ²	Passed
Occasional	: 60.1,	Allowable	: 188.9 N./mm ²	Passed
Shear	: 11.8,	Allowable	: 99.4 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[142.0/150.0, 1]$
 $= 0.947$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 180.0 - 3.0$
 $= 177.000 \text{ mm.}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 56.0 - 3.0$
 $= 53.000 \text{ mm.}$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 2844.0/2 + 3.0$
 $= 1425.000 \text{ mm.}$

Effective Length of Vessel Wall [LR]:



$$= \min(\sqrt{(\text{Reff} \cdot t)}, 2 \cdot R_n)$$

$$= \min(\sqrt{(1425.0 \cdot 95.0)}, 2 \cdot 358.6)$$

$$= 367.933 \text{ mm.}$$

Intermediate value [Cn]:

$$= \min((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min((95.0 + 0.0)/177.0)^{0.35}, 1)$$

$$= \min(0.804, 1)$$

$$= 0.804$$

Intermediate value [Cp]:

$$= 9999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t)$$

$$= \min(95.0 + 0.0 + 0.804 \cdot \sqrt{(358.6 \cdot 177.0)}, 685.5 + 95.0)$$

$$= 297.630 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A_5 \cdot \text{frp}) / LR)$$

$$= 95.0 + ((0.0 \cdot 1.0) / 367.933)$$

$$= 95.000 \text{ mm.}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$

$$= 95.0 \cdot 367.933$$

$$= 349.537 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A_{2A} + A_{2B}$$

$$= 433.65 + 79.306$$

$$= 512.956 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A_1 + \text{frn}(A_2 + A_3) + A_{41} + A_{42} + A_{43}$$

$$= 349.537 + 0.947(512.956 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 835.631 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 177.0 / \ln[1 + 177.0/358.6]$$

$$= 441.198 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= \text{teff} / \ln[1 + \text{teff}/\text{Reff}]$$

$$= 95.0 / \ln[1 + 95.0/1425.0]$$

$$= 1471.990 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 150.0 \cdot 1.0$$

$$= 225.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 190.685 \cdot 441.198 \cdot 297.63$$

$$= 255343.0 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (LR + t_n)/2$$

$$= 190.685 \cdot 1471.99 \cdot (367.933 + 177.0)/2$$

$$= 779886.4 \text{ Kgf}$$



Discontinuity Force from Internal Pressure [FY]:

$$= (P \cdot Rxs \cdot Rnc) / 2$$

$$= (190.685 \cdot 1471.99 \cdot 358.6) / 2$$

$$= 513213.6 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (779886 + 513214 + 255343.0) / 190.685$$

$$= 7963.1 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / (2 \cdot teff))$$

$$= 225.0 / (2 \cdot 7963.103 / 835.631 - 1471.99 / (2 \cdot 95.0))$$

$$= 198.9 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t/Rxs]$$

$$= 2 \cdot 150.0[95.0/1471.99]$$

$$= 193.6 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(198.9, 193.604)$$

$$= 193.604 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (255343.031 + 779886 + 513214) / 835.631$$

$$= 181.723 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / (2 \cdot teff)$$

$$= 190.685 \cdot 1471.99 / (2 \cdot 95.0)$$

$$= 147.7 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 181.723 - 147.738, 147.738)$$

$$= 215.7 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	225.00	N./mm ²
Local Primary Membrane Stress	PL	215.71	N./mm ²
Maximum Allowable Working Pressure	Pmax	193.60	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (358.6 + 177.0) / 358.6$$

$$= 1.494 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn)$$

$$= \pi / 2 \cdot (358.6 + 177.0)$$

$$= 841.318 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 177.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(513214 \cdot 1.49, 1.5 \cdot 142.0(562.123 + 0.0), \pi / 4 \cdot 190.7 \cdot 358.6^2 \cdot 1.49^2)$$

$$= 438110.781 \text{ Kgf}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$



= $438110.781 / (841.318(0.49 \cdot 7.071 + 0.6 \cdot 95.0 + 0.49 \cdot 0.0))$
= 84.460 < or = to 150.000 Weld Stress Passed

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

= $t_n / \ln[1 + t_n/R_n]$
= $177.0 / \ln[1 + 177.0/358.6]$
= 441.198 mm.

Shell Radius for Force Calculation [Rxs]:

= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
= $95.0 / \ln[1 + 95.0/1425.0]$
= 1471.990 mm.

Allowable Local Primary Membrane Stress [Sallow]:

= $1.5 \cdot S$
= $1.5 \cdot 150.0$
= 225.0 N./mm²

Force acting on the Nozzle [fN]:

= $P \cdot R_{xn} \cdot L_H$
= $1.034 \cdot 441.198 \cdot 297.63$
= 1384.9 Kgf

Force acting on the Shell [fS]:

= $P \cdot R_{xs} \cdot (L_R + t_n)/2$
= $1.034 \cdot 1471.99 \cdot (367.933 + 177.0)/2$
= 4229.8 Kgf

Discontinuity Force from External Pressure [fY]:

= $(P \cdot R_{xs} \cdot R_{nc})/2$
= $(1.034 \cdot 1471.99 \cdot 358.6)/2$
= 2783.4 Kgf

Area Resisting External Pressure [Ap]:

= $(f_S + f_Y + f_N)/P$
= $(4229.8 + 2783.4 + 1384.9)/1.034$
= 7963.1 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= $S_{allow} / (2 \cdot A_p / A_T - R_{xs} / (2 \cdot t_{eff}))$
= $225.0 / (2 \cdot 7963.103 / 835.631 - 1471.99 / (2 \cdot 95.0))$
= 198.9 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:

= $2 \cdot S[t/R_{xs}]$
= $2 \cdot 150.0[95.0/1471.99]$
= 193.6 bars

Maximum Allowable Working Pressure [Pmax]:

= min(Pmax1, Pmax2)
= min(198.9, 193.604)
= 193.604 bars

Average Primary Membrane Stress [σ_{Avg}]:

= $(f_N + f_S + f_Y) / A_T$
= $(1384.867 + 4229.756 + 2783.442) / 835.631$
= 0.986 N./mm²

General Primary Membrane Stress [σ_{Circ}]:

= $P \cdot R_{xs} / (2 \cdot t_{eff})$
= $1.034 \cdot 1471.99 / (2 \cdot 95.0)$
= 0.8 N./mm²

Maximum Local Primary Membrane Stress [PL]:

= max($2 \cdot \sigma_{Avg} - \sigma_{Circ}$, σ_{Circ})
= max($2 \cdot 0.986 - 0.801$, 0.801)
= 1.2 N./mm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	225.00	N./mm ²
Local Primary Membrane Stress	PL	1.17	N./mm ²
Maximum Allowable External Pressure	Pmax	193.60	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Post Weld Heat Treated PWHT	
Computed Minimum Design Metal Temperature	: 34 °C
User entered Minimum Design Temperature	: -10 °C
The Curve for this Material is	: A
From Figure 3.8M- with PWHT, the MDMT is	: 42 °C
Temperature reduction per Graph 3.13	: 8 °C
The computed stress ratio is	: 0.840

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (79.795 \cdot 1.0) / (98.0 - 3.0 - 0.0)$$

$$= 0.840$$

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 34 °C

The Drop for this Nozzle is : 104.7237 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 888.2238 mm.

Input Echo, WRC107/537 Item 1, Description: N228" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	2844.000	mm.
Vessel Thickness	Tv	98.000	mm.
Design Temperature	T1	200.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	175.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	150.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm.
Nozzle Diameter	Dn	711.200	mm.
Nozzle Thickness	Tn	180.000	mm.
Nozzle Material		SA-105	
Nozzle UNS Number		K03504	
Nozzle Cold S.I. Allowable	SNmc	165.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	142.00	N./mm ²
Design Internal Pressure	Dp	190.685	bars
Include Pressure Thrust		No	

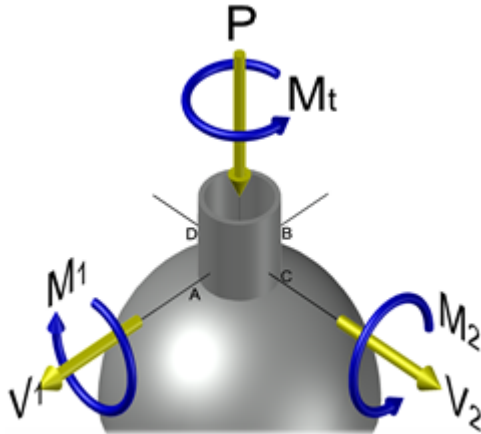
External Forces and Moments in WRC 107/537 Convention:

Radial Load	(SUS)	P	-17132.4	Kgf
Longitudinal Shear	(SUS)	(V1) V1	15143.0	Kgf
Circumferential Shear	(SUS)	(Vc) V2	15143.0	Kgf
Circumferential Moment	(SUS)	(Mc) M1	27803.9	Kg-m.
Longitudinal Moment	(SUS)	(Ml) M2	27803.9	Kg-m.
Torsional Moment	(SUS)	Mt	35964.1	Kg-m.

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 1071.2 + 2 \cdot 1.65 \cdot \sqrt{(1472.5 (98.0 - 3.0))}$$

$$= 2305.451 \text{ mm.}$$

WRC 107 Stress Calculation for SUSained loads:

Radial Load	P	-17132.4	Kgf
Circumferential Shear	(VC) V2	15143.0	Kgf
Longitudinal Shear	(VL) V1	15143.0	Kgf
Circumferential Moment	(MC) M1	27803.9	Kg-m.
Longitudinal Moment	(ML) M2	27803.9	Kg-m.
Torsional Moment	MT	35964.1	Kg-m.

Dimensionless Param: U = 1.43 TAU = 5.00 (2.53) RHO = 0.54

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.02798	(A,B,C,D)
$M(x) / P$	SP 1	0.02580	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.02855	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.03509	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.02855	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.03509	(A,B,C,D)
$N(y) * T / P$	SP 1	0.04155	(A,B,C,D)
$M(y) / P$	SP 1	0.00738	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.03880	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.01166	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.03880	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.01166	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Stress Intensity Values at



Type of Stress	Load	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Rad.	Bend. P	2.9	-2.9	2.9	-2.9	2.9	-2.9	2.9	-2.9
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-2.3	-2.3	2.3	2.3
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-17.0	17.0	17.0	-17.0
Rad.	Memb. ML	-2.3	-2.3	2.3	2.3	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-17.0	17.0	17.0	-17.0	0.0	0.0	0.0	0.0
Tot. Rad.	Str.	-15.9	12.3	22.7	-17.1	-15.9	12.3	22.7	-17.1
Tang.	Memb. P	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
Tang.	Bend. P	0.8	-0.8	0.8	-0.8	0.8	-0.8	0.8	-0.8
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-3.1	-3.1	3.1	3.1
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-5.6	5.6	5.6	-5.6
Tang.	Memb. ML	-3.1	-3.1	3.1	3.1	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-5.6	5.6	5.6	-5.6	0.0	0.0	0.0	0.0
Tot. Tang.	Str.	-7.2	2.5	10.4	-2.6	-7.2	2.5	10.4	-2.6
	Shear VC	0.9	0.9	-0.9	-0.9	0.0	0.0	0.0	0.0
	Shear VL	0.0	0.0	0.0	0.0	-0.9	-0.9	0.9	0.9
	Shear MT	2.1	2.1	2.1	2.1	2.1	2.1	2.1	2.1
	Tot. Shear	3.0	3.0	1.1	1.1	1.1	1.1	3.0	3.0
	Str. Int.	16.8	13.2	22.8	17.1	16.1	12.5	23.4	17.7

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	147.7	147.7	147.7	147.7	147.7	147.7	147.7	147.7
Rad.	Pl (SUS)	-1.8	-1.8	2.8	2.8	-1.8	-1.8	2.8	2.8
Rad.	Q (SUS)	-14.1	14.1	19.9	-19.9	-14.1	14.1	19.9	-19.9
Long.	Pm (SUS)	147.7	147.7	147.7	147.7	147.7	147.7	147.7	147.7
Long.	Pl (SUS)	-2.4	-2.4	3.9	3.9	-2.4	-2.4	3.9	3.9
Long.	Q (SUS)	-4.8	4.8	6.5	-6.5	-4.8	4.8	6.5	-6.5
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.9	0.9	-0.9	-0.9	-0.9	-0.9	0.9	0.9
Shear	Q (SUS)	2.1	2.1	2.1	2.1	2.1	2.1	2.1	2.1
	Pm (SUS)	147.7	147.7	147.7	147.7	147.7	147.7	147.7	147.7
	Pm+Pl (SUS)	146.6	146.6	152.2	152.2	146.6	146.6	152.2	152.2
	Pm+Pl+Q (Total)	141.5	160.9	170.6	145.3	140.7	160.2	171.1	145.8

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	147.74	150.00	Passed
Pm+Pl (SUS)	152.18	225.00	Passed
Pm+Pl+Q (TOTAL)	171.14	450.02	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.





Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

- ** Warning: One or more Nozzles failed Code Requirements!**
- ** Warning: One or more nozzles do not meet proper Spacing Requirements!**
- ** Error: Insufficient bolt area on one or more flanges!**
- ** Error: Skirt Overstress Failure!**
- ** Warning: The computed vessel MAWP seems to be less than the design pressure!**

Please review all reports carefully!

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PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: